



CHAPTE 3

MATRICES AND DETERMINANTS

Matrix:

A rectangular array of numbers enclosed by a pair of brackets e.g $\begin{bmatrix} 2 & -1 & 3 \\ -5 & 4 & 7 \end{bmatrix}$ is called matrix.

Points to Remember:

- Horizontal lines of numbers are called rows.
- Vertical lines of numbers are called columns.
- If there are m rows and n columns then order of matrix is defined as $m \times n$.

Row Matrix or Row Vector:

A matrix, which has only one row, i.e, a $1 \times n$ matrix of the form $[a_{i1} \ a_{i2} \ a_{i3} \ \dots \ a_{in}]$ is said to be a row matrix or row vector. For example $[1 \ -1 \ 3 \ 4]$ is a row matrix having 4 columns.

Column Matrix or Column Vector:

A matrix which has only one column i.e $m \times 1$ matrix of the form $\begin{bmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \\ a_{mj} \end{bmatrix}$ is said to be a

column matrix or a column vector. For example $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ is a column matrix having 3 rows.

Rectangular Matrix:

If $m \neq n$, then the matrix is called a rectangular matrix of order $m \times n$, that is, the matrix in which number of rows are not equal to the number of columns, is said to be a rectangular matrix for example; $\begin{bmatrix} 2 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$ is rectangular matrix of order 2×3 .

Square Matrix:

If $m = n$, then the matrix of the order $m \times n$ is said to be a square matrix of order n or m . i.e., the matrix which has same number of rows and columns is called a square matrix. For example;

$$\begin{bmatrix} 2 & 5 \\ -1 & 6 \end{bmatrix} \text{ is a square matrix of order 2.}$$

Points to Remember:

- Let $A = [a_{ij}]$ be a square matrix of order n , then the entries $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ form a **principal diagonal** for the matrix A and the entries $a_{1n}, a_{2n-1}, a_{3n-2}, \dots, a_{n-1,2}, a_{n1}$ form secondary diagonal for the matrix A .
- The principal diagonal is also named as **leading diagonal** or **main diagonal**.
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Diagonal Matrix:

Let $A = [a_{ij}]$ be a square matrix of order ' n '. If $a_{ij} = 0 \forall i \neq j$ and at least one $a_{ij} \neq 0$ for $i = j$, that is, some elements of the principal diagonal of A may be zero but not all, then the matrix A is called a **diagonal matrix**. The matrices

$$[7], \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \text{ are diagonal matrices.}$$

Scalar Matrix:

Let $A = [a_{ij}]$ be a square matrix of order ' n '. If $a_{ij} = 0 \forall i \neq j$ and $a_{ij} = k$ (some non-zero scalar) $\forall i = j$, then the matrix is called a **scalar matrix** of order ' n '. For example

$$\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}, \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \text{ are scalar matrices of order 2 and 4.}$$

Unit Matrix or Identity Matrix:

Let $A = [a_{ij}]$ be a square matrix of order n . If $a_{ij} = 0 \forall i \neq j$ and $a_{ij} = 1 \forall i = j$, then the matrix A is called a **unit matrix** or **identity matrix** of order n denoted by I_n . The

$$\text{identity matrix of order is 3 is denoted by } I_3, \text{ that is, } I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Null Matrix or Zero Matrix:

A square or rectangular matrix whose each element is zero, is called a **null** or **zero** matrix. An $m \times n$ matrix with all its elements equal to zero is denoted by $O_{m \times n}$. Here are some example

$$[0], [0 \ 0 \ 0], \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

O may be used to denote null matrix of any order if there is no confusion.

Equal Matrices:

Two matrices of the same order are said to be **equal** if their corresponding entries are

equal e.g. $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ are equal i.e., $A = B$ iff $a_{ij} = b_{ij}$ for

$i = 1, 2, 3, \dots, m, j = 1, 2, 3, \dots, n$.

Points to Remember:

- Two matrices are confirmable for addition if they are of the same order.
- If $A = [a_{ij}]$ is $m \times n$ matrix and k is a scalar then $kA = [ka_{ij}]$.
- Two matrices A and B are said to be confirmable for the product AB if the number of columns of A is equal to the number of rows of B .

If $O(A) = 2 \times 3$ and $O(B) = 3 \times 2$

$\Rightarrow O(AB) = 2 \times 2$

EXERCISE 3.1

Q.1 If $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 7 \\ 6 & 4 \end{bmatrix}$,

then show that

(i) $4A - 3A = A$

Solution:

$$4A - 3A = A$$

$$\text{L.H.S} = 4A - 3A$$

$$= 4 \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} - 3 \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 12 \\ 4 & 20 \end{bmatrix} - \begin{bmatrix} 6 & 9 \\ 3 & 15 \end{bmatrix}$$

$$= \begin{bmatrix} 8-6 & 12-9 \\ 4-3 & 20-15 \end{bmatrix}$$

$$\text{L.H.S} = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} = A = \text{R.H.S}$$

$$\text{L.H.S} = \text{R.H.S}$$

$$\text{Hence } 4A - 3A = A$$

(ii) $3B - 3A = 3(B - A)$

Solution: $\text{L.H.S} = 3B - 3A$

$$= 3 \begin{bmatrix} 1 & 7 \\ 6 & 4 \end{bmatrix} - 3 \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 21 \\ 18 & 12 \end{bmatrix} - \begin{bmatrix} 6 & 9 \\ 3 & 15 \end{bmatrix}$$

$$= \begin{bmatrix} 3-6 & 21-9 \\ 18-3 & 12-15 \end{bmatrix}$$

$$\text{L.H.S} = \begin{bmatrix} -3 & 12 \\ 15 & -3 \end{bmatrix} \quad (\text{i})$$

$$\text{R.H.S} = 3(B - A)$$

$$= 3 \left(\begin{bmatrix} 1 & 7 \\ 6 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \right)$$

$$= 3 \left(\begin{bmatrix} 1-2 & 7-3 \\ 6-1 & 4-5 \end{bmatrix} \right)$$

$$= 3 \begin{bmatrix} -1 & 4 \\ 5 & -1 \end{bmatrix}$$

$$\text{R.H.S} = \begin{bmatrix} -3 & 12 \\ 15 & -3 \end{bmatrix} \quad (\text{ii})$$

From (i) and (ii)

$$\text{L.H.S} = \text{R.H.S}$$

$$3B - 3A = 3(B - A)$$

Q.2 If $A = \begin{bmatrix} i & 0 \\ 1 & -i \end{bmatrix}$ show that $A^4 = I_2$

Solution: We know that

$$A^2 = A.A$$

$$= \begin{bmatrix} i & 0 \\ 1 & -i \end{bmatrix} \cdot \begin{bmatrix} i & 0 \\ 1 & -i \end{bmatrix}$$

$$= \begin{bmatrix} i \times i + 0 \times 1 & i \times 0 + 0 \times (-i) \\ 1 \times i + (-i \times 1) & 1 \times 0 + (-i) \times (-i) \end{bmatrix}$$

$$= \begin{bmatrix} i^2 + 0 & 0 + 0 \\ i - i & 0 + i^2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} i^2 & 0 \\ 0 & i^2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^4 = A^2.A^2$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} (-1 \times (-1)) + (0 \times 0) & (-1 \times 0) + (0 \times (-1)) \\ (0 \times (-1)) + (-1 \times 0) & (0 \times 0) + (-1 \times (-1)) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I_2$$

$$\text{Hence } A^4 = I_2$$

Q.3 Find x and y if

$$(i) \begin{bmatrix} x+3 & 1 \\ -3 & 3y-4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} x+3 & 1 \\ -3 & 3y-4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$$

By definition of equal matrices, if two matrices are equal, then their corresponding elements must be equal.

$$x+3=2 \quad \text{and} \quad 3y-4=2$$

$$\Rightarrow x=2-3, \quad \Rightarrow 3y=6$$

$$\Rightarrow x=-1 \quad \Rightarrow y=2$$

Hence $x=-1$ and $y=2$ are the required values of x and y .

$$(ii) \begin{bmatrix} x+3 & 1 \\ -3 & 3y-4 \end{bmatrix} = \begin{bmatrix} y & 1 \\ -3 & 2x \end{bmatrix}$$

Solution:

By definition of equal matrices, if two matrices are equal, then their

corresponding elements must be equal.

$$x+3=y \quad \text{and} \quad 3y-4=2x$$

$$x-y=-3 \quad (i)$$

$$2x-3y=-4 \quad (ii)$$

Multiply 2 to equation (i) and subtracting from (ii)

$$2x-3y=-4$$

$$\underline{-2x+2y=-6}$$

$$-y=2$$

$$\Rightarrow y=-2$$

Put the value of y in (i)

$$x-(-2)=-3$$

$$x+2=-3$$

$$\Rightarrow x=-5$$

Hence $x=-5$ and $y=-2$ are the required values

$$\text{Q.4 If } A = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 3 & 2 \\ 1 & -1 & 2 \end{bmatrix}$$

Find the following matrices,

$$(i) \quad 4A-3B$$

Solution:

$$\begin{aligned} 4A-3B &= 4 \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \end{bmatrix} - 3 \begin{bmatrix} 0 & 3 & 2 \\ 1 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 8 & 12 \\ 4 & 0 & 8 \end{bmatrix} - \begin{bmatrix} 0 & 9 & 6 \\ 3 & -3 & 6 \end{bmatrix} \\ &= \begin{bmatrix} -4-0 & 8-9 & 12-6 \\ 4-3 & 0+3 & 8-6 \end{bmatrix} \\ &= \begin{bmatrix} -4 & -1 & 6 \\ 1 & 3 & 2 \end{bmatrix} \end{aligned}$$

Hence

$$4A - 3B = \begin{bmatrix} -4 & -1 & 6 \\ 1 & 3 & 2 \end{bmatrix}$$

(ii) $A + 3(B - A)$

Solution:

$$\begin{aligned} \text{Consider } B - A &= \begin{bmatrix} 0 & 3 & 2 \\ 1 & -1 & 2 \end{bmatrix} - \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0+1 & 3-2 & 2-3 \\ 1-1 & -1-0 & 2-2 \end{bmatrix} \end{aligned}$$

$$B - A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Now $A + 3(B - A)$

$$\begin{aligned} &= \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \end{bmatrix} + 3 \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 3 & -3 \\ 0 & -3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -1+3 & 2+3 & 3-3 \\ 1+0 & 0-3 & 2+0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 5 & 0 \\ 1 & -3 & 2 \end{bmatrix} \end{aligned}$$

Hence

$$A + 3(B - A) = \begin{bmatrix} 2 & 5 & 0 \\ 1 & -3 & 2 \end{bmatrix}$$

Q.5 Find x and y if $\begin{bmatrix} 2 & 0 & x \\ 1 & y & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & x & y \\ 0 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 6 & 1 \end{bmatrix}$

Solution: $\begin{bmatrix} 2 & 0 & x \\ 1 & y & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & x & y \\ 0 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 6 & 1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 0 & x \\ 1 & y & 3 \end{bmatrix} + \begin{bmatrix} 2 & 2x & 2y \\ 0 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 6 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2+2 & 0+2x & x+2y \\ 1+0 & y+4 & 3-2 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 6 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 2x & x+2y \\ 1 & y+4 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 3 \\ 1 & 6 & 1 \end{bmatrix}$$

By definition of equal matrices, if two matrices are equal, then their corresponding elements must be equal.

$$2x = -2 \quad \text{and} \quad y + 4 = 6$$

$$\Rightarrow x = -1 \quad \text{and} \quad y = 2$$

So,

$$x = -1 \text{ and } y = 2$$

Are the required values of x and y .

Q.6 If $A = [a_{ij}]_{3 \times 3}$, show that

$$(i) \quad \lambda(\mu A) = (\lambda\mu)A$$

Solution: We have

$$A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

If μ is a scalar then by scalar multiplication

$$\begin{aligned} \mu A &= \mu \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\ \mu A &= \begin{bmatrix} \mu a_{11} & \mu a_{12} & \mu a_{13} \\ \mu a_{21} & \mu a_{22} & \mu a_{23} \\ \mu a_{31} & \mu a_{32} & \mu a_{33} \end{bmatrix} \end{aligned}$$

Now consider λ is another scalar then

$$\begin{aligned} \lambda(\mu A) &= \lambda \begin{bmatrix} \mu a_{11} & \mu a_{12} & \mu a_{13} \\ \mu a_{21} & \mu a_{22} & \mu a_{23} \\ \mu a_{31} & \mu a_{32} & \mu a_{33} \end{bmatrix} \\ \lambda(\mu A) &= \begin{bmatrix} \lambda \mu a_{11} & \lambda \mu a_{12} & \lambda \mu a_{13} \\ \lambda \mu a_{21} & \lambda \mu a_{22} & \lambda \mu a_{23} \\ \lambda \mu a_{31} & \lambda \mu a_{32} & \lambda \mu a_{33} \end{bmatrix} \\ &= (\lambda\mu) \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \because \text{by taking common } \lambda\mu \\ &= (\lambda\mu)A \end{aligned}$$

Hence $\lambda(\mu A) = (\lambda\mu)A$

$$(ii) \quad (\lambda + \mu)A = \lambda A + \mu A$$

Solution: Consider R.H.S = $\lambda A + \mu A$

$$= \lambda \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \mu \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \lambda a_{13} \\ \lambda a_{21} & \lambda a_{22} & \lambda a_{23} \\ \lambda a_{31} & \lambda a_{32} & \lambda a_{33} \end{bmatrix} + \begin{bmatrix} \mu a_{11} & \mu a_{12} & \mu a_{13} \\ \mu a_{21} & \mu a_{22} & \mu a_{23} \\ \mu a_{31} & \mu a_{32} & \mu a_{33} \end{bmatrix} \\
&= \begin{bmatrix} \lambda a_{11} + \mu a_{11} & \lambda a_{12} + \mu a_{12} & \lambda a_{13} + \mu a_{13} \\ \lambda a_{21} + \mu a_{21} & \lambda a_{22} + \mu a_{22} & \lambda a_{23} + \mu a_{23} \\ \lambda a_{31} + \mu a_{31} & \lambda a_{32} + \mu a_{32} & \lambda a_{33} + \mu a_{33} \end{bmatrix} \\
&= \begin{bmatrix} (\lambda + \mu)a_{11} & (\lambda + \mu)a_{12} & (\lambda + \mu)a_{13} \\ (\lambda + \mu)a_{21} & (\lambda + \mu)a_{22} & (\lambda + \mu)a_{23} \\ (\lambda + \mu)a_{31} & (\lambda + \mu)a_{32} & (\lambda + \mu)a_{33} \end{bmatrix} \\
&= (\lambda + \mu) \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\
&= (\lambda + \mu)A = \text{L.H.S}
\end{aligned}$$

Hence $(\lambda + \mu)A = \lambda A + \mu A$

(iii) $\lambda A - A = (\lambda - 1)A$

Solution: Consider L.H.S = $\lambda A - A$

$$\begin{aligned}
&= \lambda \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} - \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\
&= \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \lambda a_{13} \\ \lambda a_{21} & \lambda a_{22} & \lambda a_{23} \\ \lambda a_{31} & \lambda a_{32} & \lambda a_{33} \end{bmatrix} - \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\
&= \begin{bmatrix} \lambda a_{11} - a_{11} & \lambda a_{12} - a_{12} & \lambda a_{13} - a_{13} \\ \lambda a_{21} - a_{21} & \lambda a_{22} - a_{22} & \lambda a_{23} - a_{23} \\ \lambda a_{31} - a_{31} & \lambda a_{32} - a_{32} & \lambda a_{33} - a_{33} \end{bmatrix} \\
&= \begin{bmatrix} (\lambda - 1)a_{11} & (\lambda - 1)a_{12} & (\lambda - 1)a_{13} \\ (\lambda - 1)a_{21} & (\lambda - 1)a_{22} & (\lambda - 1)a_{23} \\ (\lambda - 1)a_{31} & (\lambda - 1)a_{32} & (\lambda - 1)a_{33} \end{bmatrix} \\
&= (\lambda - 1) \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\
&= (\lambda - 1)A \\
&= \text{R.H.S}
\end{aligned}$$

$$\text{L.H.S} = \text{R.H.S}$$

$$\text{So } \lambda A - A = (\lambda - 1)A$$

Q.7 If $A = [a_{ij}]_{2 \times 3}$ and $B = [b_{ij}]_{2 \times 3}$, show that $\lambda(A + B) = \lambda A + \lambda B$

Solution: Let $A = [a_{ij}]_{2 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ and $B = [b_{ij}]_{2 \times 3} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$

$$\text{Consider R.H.S} = \lambda A + \lambda B$$

$$\begin{aligned} &= \lambda \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \lambda \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ &= \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \lambda a_{13} \\ \lambda a_{21} & \lambda a_{22} & \lambda a_{23} \end{bmatrix} + \begin{bmatrix} \lambda b_{11} & \lambda b_{12} & \lambda b_{13} \\ \lambda b_{21} & \lambda b_{22} & \lambda b_{23} \end{bmatrix} \\ &= \begin{bmatrix} \lambda a_{11} + \lambda b_{11} & \lambda a_{12} + \lambda b_{12} & \lambda a_{13} + \lambda b_{13} \\ \lambda a_{21} + \lambda b_{21} & \lambda a_{22} + \lambda b_{22} & \lambda a_{23} + \lambda b_{23} \end{bmatrix} \\ &= \begin{bmatrix} \lambda(a_{11} + b_{11}) & \lambda(a_{12} + b_{12}) & \lambda(a_{13} + b_{13}) \\ \lambda(a_{21} + b_{21}) & \lambda(a_{22} + b_{22}) & \lambda(a_{23} + b_{23}) \end{bmatrix} \\ &= \lambda \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{bmatrix} \end{aligned} \tag{i}$$

Now consider

$$\begin{aligned} A + B &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ A + B &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{bmatrix} \end{aligned} \tag{ii}$$

So by using (ii), (i) becomes

$$\lambda(A + B) = \text{L.H.S}$$

$$\text{So L.H.S} = \text{R.H.S}$$

$$\text{Hence } \lambda(A + B) = \lambda A + \lambda B$$

Q.8 If $A = \begin{bmatrix} 1 & 2 \\ a & b \end{bmatrix}$ and $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, find the values of a and b.

Solution: We know that

$$\begin{aligned} A^2 &= A \cdot A \\ &= \begin{bmatrix} 1 & 2 \\ a & b \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ a & b \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 1 \times 1 + 2 \times a & 1 \times 2 + 2 \times b \\ a \times 1 + b \times a & a \times 2 + b \times b \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 + 2a & 2 + 2b \\ a + ab & 2a + b^2 \end{bmatrix}$$

Now by definition of equal matrices

$$1 + 2a = 0 \quad \text{and} \quad 2 + 2b = 0$$

$$\Rightarrow 2a = -1$$

$$\Rightarrow a = -\frac{1}{2} \quad \text{and} \quad 2b = -2$$

$$\Rightarrow b = -1$$

So, $a = -\frac{1}{2}$ and $b = -1$ are required values of a and b .

Q.9 If $A = \begin{bmatrix} 1 & -1 \\ a & b \end{bmatrix}$ and $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, Find the value of a and b .

Solution: We know that

$$\begin{aligned} A^2 &= A.A \\ &= \begin{bmatrix} 1 & -1 \\ a & b \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ a & b \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 1 + (-1) \times a & 1 \times (-1) + (-1) \times b \\ a \times 1 + b \times a & a \times (-1) + b \times b \end{bmatrix} \\ &= \begin{bmatrix} 1 - a & -1 - b \\ a + ab & -a + b^2 \end{bmatrix} \\ &= \begin{bmatrix} 1 - a & -1 - b \\ a + ab & -a + b^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Now by definition of equal matrices

$$1 - a = 1 \quad \text{and} \quad -1 - b = 0$$

$$\Rightarrow -a = 0 \quad \Rightarrow b = -1$$

$$\Rightarrow a = 0$$

Hence $a = 0$ and $b = -1$ are required values of a and b .

Q.10 If $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 2 & -1 \end{bmatrix}$, then show that $(A + B)^t = A^t + B^t$

Solution: As $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}$, then $A^t = \begin{bmatrix} 1 & 0 \\ -1 & 3 \\ 2 & 1 \end{bmatrix}$

$B = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 2 & -1 \end{bmatrix}$, then $B^t = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 0 & -1 \end{bmatrix}$

Now

$$\begin{aligned}
 A^t + B^t &= \begin{bmatrix} 1 & 0 \\ -1 & 3 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 0 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} 1+2 & 0+1 \\ -1+3 & 3+2 \\ 2+0 & 1+(-1) \end{bmatrix} \\
 &= \begin{bmatrix} 3 & 1 \\ 2 & 5 \\ 2 & 0 \end{bmatrix} \quad \text{(i)}
 \end{aligned}$$

Now consider

$$\begin{aligned}
 A+B &= \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 0 \\ 1 & 2 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} 1+2 & -1+3 & 2+0 \\ 0+1 & 3+2 & 1-1 \end{bmatrix} \\
 A+B &= \begin{bmatrix} 3 & 2 & 2 \\ 1 & 5 & 0 \end{bmatrix}
 \end{aligned}$$

By taking transpose

$$\begin{aligned}
 (A+B)^t &= \begin{bmatrix} 3 & 2 & 2 \\ 1 & 5 & 0 \end{bmatrix}^t \\
 &= \begin{bmatrix} 3 & 1 \\ 2 & 5 \\ 2 & 0 \end{bmatrix} \text{———(ii)}
 \end{aligned}$$

From (i) and (ii)

$$(A+B)^t = A^t + B^t$$

Q.11 Find A^3 if $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

Solution: We know that

$$\begin{aligned}
 A^2 &= A \cdot A \\
 &= \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \times 1 + 1 \times 5 + 3 \times (-2) & 1 \times 1 + 1 \times 2 + 3 \times (-1) & 1 \times 3 + 1 \times 6 + 3 \times (-3) \\ 5 \times 1 + 2 \times 5 + 6 \times (-2) & 5 \times 1 + 2 \times 2 + 6 \times (-1) & 5 \times 3 + 2 \times 6 + 6 \times (-3) \\ -2 \times 1 + (-1) \times 5 + (-3) \times (-2) & -2 \times 1 + (-1) \times 2 + (-3) \times (-1) & -2 \times 3 + (-1) \times 6 + (-3) \times (-3) \end{bmatrix}
 \end{aligned}$$

$$= \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

Now $A^3 = A^2.A$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \times 1 + 0 \times 5 + 0 \times (-2) & 0 \times 1 + 0 \times 2 + 0 \times (-1) & 0 \times 3 + 0 \times 6 + 0 \times (-3) \\ 3 \times 1 + 3 \times 5 + 9 \times (-2) & 3 \times 1 + 3 \times 2 + 9 \times (-1) & 3 \times 3 + 3 \times 6 + 9 \times (-3) \\ -1 \times 1 + (-1) \times 5 + (-3) \times (-2) & (-1) \times 1 + (-1) \times 2 + (-3) \times (-1) & -1 \times 3 + (-1) \times 6 + (-3) \times (-3) \end{bmatrix}$$

$$= \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 3+15-18 & 3+6-9 & 9+18-27 \\ -1-5+6 & -1-2+3 & -3-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence

$$A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O_3$$

Q.12 Find the matrix X if

$$(i) \quad X \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 12 & 3 \end{bmatrix}$$

Let

$$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 12 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5a-2b & 2a+b \\ 5c-2d & 2c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 12 & 3 \end{bmatrix}$$

Now by definition of equal matrices

$$5a - 2b = -1 \quad (i)$$

$$2a + b = 5 \quad (ii)$$

$$5c - 2d = 12 \quad (iii)$$

$$2c + d = 3 \quad (iv)$$

Multiply 2 to equation (ii) and adding in (i) multiply 2 to equation (iv) and adding in (iii)

$$5a - 2b = -1$$

$$5c - 2d = 12$$

$$\underline{4a + 2b = 10}$$

$$\underline{4c + 2d = 6}$$

$$9a = 9$$

$$9c = 18$$

$$\Rightarrow a = 1$$

$$\Rightarrow c = 2$$

Substituting $a = 1$ in (ii) we get

Substituting $c = 2$ in (iv) we get

$$2(1) + b = 5$$

$$2(2) + d = 3$$

$$\Rightarrow b = 3$$

$$\Rightarrow d = -1$$

Hence $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ is the enquired matrix

$$(ii) \quad \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} X = \begin{bmatrix} 2 & 1 \\ 5 & 10 \end{bmatrix}$$

Let

$$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Then } \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5a + 2c & 5b + 2d \\ -2a + c & -2b + d \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 10 \end{bmatrix}$$

By definition of equal matrices

$$5a + 2c = 2 \quad (i)$$

$$-2a + c = 5 \quad (ii)$$

$$5b + 2d = 1 \quad (iii)$$

$$-2b + d = 10 \quad (iv)$$

Multiply 2 to equation (ii) and subtracting from (i) we get

multiply 2 to equation (iv) and subtracting from (iii) we get

$$5a + 2c = 2$$

$$5b + 2d = 1$$

$$\underline{-4a + 2c = -10}$$

$$\underline{-4b + 2d = -20}$$

$$9a = -8$$

$$9b = -19$$

$$\Rightarrow a = -\frac{8}{9}$$

Substituting the value of a in (ii) we get

$$-2\left(-\frac{8}{9}\right) + c = 5$$

$$\frac{16}{9} + c = 5$$

$$c = 5 - \frac{16}{9}$$

$$= \frac{45-16}{9}$$

$$\Rightarrow c = \frac{29}{9}$$

$$\Rightarrow b = \frac{-19}{9}$$

Substituting the value of b in (iv) we get

$$-2\left(-\frac{19}{9}\right) + d = 10$$

$$\frac{38}{9} + d = 10$$

$$d = 10 - \frac{38}{9}$$

$$= \frac{90-38}{9}$$

$$\Rightarrow d = \frac{52}{9}$$

$$\text{Hence } X = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -\frac{8}{9} & -\frac{19}{9} \\ \frac{29}{9} & \frac{52}{9} \end{bmatrix}$$

Q.13 Find the matrix A if,

$$(i) \quad \begin{bmatrix} 5 & -1 \\ 0 & 0 \\ 3 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & -7 \\ 0 & 0 \\ 7 & 2 \end{bmatrix}$$

Solution:

$$\text{Since } \begin{bmatrix} 5 & -1 \\ 0 & 0 \\ 3 & 1 \end{bmatrix}_{3 \times 2} A_{2 \times 2} = \begin{bmatrix} 3 & -7 \\ 0 & 0 \\ 7 & 2 \end{bmatrix}_{3 \times 2}$$

$$\text{So let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then}$$

$$\begin{bmatrix} 5 & -1 \\ 0 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3 & -7 \\ 0 & 0 \\ 7 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5a-c & 5b-d \\ 0+0 & 0+0 \\ 3a+c & 3b+d \end{bmatrix} = \begin{bmatrix} 3 & -7 \\ 0 & 0 \\ 7 & 2 \end{bmatrix}$$

By definition of equal matrices, we get

$$5a - c = 3 \quad \text{(i)}$$

$$3a + c = 7 \quad \text{(ii)}$$

$$5b - d = -7 \quad \text{(iii)}$$

$$3b + d = 2 \quad \text{(iv)}$$

Adding (i) and (ii)

$$8a = 10$$

$$\Rightarrow a = \frac{5}{4}$$

Substituting the value of a in (ii) we get

$$3\left(\frac{5}{4}\right) + c = 7$$

$$\Rightarrow c = 7 - \frac{15}{4}$$

$$c = \frac{13}{4}$$

Hence

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \frac{5}{4} & -\frac{5}{8} \\ \frac{13}{4} & \frac{31}{8} \end{bmatrix} \text{ is the required matrix.}$$

$$\text{(ii)} \quad \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} A = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 3 & -7 \end{bmatrix}$$

Solution:

$$\text{Since } \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}_{2 \times 2} A_{2 \times 3} = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 3 & -7 \end{bmatrix}_{2 \times 3}$$

$$\text{So let } A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}, \text{ then}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 3 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 2a-d & 2b-e & 2c-f \\ -a+2d & -b+2e & -c+2f \end{bmatrix} = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 3 & -7 \end{bmatrix}$$

By definition of equal matrices, we get

$$2a - d = 0 \quad \text{(i)}$$

$$-a + 2d = 3 \quad \text{(ii)}$$

$$2b - e = -3 \quad \text{(iii)}$$

$$-b + 2e = 3 \quad \text{(iv)}$$

$$2c - f = 8 \quad \text{(v)}$$

$$-c + 2f = -7 \quad \text{(vi)}$$

Multiply 2 to equation (i) and adding in (ii)

$$4a - 2d = 0$$

$$\underline{-a + 2d = 3}$$

$$3a = 3$$

$$\Rightarrow a = 1$$

Substituting $a = 1$ in (i) we get

$$2(1) - d = 0$$

$$\Rightarrow d = 2$$

Multiply 2 to equation (iii) and adding in (iv)

$$4b - 2e = -6$$

$$\underline{-b + 2e = 3}$$

$$3b = -3$$

$$\Rightarrow b = -1$$

Substituting $b = -1$, in (iv) we get

$$-(-1) + 2e = 3$$

$$\Rightarrow 2e = 2$$

$$\Rightarrow e = 1$$

Multiply 2 to equation (v) and adding in (vi) we get

$$4c - 2f = 16$$

$$\underline{-c + 2f = -7}$$

$$3c = 9$$

$$\Rightarrow c = 3$$

Substituting $c = 3$ in (vi) we get,

$$-(3) + 2f = -7$$

$$2f = -4$$

$$\Rightarrow f = -2$$

$$\text{Hence } A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 1 & -2 \end{bmatrix}$$

Q.14 Show that

$$\begin{bmatrix} r\cos\phi & 0 & -\sin\phi \\ 0 & r & 0 \\ r\sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -r\sin\phi & 0 & r\cos\phi \end{bmatrix} = rI_3$$

Proof: Consider L.H.S = $\begin{bmatrix} r\cos\phi & 0 & -\sin\phi \\ 0 & r & 0 \\ r\sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -r\sin\phi & 0 & r\cos\phi \end{bmatrix}$

$$= \begin{bmatrix} r\cos^2\phi + 0 + r\sin^2\phi & 0 + 0 + 0 & r\cos\phi\sin\phi + 0 - r\cos\phi\sin\phi \\ 0 + 0 + 0 & 0 + r + 0 & 0 + 0 + 0 \\ r\sin\phi\cos\phi + 0 + (-r\sin\phi)\cos\phi & 0 + 0 + 0 & r\sin^2\phi + 0 + r\cos^2\phi \end{bmatrix}$$

$$= \begin{bmatrix} r(\cos^2\phi + \sin^2\phi) & 0 & r\sin\phi\cos\phi - r\sin\phi\cos\phi \\ 0 & r & 0 \\ r\sin\phi - r\sin\phi & 0 & r(\sin^2\phi + \cos^2\phi) \end{bmatrix}$$

$$= \begin{bmatrix} r(1) & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r(1) \end{bmatrix}$$

$$= r \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= rI_3 = \text{R.H.S}$$

Since L.H.S = R.H.S hence proved.

Properties of Matrix Addition, Scalar Multiplication and Matrix Multiplication:

If A, B and C are $n \times n$ matrices and c, d are scalars, the following properties are true:

1. Commutative property w.r.t. addition: $A + B = B + A$
2. Associative property w.r.t addition: $(A + B) + C = A + (B + C)$
3. Associative property of scalar multiplication: $(cd)A = c(dA)$.
4. Existence of additive identity: $A + O = O + A = A$ (O is null matrix)
5. Existence of multiplicative identity: $IA = AI = A$ (I is unit/identity matrix)
6. Distributive property w.r.t scalar multiplication:
(a) $c(A + B) = cA + cB$ (b) $(c + d)A = cA + dA$
7. Associative property w.r.t multiplication: $A(BC) = (AB)C$
8. Left distributive property: $A(B + C) = AB + AC$
9. Right distributive property: $(A + B)C = AC + BC$
10. $c(AB) = (cA)B = A(cB)$.

EXERCISE 3.2

Q.1 If $A = [a_{ij}]_{3 \times 4}$ then show that

i) $I_3 A = A$ **ii)** $A I_4 = A$

We have

$$A = [a_{ij}]_{3 \times 4}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(i) $I_3 A = A$

Consider L.H.S = $I_3 A$

$$\begin{aligned} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \\ &= \begin{bmatrix} 1 \times a_{11} + 0 \times a_{21} + 0 \times a_{31} & 1 \times a_{12} + 0 \times a_{22} + 0 \times a_{32} & 1 \times a_{13} + 0 \times a_{23} + 0 \times a_{33} & 1 \times a_{14} + 0 \times a_{24} + 0 \times a_{34} \\ 0 \times a_{11} + 1 \times a_{21} + 0 \times a_{31} & 0 \times a_{12} + 1 \times a_{22} + 0 \times a_{32} & 0 \times a_{13} + 1 \times a_{23} + 0 \times a_{33} & 0 \times a_{14} + 1 \times a_{24} + 0 \times a_{34} \\ 0 \times a_{11} + 0 \times a_{21} + 1 \times a_{31} & 0 \times a_{12} + 0 \times a_{22} + 1 \times a_{32} & 0 \times a_{13} + 0 \times a_{23} + 1 \times a_{33} & 0 \times a_{14} + 0 \times a_{24} + 1 \times a_{34} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \end{aligned}$$

$$= A = \text{R.H.S}$$

$$\text{L.H.S} = \text{R.H.S}$$

Hence

$$I_3 A = A$$

(ii) $A I_4 = A$

Consider L.H.S = $A I_4$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} a_{11} \times 1 + a_{12} \times 0 + a_{13} \times 0 + a_{14} \times 0 & a_{11} \times 0 + a_{12} \times 1 + a_{13} \times 0 + a_{14} \times 0 \\ a_{21} \times 1 + a_{22} \times 0 + a_{23} \times 0 + a_{24} \times 0 & a_{21} \times 0 + a_{22} \times 1 + a_{23} \times 0 + a_{24} \times 0 \\ a_{31} \times 1 + a_{32} \times 0 + a_{33} \times 0 + a_{34} \times 0 & a_{31} \times 0 + a_{32} \times 1 + a_{33} \times 0 + a_{34} \times 0 \\ a_{11} \times 0 + a_{12} \times 0 + a_{13} \times 1 + a_{14} \times 0 & a_{11} \times 0 + a_{12} \times 0 + a_{13} \times 0 + a_{14} \times 1 \\ a_{21} \times 0 + a_{22} \times 0 + a_{23} \times 1 + a_{24} \times 0 & a_{21} \times 0 + a_{22} \times 0 + a_{23} \times 0 + a_{24} \times 1 \\ a_{31} \times 0 + a_{32} \times 0 + a_{33} \times 1 + a_{34} \times 0 & a_{31} \times 0 + a_{32} \times 0 + a_{33} \times 0 + a_{34} \times 1 \end{bmatrix} \\
&= \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}
\end{aligned}$$

$$= A = \text{R.H.S}$$

$$\text{L.H.S} = \text{R.H.S}$$

$$\text{Hence } AI_4 = A$$

Q.2 Find the inverse of the following matrices.

(i) $\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}$

Solution: Let $A = \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}$, then inverse of A will be A^{-1} given by

$$A^{-1} = \frac{\text{Adj}A}{|A|}, |A| \neq 0$$

$$\text{Now } \text{Adj}A = \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$$

$$\begin{aligned}
\text{And } |A| &= \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix} = 3 \times 1 - (2)(-1) \\
&= 3 + 2 \\
|A| &= 5 \neq 0
\end{aligned}$$

$$\text{Now } A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{\begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}}{5}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

$$\text{Hence inverse of } \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} \text{ is } \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

(ii) $\begin{bmatrix} -2 & 3 \\ -4 & 5 \end{bmatrix}$

Solution: Let $B = \begin{bmatrix} -2 & 3 \\ -4 & 5 \end{bmatrix}$, then inverse of B is given by

$$B^{-1} = \frac{AdjB}{|B|}, |B| \neq 0$$

Now $|B| = \begin{vmatrix} -2 & 3 \\ -4 & 5 \end{vmatrix} = (-2 \times 5) - (-4) \times 3$

$$= -10 + 12$$

$$|B| = 2 \neq 0$$

Now $AdjB = \begin{bmatrix} 5 & -3 \\ 4 & -2 \end{bmatrix}$

$$B^{-1} = \frac{AdjB}{|B|} = \frac{\begin{bmatrix} 5 & -3 \\ 4 & -2 \end{bmatrix}}{2}$$

$$= \begin{bmatrix} \frac{5}{2} & \frac{-3}{2} \\ 2 & -1 \end{bmatrix}$$

Hence inverse of $\begin{bmatrix} -2 & 3 \\ -4 & 5 \end{bmatrix}$ is $\begin{bmatrix} \frac{5}{2} & \frac{-3}{2} \\ 2 & -1 \end{bmatrix}$

(iii) $\begin{bmatrix} 2i & i \\ i & -i \end{bmatrix}$

Solution: Let $C = \begin{bmatrix} 2i & i \\ i & -i \end{bmatrix}$, then inverse of C is given by

$$C^{-1} = \frac{AdjC}{|C|}, |C| \neq 0$$

Now $|C| = \begin{vmatrix} 2i & i \\ i & -i \end{vmatrix} = 2i \times (-i) - (i \times i)$

$$= -2i^2 - i^2$$

$$= -2(-1) - (-1)$$

$$= 2 + 1$$

$$= 3$$

$$\text{Now } AdjC = \begin{bmatrix} -i & -i \\ -i & 2i \end{bmatrix}$$

$$C^{-1} = \frac{AdjC}{|C|}$$

$$= \frac{\begin{bmatrix} -i & -i \\ -i & 2i \end{bmatrix}}{3}$$

$$C^{-1} = \begin{bmatrix} \frac{-i}{3} & \frac{-i}{3} \\ \frac{-i}{3} & \frac{2i}{3} \end{bmatrix}$$

$$\text{Hence inverse of } \begin{bmatrix} 2i & i \\ i & -i \end{bmatrix} \text{ is } \begin{bmatrix} \frac{-i}{3} & \frac{-i}{3} \\ \frac{-i}{3} & \frac{2i}{3} \end{bmatrix}$$

$$\text{(iv)} \quad \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$$

Solution: Let $D = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$, then inverse of D is given by

$$D^{-1} = \frac{AdjD}{|D|}, |D| \neq 0$$

$$\text{Now } |D| = \begin{vmatrix} 2 & 1 \\ 6 & 3 \end{vmatrix}$$

$$= 2 \times 3 - 6 \times 1$$

$$= 6 - 6$$

Since $|D| = 0$ so inverse of $D = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$, is not possible

Q.3 Solve the following system of linear equations.

Hence equations

$$\text{(i)} \quad \begin{cases} 2x_1 - 3x_2 = 5 \\ 5x_1 + x_2 = 4 \end{cases}$$

Solution: The matrix form of the system $\begin{cases} 2x_1 - 3x_2 = 5 \\ 5x_1 + x_2 = 4 \end{cases}$ is

$$\begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow X = A^{-1}B \dots \text{(i)}$$

Where $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$

Now

$$|A| = \begin{vmatrix} 2 & -3 \\ 5 & 1 \end{vmatrix} = (2 \times 1) - (-5 \times (-3))$$

$$= 2 + 15$$

$$|A| = 17 \neq 0$$

And $\text{Adj}A = \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix}$, therefore

$$A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{\begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix}}{17}$$

Put the values of X , A^{-1} and B in (i)

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{17} \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$= \frac{1}{17} \begin{bmatrix} 1 \times 5 + 3 \times 4 \\ -5 \times 5 + 2 \times 4 \end{bmatrix}$$

$$= \frac{1}{17} \begin{bmatrix} 5 + 12 \\ -25 + 8 \end{bmatrix}$$

$$= \frac{1}{17} \begin{bmatrix} 17 \\ -17 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow x_1 = 1 \quad \text{and} \quad x_2 = -1$$

$$(ii) \quad \begin{cases} 4x_1 + 3x_2 = 5 \\ 3x_1 - x_2 = 7 \end{cases}$$

Solution: The matrix form of the system $\begin{cases} 4x_1 + 3x_2 = 5 \\ 3x_1 - x_2 = 7 \end{cases}$ is

$$\begin{bmatrix} 4 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow X = A^{-1}B \dots (i)$$

Where

$$A = \begin{bmatrix} 4 & 3 \\ 3 & -1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 4 & 3 \\ 3 & -1 \end{vmatrix} = (-1 \times 4) - (3 \times 3) \\ = -4 - 9$$

$$|A| = -13 \neq 0$$

$$\text{And } \text{Adj}A = \begin{bmatrix} -1 & -3 \\ -3 & 4 \end{bmatrix}, \text{ therefore}$$

$$A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{\begin{bmatrix} -1 & -3 \\ -3 & 4 \end{bmatrix}}{-13}$$

Put the values of X , A^{-1} and B in (i)

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{13} \begin{bmatrix} -1 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{13} \begin{bmatrix} -1 \times 5 + (-3) \times 7 \\ -3 \times 5 + 4 \times 7 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{13} \begin{bmatrix} -5 - 21 \\ -15 + 28 \end{bmatrix} \\ = \frac{-1}{13} \begin{bmatrix} -26 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow x_1 = 2, \quad x_2 = -1$$

$$\text{(iii)} \quad \left. \begin{array}{l} 3x - 5y = 1 \\ -2x + y = -3 \end{array} \right\}$$

Solution: The matrix form of the system

$$\left. \begin{array}{l} 3x - 5y = 1 \\ -2x + y = -3 \end{array} \right\} \text{ is}$$

$$\begin{bmatrix} 3 & -5 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$AX = B$$

$$\Rightarrow X = A^{-1}B \dots \text{(i)}$$

$$\text{Where } A = \begin{bmatrix} 3 & -5 \\ -2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & -5 \\ -2 & 1 \end{vmatrix} = 3 \times 1 - (-2)(-5) \\ = 3 - 10 \\ |A| = -7 \neq 0$$

$$\text{And } \text{Adj}A = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{\begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}}{-7}$$

Put the values of X , A^{-1} and B in (i)

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{-7} \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \end{bmatrix} \\ = \frac{-1}{7} \begin{bmatrix} 1 \times 1 + 5 \times (-3) \\ 2 \times 1 + 3 \times (-3) \end{bmatrix} \\ = \frac{-1}{7} \begin{bmatrix} 1 - 15 \\ 2 - 9 \end{bmatrix} \\ = \frac{-1}{7} \begin{bmatrix} -14 \\ -7 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow x = 2 \quad \text{and} \quad y = 1$$

Q.4 If $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 5 \\ -1 & 0 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 4 \\ -1 & 2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 0 \\ 3 & 4 & -1 \end{bmatrix}$, then find

i) $A - B$ ii) $B - A$ iii) $(A - B) - C$ iv) $A - (B - C)$

(i) $A - B$

Solution:

$$A - B = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 5 \\ -1 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 4 \\ -1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-2 & -1-1 & 2+1 \\ 3-1 & 2-3 & 5-4 \\ -1+1 & 0-2 & 4-1 \end{bmatrix}$$

$$A - B = \begin{bmatrix} -1 & -2 & 3 \\ 2 & -1 & 1 \\ 0 & -2 & 3 \end{bmatrix}$$

(ii) B – A**Solution:**

$$\begin{aligned} B - A &= \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 4 \\ -1 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 5 \\ -1 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 2-1 & 1+1 & -1-2 \\ 1-3 & 3-2 & 4-5 \\ -1+1 & 2-0 & 1-4 \end{bmatrix} \end{aligned}$$

$$B - A = \begin{bmatrix} 1 & 2 & -3 \\ -2 & 1 & -1 \\ 0 & 2 & -3 \end{bmatrix}$$

(iii) (A – B) – C

Consider

$$\begin{aligned} A - B &= \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 5 \\ -1 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 4 \\ -1 & 2 & 1 \end{bmatrix} \\ A - B &= \begin{bmatrix} -1 & -2 & 3 \\ 2 & -1 & 1 \\ 0 & -2 & 3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Now } (A - B) - C &= \begin{bmatrix} -1 & -2 & 3 \\ 2 & -1 & 1 \\ 0 & -2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 0 \\ 3 & 4 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -1-1 & -2-3 & 3+2 \\ 2+1 & -1-2 & 1-0 \\ 0-3 & -2-4 & 3+1 \end{bmatrix} \end{aligned}$$

$$(A - B) - C = \begin{bmatrix} -2 & -5 & 5 \\ 3 & -3 & 1 \\ -3 & -6 & 4 \end{bmatrix}$$

Q.5 If $A = \begin{bmatrix} i & 2i \\ 1 & -i \end{bmatrix}$, $B = \begin{bmatrix} -i & 1 \\ 2i & i \end{bmatrix}$ and $C = \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix}$ then show that

(i) $(AB)C = A(BC)$

(ii) $(A+B)C = AC + BC$

(i) $(AB)C = A(BC)$

Consider $AB = \begin{bmatrix} i & 2i \\ 1 & -i \end{bmatrix} \begin{bmatrix} -i & 1 \\ 2i & i \end{bmatrix}$

$$= \begin{bmatrix} i \times (-i) + 2i \times 2i & i \times 1 + 2i \times i \\ 1 \times (-i) + (-i) \times (2i) & 1 \times 1 + (-i) \times i \end{bmatrix}$$

$$= \begin{bmatrix} -i^2 + 4i^2 & i + 2i^2 \\ -i - 2i^2 & 1 - i^2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 4(-1) & i + 2(-1) \\ -i - 2(-1) & 1 - (-1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 - 4 & i - 2 \\ -i + 2 & 1 + 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -3 & i - 2 \\ -i + 2 & 2 \end{bmatrix}$$

Now

$$\text{L.H.S} = (AB)C$$

$$= \begin{bmatrix} -3 & i - 2 \\ -i + 2 & 2 \end{bmatrix} \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix}$$

$$= \begin{bmatrix} (-3 \times 2i) + ((i - 2) \times (-i)) & (-3) \times (-1) + (i - 2) \times i \\ (-i + 2) \times 2i + 2 \times (-i) & (-i + 2) \times (-1) + 2 \times i \end{bmatrix}$$

$$= \begin{bmatrix} -6i - i^2 + 2i & 3 + i^2 - 2i \\ -2i^2 + 2i & i - 2 + 2i \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 1 - 4i & 2 - 2i \\ 2 + 2i & 3i - 2 \end{bmatrix} \quad (i)$$

Now Consider $BC = \begin{bmatrix} -i & 1 \\ 2i & i \end{bmatrix} \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix}$

$$= \begin{bmatrix} -i \times 2i + 1 \times (-i) & (-i) \times (-1) + 1 \times i \\ 2i \times 2i + i \times (-i) & 2i \times (-1) + i \times i \end{bmatrix}$$

$$= \begin{bmatrix} -2i^2 - i & i + i \\ 4i^2 - i^2 & -2i + i^2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 - i & 2i \\ -3 & -2i - 1 \end{bmatrix}$$

Now R.H.S = $A(BC)$

$$= \begin{bmatrix} i & 2i \\ 1 & -i \end{bmatrix} \begin{bmatrix} 2 - i & 2i \\ -3 & -2i - 1 \end{bmatrix}$$

$$= \begin{bmatrix} i \times (2 - i) + 2i \times (-3) & i \times 2i + 2i(-2i - 1) \\ 1 \times (2 - i) + (-i)(-3) & 1 \times 2i + (-i)(-2i - 1) \end{bmatrix}$$

$$= \begin{bmatrix} 2i - i^2 - 6i & 2i^2 - 4i^2 - 2i \\ 2 - i + 3i & 2i + 2i^2 + i \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} -4i + 1 & 2 - 2i \\ 2 + 2i & 3i - 2 \end{bmatrix} \quad \text{(ii)}$$

From (i) and (ii)

L.H.S = R.H.S

So $(AB)C = A(BC)$

(ii) $(A + B)C = AC + BC$

$$A + B = \begin{bmatrix} i & 2i \\ 1 & -i \end{bmatrix} + \begin{bmatrix} -i & 1 \\ 2i & i \end{bmatrix}$$

$$= \begin{bmatrix} i - i & 2i + 1 \\ 1 + 2i & -i + i \end{bmatrix}$$

$$A + B = \begin{bmatrix} 0 & 2i + 1 \\ 1 + 2i & 0 \end{bmatrix}$$

Now L.H.S

$= (A + B)C$

$$= \begin{bmatrix} 0 & 1 + 2i \\ 1 + 2i & 0 \end{bmatrix} \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \times 2i + (1 + 2i)(-i) & 0 \times (-1) + (1 + 2i) \times i \\ (1 + 2i) \times 2i + 0 \times (-i) & (1 + 2i) \times (-1) + 0 \times i \end{bmatrix}$$

$$= \begin{bmatrix} 0 - i - 2i^2 & 0 + i + 2i^2 \\ 2i + 4i^2 + 0 & -1 - 2i \end{bmatrix}$$

$$\text{L.H.S} = \begin{bmatrix} 2 - i & i - 2 \\ 2i - 4 & -1 - 2i \end{bmatrix} \quad \text{(i)}$$

Now consider R.H.S = $AC + BC$

$$\begin{aligned}
&= \begin{bmatrix} i & 2i \\ 1 & -i \end{bmatrix} \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix} + \begin{bmatrix} -i & 1 \\ 2i & i \end{bmatrix} \begin{bmatrix} 2i & -1 \\ -i & i \end{bmatrix} \\
&= \begin{bmatrix} i \times 2i + 2i \times (-i) & i \times (-1) + 2i \times i \\ 1 \times 2i + (-i) \times (-i) & 1 \times (-1) + (-i) \times i \end{bmatrix} \\
&\quad + \begin{bmatrix} -i \times 2i + 1 \times (-i) & -i \times (-1) + 1 \times i \\ 2i \times 2i + i \times (-i) & 2i \times (-1) + i \times i \end{bmatrix} \\
&= \begin{bmatrix} 2i^2 - 2i^2 & -i + 2i^2 \\ 2i + i^2 & -1 - i^2 \end{bmatrix} + \begin{bmatrix} -2i^2 - i & i + i \\ 4i^2 - i^2 & -2i + i^2 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -i - 2 \\ 2i - 1 & -1 + 1 \end{bmatrix} + \begin{bmatrix} 2 - i & 2i \\ -3 & -2i - 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 + 2 - i & -i - 2 + 2i \\ 2i - 1 - 3 & 0 - 2i - 1 \end{bmatrix} \\
\text{R.H.S} &= \begin{bmatrix} 2 - i & i - 2 \\ 2i - 4 & -2i - 1 \end{bmatrix} \tag{ii}
\end{aligned}$$

Since from (i) and (ii) L.H.S = R.H.S

Hence $(A + B)C = AC + BC$

Q.6 If A and B are square matrices of the same order, then explain why in general

i) $(A + B)^2 \neq A^2 + 2AB + B^2$

ii) $(A - B)^2 \neq A^2 - 2AB + B^2$

iii) $(A + B)(A - B) \neq A^2 - B^2$

(i) $(A + B)^2 \neq A^2 + 2AB + B^2$

$$\begin{aligned}
\text{L.H.S} &= (A + B)^2 \\
&= (A + B)(A + B) \\
&= A.A + A.B + B.A + B.B \\
&= A^2 + AB + BA + B^2 \\
&= A^2 + (AB + BA) + B^2
\end{aligned}$$

In matrices $AB \neq BA$ in general

Therefore $AB + BA \neq 2AB$

$$\begin{aligned}
(A + B)^2 &= A^2 + (AB + BA) + B^2 \\
&\neq A^2 + 2AB + B^2
\end{aligned}$$

Hence

$$(A + B)^2 \neq A^2 + 2AB + B^2$$

(ii) $(A-B)^2 \neq A^2 - 2AB + B^2$

Consider

$$\begin{aligned}(A-B)^2 &= (A-B)(A-B) \\ &= A.A - AB - BA + B.B \\ &= A^2 - (AB + BA) + B^2\end{aligned}$$

In matrices $AB \neq BA$ in general

Therefore $AB + BA \neq 2AB$

So $(A-B)^2 \neq A^2 - 2AB + B^2$

Hence

$$(A-B)^2 \neq A^2 - 2AB + B^2$$

(iii) $(A+B)(A-B) \neq A^2 - B^2$

Consider $(A+B)(A-B)$

$$\begin{aligned}&= A.A - AB + BA - B.B \\ &= A^2 - AB + BA - B^2\end{aligned}$$

In matrices $AB \neq BA$ in general

Therefore $-AB + BA \neq 0$

So, $(A+B)(A-B) = A^2 - AB + BA - B^2$
 $\neq A^2 - B^2$

Hence $(A+B)(A-B) \neq A^2 - B^2$

Q.7 If $A = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 4 & -2 \\ -3 & 5 & 2 & -1 \end{bmatrix}$, then find AA^t and A^tA

We have $A = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 4 & -2 \\ -3 & 5 & 2 & -1 \end{bmatrix}$, then

$$A^t = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 4 & -2 \\ -3 & 5 & 2 & -1 \end{bmatrix}^t$$

$$A^t = \begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 5 \\ 3 & 4 & 2 \\ 0 & -2 & -1 \end{bmatrix}$$

Now $AA^t = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 4 & -2 \\ -3 & 5 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 5 \\ 3 & 4 & 2 \\ 0 & -2 & -1 \end{bmatrix}$

$$\begin{aligned}
&= \begin{bmatrix} 2 \times 2 + (-1) \times (-1) + 3 \times 3 + 0 \times 0 & 2 \times 1 + (-1) \times 0 + 3 \times 4 + 0 \times (-2) \\ 1 \times 2 + 0 \times (-1) + 4 \times 3 + (-2) \times 0 & 1 \times 1 + 0 \times 0 + 4 \times 4 + (-2) \times (-2) \\ -3 \times 2 + 5 \times (-1) + 2 \times 3 + (-1) \times 0 & -3 \times 1 + 5 \times (0) + 2 \times 4 + (-1) \times (-2) \\ 2 \times (-3) + (-1) \times 5 + 3 \times 2 + 0 \times (-1) \\ 1 \times (-3) + 0 \times 5 + 4 \times 2 + (-2) \times (-1) \\ -3 \times (-3) + 5 \times 5 + 2 \times 2 + (-1) \times (-1) \end{bmatrix} \\
&= \begin{bmatrix} 4+1+9+0 & 2+0+12+0 & -6-5+6+0 \\ 2+0+12+0 & 1+0+16+4 & -3+0+8+2 \\ -6-5+6+0 & -3+0+8+2 & 9+25+4+1 \end{bmatrix} \\
AA^t &= \begin{bmatrix} 14 & 14 & -5 \\ 14 & 21 & 7 \\ -5 & 7 & 39 \end{bmatrix}
\end{aligned}$$

Now

$$\begin{aligned}
A^t A &= \begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 5 \\ 3 & 4 & 2 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 4 & -2 \\ -3 & 5 & 2 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 2 \times 2 + 1 \times 1 + (-3) \times (-3) & 2 \times (-1) + 1 \times 0 + (-3) \times (5) \\ -1 \times 2 + 0 \times 1 + 5 \times (-3) & -1 \times (-1) + 0 \times 0 + (5) \times (5) \\ 3 \times 2 + 4 \times 1 + 2 \times (-3) & 3 \times (-1) + 4 \times 0 + (2) \times (5) \\ 0 \times 2 + (-2) \times 1 + (-1) \times (-3) & 0 \times (-1) + (-2) \times 0 + (-1) \times 5 \\ 2 \times 3 + 1 \times 4 + (-3) \times (2) & 2 \times 0 + 1 \times (-2) + (-3) \times (-1) \\ -1 \times 3 + 0 \times 4 + 5 \times (2) & -1 \times 0 + 0 \times (-2) + 5 \times (-1) \\ 3 \times 3 + 4 \times 4 + 2 \times 2 & 3 \times 0 + 4 \times (-2) + 2 \times (-1) \\ 0 \times 3 + (-2) \times 4 + (-1) \times 2 & 0 \times 0 + (-2) \times (-2) + (-1) \times (-1) \end{bmatrix} \\
&= \begin{bmatrix} 4+1+9 & -2+0-15 & 6+4-6 & 0-2+3 \\ -2+0-15 & 1+0+25 & -3+0+10 & 0+0-5 \\ 6+4-6 & -3+0+10 & 9+16+4 & 0-8-2 \\ 0-2+3 & 0+0-5 & 0-8-2 & 0+4+1 \end{bmatrix} \\
A^t A &= \begin{bmatrix} 14 & -17 & 4 & 1 \\ -17 & 26 & 7 & -5 \\ 4 & 7 & 29 & -10 \\ 1 & -5 & -10 & 5 \end{bmatrix}
\end{aligned}$$

Q.8 Solve the following equations for X .

(i) $3X - 2A = B$ if $A = \begin{bmatrix} 2 & 3 & -2 \\ -1 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 & 1 \\ 5 & 4 & -1 \end{bmatrix}$

Solution:

$$3X - 2A = B$$

Put the values of A and B

$$\Rightarrow 3X - 2 \begin{bmatrix} 2 & 3 & -2 \\ -1 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 1 \\ 5 & 4 & -1 \end{bmatrix}$$

$$\Rightarrow 3X - \begin{bmatrix} 4 & 6 & -4 \\ -2 & 2 & 10 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 1 \\ 5 & 4 & -1 \end{bmatrix}$$

$$\Rightarrow 3X = \begin{bmatrix} 2 & -3 & 1 \\ 5 & 4 & -1 \end{bmatrix} + \begin{bmatrix} 4 & 6 & -4 \\ -2 & 2 & 10 \end{bmatrix}$$

$$\Rightarrow 3X = \begin{bmatrix} 2+4 & -3+6 & 1-4 \\ 5-2 & 4+2 & -1+10 \end{bmatrix}$$

$$3X = \begin{bmatrix} 6 & 3 & -3 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\Rightarrow X = \frac{1}{3} \begin{bmatrix} 6 & 3 & -3 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\Rightarrow X = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$$

(ii) $2X - 3A = B$ if $A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$

Solution: $2X - 3A = B$

Put the values of A and B

$$2X - 3 \begin{bmatrix} 1 & -1 & 2 \\ -2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow 2X - \begin{bmatrix} 3 & -3 & 6 \\ -6 & 12 & 15 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow 2X = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 3 & -3 & 6 \\ -6 & 12 & 15 \end{bmatrix}$$

$$\Rightarrow 2X = \begin{bmatrix} 3+3 & -1-3 & 0+6 \\ 4-6 & 2+12 & 1+15 \end{bmatrix}$$

$$\Rightarrow 2X = \begin{bmatrix} 6 & -4 & 6 \\ -2 & 14 & 16 \end{bmatrix}$$

$$\Rightarrow X = \frac{1}{2} \begin{bmatrix} 6 & -4 & 6 \\ -2 & 14 & 16 \end{bmatrix}$$

$$\Rightarrow X = \begin{bmatrix} 3 & -2 & 3 \\ -1 & 7 & 8 \end{bmatrix}$$

Q.9 Solve the following matrix equations for A

$$(i) \quad \begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} A - \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -1 & -4 \\ 3 & 6 \end{bmatrix}$$

$$\text{Solution:} \quad \begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} A - \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -1 & -4 \\ 3 & 6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} A = \begin{bmatrix} -1 & -4 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} A = \begin{bmatrix} -1+2 & -4+3 \\ 3-1 & 6-2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} \quad (i)$$

Now let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then (i) becomes

$$\begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4a+3c & 4b+3d \\ 2a+2c & 2b+2d \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

Now by definition of equal matrices

$$4a+3c=1 \quad (ii)$$

$$2a+2c=2 \quad (iii)$$

$$4b+3d=-1 \quad (iv)$$

$$2b+2d=4 \quad (v)$$

Multiply 2 to equation (iii) and Subtracting from (ii)

$$4a+3c=1$$

$$\underline{-4a+4c=-4}$$

$$-c=-3$$

$$\Rightarrow c=3$$

substituting the value of c in (iii)

$$2a+2(3)=2$$

$$\Rightarrow 2a=-4$$

$$a=-2$$

Multiply 2 to equation (v) and subtracting from (iv)

$$4b+3d=-1$$

$$\underline{-4b+4d=-8}$$

$$-d = -9$$

$$\Rightarrow d = 9$$

Substituting the value of d in (v) we get

$$2b + 2(9) = 4$$

$$\Rightarrow 2b = -14$$

$$\Rightarrow b = -7$$

So,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A = \begin{bmatrix} -2 & -7 \\ 3 & 9 \end{bmatrix} \text{ is the square matrix.}$$

$$(ii) \quad A \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix}$$

$$\Rightarrow A \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -1 & 5 \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$$

$$\Rightarrow A \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2-1 & 0+2 \\ -1+3 & 5+1 \end{bmatrix}$$

$$\Rightarrow A \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \quad (i)$$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then (i) becomes

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3a+4b & a+2b \\ 3c+4d & c+2d \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$$

Now by definition of equal matrices

$$3a + 4b = 1 \quad (ii)$$

$$3c + 4d = 2 \quad (iii)$$

$$a + 2b = 2 \quad (iv)$$

$$c + 2d = 6 \quad (v)$$

Multiply 2 to equation (iv) and subtracting from (ii)

$$3a + 4b = 1$$

$$\underline{-2a + 4b = -4}$$

$$a = -3$$

Substituting the value of a in (iv) we get

$$-3 + 2b = 2$$

$$\Rightarrow 2b = 5$$

$$\Rightarrow b = \frac{5}{2}$$

Multiply 2 to equation (v) and subtracting from (iii)

$$3c + 4d = 2$$

$$\underline{-2c + 4d = -12}$$

$$c = -10$$

Substituting the value of c in (v) we get

$$-10 + 2d = 6$$

$$\Rightarrow 2d = 16$$

$$\Rightarrow d = 8$$

So,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \begin{bmatrix} -3 & \frac{5}{2} \\ -10 & 8 \end{bmatrix} \text{ is the required matrix.}$$

Singular and Non Singular Matrices:

A square matrix A is said to be **singular** if $|A| = 0$ and if $|A| \neq 0$ then A is said to be non singular.

Minor and Cofactor of an Element of a Matrix:

Minor:

Let us consider a square matrix A of order n then minor of an element a_{ij} denoted by M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix formed by deleting the i th row and j th column of A .

Cofactor:

The cofactor of an element a_{ij} denoted by A_{ij} is defined by $A_{ij} = (-1)^{i+j} \times M_{ij}$.

Properties of Determinants:

1. For a square matrix A , $|A| = |A^t|$.
2. If in a square matrix A , two rows or two columns are interchanged, the determinant of the resulting matrix is $-|A|$.
3. If a square matrix A has two identical rows or two identical columns, then $|A| = 0$.
4. If all the entries of a row (or a column) of a square matrix A are zero, then $|A| = 0$.
5. If the entries of a row (or a column) in a square matrix A are multiplied by a number $k \in \mathbb{R}$, then the determinant of the resulting matrix is $k|A|$.
6. If each entry of a row (or a column) of a square matrix consists of two terms, then its determinant can be written as the sum of two determinants, i.e., if

$$B = \begin{bmatrix} a_{11} + b_{11} & a_{12} & a_{13} \\ a_{21} + b_{21} & a_{22} & a_{23} \\ a_{31} + b_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ then}$$

$$|B| = \begin{vmatrix} a_{11} + b_{11} & a_{12} & a_{13} \\ a_{21} + b_{21} & a_{22} & a_{23} \\ a_{31} + b_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_{11} & a_{12} & a_{13} \\ b_{21} & a_{22} & a_{23} \\ b_{31} & a_{32} & a_{33} \end{vmatrix}$$

7. If to each entry of a row (or a column) of a square matrix A is added a non-zero multiple of the corresponding entry of another row (or column), then the determinant of the resulting matrix is $|A|$.
8. If a matrix is in triangular form, then the value of its determinant is the product of all the entries on its main diagonal.

EXERCISE 3.3**Q.1 Evaluate the determinants.**

$$(i) \begin{vmatrix} 5 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 1 & 2 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 5 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 1 & 2 \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= 5 \begin{vmatrix} -1 & -3 \\ 1 & 2 \end{vmatrix} - (-2) \begin{vmatrix} 3 & -3 \\ -2 & 2 \end{vmatrix} - 4 \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} \\ &= 5((-1) \times 2 - 1 \times (-3)) + 2(3 \times 2 - (-2) \times (-3)) - 4(3 \times 1 - (-2)(-1)) \\ &= 5(-2 + 3) + 2(6 - 6) - 4(3 - 2) \\ &= 5(1) + 2(0) - 4(1) \\ &= 5 + 0 - 4 \\ &= 1 \end{aligned}$$

$$\text{Hence } \begin{vmatrix} 5 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 1 & 2 \end{vmatrix} = 1$$

$$(ii) \begin{vmatrix} 5 & 2 & -3 \\ 3 & -1 & 1 \\ -2 & 1 & -2 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 5 & 2 & -3 \\ 3 & -1 & 1 \\ -2 & 1 & -2 \end{vmatrix}$$

Expanding from R_1 we set

$$\begin{aligned} &= 5 \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix} - 2 \begin{vmatrix} 3 & 1 \\ -2 & -2 \end{vmatrix} - 3 \begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix} \\ &= 5((-1) \times (-2) - (1)(1)) - 2(3 \times (-2) - (-2)(1)) - 3(3 \times 1 - (-2)(-1)) \\ &= 5(2 - 1) - 2(-6 + 2) - 3(3 - 2) \end{aligned}$$

$$= 5(1) - 2(-4) - 3(1)$$

$$= 5 + 8 - 3$$

$$= 10$$

$$\text{Hence } \begin{vmatrix} 5 & 2 & -3 \\ 3 & -1 & 1 \\ -2 & 1 & -2 \end{vmatrix} = 10$$

$$\text{(iii)} \quad \begin{vmatrix} 1 & 2 & -3 \\ -1 & 3 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 1 & 2 & -3 \\ -1 & 3 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= 1 \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} - 2 \begin{vmatrix} -1 & 4 \\ -2 & 6 \end{vmatrix} - 3 \begin{vmatrix} -1 & 3 \\ -2 & 5 \end{vmatrix} \\ &= 1(18 - 20) - 2(-6 + 8) - 3(-5 + 6) \\ &= 1(-2) - 2(2) - 3(1) \\ &= -2 - 4 - 3 \\ &= -9 \end{aligned}$$

$$\text{Hence } \begin{vmatrix} 1 & 2 & -3 \\ -1 & 3 & 4 \\ -2 & 5 & 6 \end{vmatrix} = -9$$

$$\text{(iv)} \quad \begin{vmatrix} a+l & a-l & a \\ a & a+l & a-l \\ a-l & a & a+l \end{vmatrix}$$

Solution:

$$\begin{vmatrix} a+l & a-l & a \\ a & a+l & a-l \\ a-l & a & a+l \end{vmatrix}$$

Expanding from R_1 we get

$$= (a+l) \begin{vmatrix} a+l & a-l \\ a & a+l \end{vmatrix} - (a-l) \begin{vmatrix} a & a-l \\ a-l & a+l \end{vmatrix} + a \begin{vmatrix} a & a+l \\ a-l & a \end{vmatrix}$$

$$\begin{aligned}
&= (a+l)\left((a+l)^2 - a(a-l)\right) - (a-l)\left(a(a+l) - (a-l)^2\right) + a\left(a^2 - (a+l)(a-l)\right) \\
&= (a+l)\left(a^2 + l^2 + 2al - a^2 + al\right) - (a-l)\left(a^2 + al - a^2 - l^2 + 2al\right) + a\left(a^2 - a^2 + l^2\right) \\
&= (a+l)\left(a^2 + l^2 + 2al - a^2 + al\right) - (a-l)\left(a^2 + al - a^2 - l^2 + 2al\right) + a\left(a^2 - a^2 + l^2\right) \\
&= (a+l)\left(l^2 + 3al\right) - (a-l)\left(3al - l^2\right) + al^2 \\
&= al^2 + 3a^2l + l^3 + 3al^2 - \left(3a^2l - al^2 - 3al^2 + l^3\right) + al^2 \\
&= 4al^2 + 3a^2l + l^3 - 3a^2l + 4al^2 - l^3 + al^2 \\
&= 9al^2
\end{aligned}$$

Hence
$$\begin{vmatrix} a+l & a-l & a \\ a & a+l & a-l \\ a-l & a & a+l \end{vmatrix} = 9al^2$$

(v)
$$\begin{vmatrix} 1 & 2 & -2 \\ -1 & 1 & -3 \\ 2 & 4 & -1 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 1 & 2 & -2 \\ -1 & 1 & -3 \\ 2 & 4 & -1 \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned}
&= 1 \begin{vmatrix} 1 & -3 \\ 4 & -1 \end{vmatrix} - 2 \begin{vmatrix} -1 & -3 \\ 2 & -1 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ 2 & 4 \end{vmatrix} \\
&= 1(1 \times (-1) - 4(-3)) - 2((-1) \times (-1) - (2)(-3)) - 2((-1) \times 4 - 2 \times 1) \\
&= 1(-1 + 12) - 2(1 + 6) - 2(-4 - 2) \\
&= 1(11) - 2(7) - 2(-6) \\
&= 11 - 14 + 12 \\
&= 9
\end{aligned}$$

Hence
$$\begin{vmatrix} 1 & 2 & -2 \\ -1 & 1 & -3 \\ 2 & 4 & -1 \end{vmatrix} = 9$$

$$(vi) \begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= 2a \begin{vmatrix} 2b & b \\ c & 2c \end{vmatrix} - a \begin{vmatrix} b & b \\ c & 2c \end{vmatrix} + a \begin{vmatrix} b & 2b \\ c & c \end{vmatrix} \\ &= 2a(4bc - bc) - a(2bc - bc) + a(bc - 2bc) \\ &= 2a(3bc) - a(bc) + a(-bc) \\ &= 6abc - abc - abc \\ &= 4abc \end{aligned}$$

$$\text{Hence } \begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix} = 4abc$$

Q.2 Without expansion show that

$$(i) \begin{vmatrix} 6 & 7 & 8 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

Solution:

$$\text{Consider L.H.S} = \begin{vmatrix} 6 & 7 & 8 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 6 & 1 & 1 \\ 3 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$= 0 = \text{R.H.S}$$

$$\text{By } C_3 - C_2 \rightarrow C'_3 \text{ and } C_2 - C_1 \rightarrow C'_2$$

$$\therefore C_2 \text{ and } C_3 \text{ are identical}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 6 & 7 & 8 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

$$(ii) \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} = 0$$

Solution:

$$\begin{aligned} \text{Consider L.H.S} &= \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} \\ &= \begin{vmatrix} 2 & 3 & -1+3 \\ 1 & 1 & 0+1 \\ 2 & -3 & 5-3 \end{vmatrix} \\ &= \begin{vmatrix} 2 & 3 & 2 \\ 1 & 1 & 1 \\ 2 & -3 & 2 \end{vmatrix} \\ &= 0 = \text{R.H.S} \end{aligned}$$

By $C_3 + C_2 \rightarrow C'_3$

$\therefore C_1$ and C_3 are identical.

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} = 0$$

$$(iii) \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$$

Solution:

$$\begin{aligned} \text{Consider L.H.S} &= \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 2-1 & 3-2 \\ 4 & 5-4 & 6-5 \\ 7 & 8-7 & 9-8 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & 1 \\ 4 & 1 & 1 \\ 7 & 1 & 1 \end{vmatrix} \\ &= 0 = \text{R.H.S} \end{aligned}$$

By $C_3 - C_2 \rightarrow C'_3$ and $C_2 - C_1 \rightarrow C'_2$

$\therefore C_2$ and C_3 are identical.

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$$

Q.3 Show that

$$(i) \quad \begin{vmatrix} a_{11} & a_{12} & a_{13} + \alpha_{13} \\ a_{21} & a_{22} & a_{23} + \alpha_{23} \\ a_{31} & a_{32} & a_{33} + \alpha_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \alpha_{13} \\ a_{21} & a_{22} & \alpha_{23} \\ a_{31} & a_{32} & \alpha_{33} \end{vmatrix}$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} a_{11} & a_{12} & a_{13} + \alpha_{13} \\ a_{21} & a_{22} & a_{23} + \alpha_{23} \\ a_{31} & a_{32} & a_{33} + \alpha_{33} \end{vmatrix}$$

Expanding from C_3 we get

$$\begin{aligned} &= (a_{13} + \alpha_{13}) \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - (a_{23} + \alpha_{23}) \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + (a_{33} + \alpha_{33}) \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\ &= a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} + \alpha_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} - \alpha_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \alpha_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\ &= \left(a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \right) + \left(\alpha_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - \alpha_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + \alpha_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \right) \quad (i) \\ &\text{Now consider R.H.S} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \alpha_{13} \\ a_{21} & a_{22} & \alpha_{23} \\ a_{31} & a_{32} & \alpha_{33} \end{vmatrix} \end{aligned}$$

Expanding both the determinants from C_3

$$= \left(a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \right) + \left(\alpha_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - \alpha_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + \alpha_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \right) \quad (ii)$$

From (i) and (ii) L.H.S = R.H.S

$$\text{Hence} \quad \begin{vmatrix} a_{11} & a_{12} & a_{13} + \alpha_{13} \\ a_{21} & a_{22} & a_{23} + \alpha_{23} \\ a_{31} & a_{32} & a_{33} + \alpha_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \alpha_{13} \\ a_{21} & a_{22} & \alpha_{23} \\ a_{31} & a_{32} & \alpha_{33} \end{vmatrix}$$

$$(ii) \quad \begin{vmatrix} 2 & 3 & 0 \\ 3 & 9 & 6 \\ 2 & 15 & 1 \end{vmatrix} = 9 \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 2 & 5 & 1 \end{vmatrix}$$

Proof:

$$\begin{aligned} \text{Consider L.H.S} &= \begin{vmatrix} 2 & 3 & 0 \\ 3 & 9 & 6 \\ 2 & 15 & 1 \end{vmatrix} \\ &= 3 \begin{vmatrix} 2 & 3 & 0 \\ 1 & 3 & 2 \\ 2 & 15 & 1 \end{vmatrix} \end{aligned}$$

By taking 3 common from R_2

$$= 3 \times 3 \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 2 & 5 & 1 \end{vmatrix}$$

By taking 3 common from C_2

$$= 9 \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 2 & 5 & 1 \end{vmatrix} = \text{R.H.S}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 2 & 3 & 0 \\ 3 & 9 & 6 \\ 2 & 15 & 1 \end{vmatrix} = 9 \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 2 & 5 & 1 \end{vmatrix}$$

$$\text{(iii)} \quad \begin{vmatrix} a+l & a & a \\ a & a+l & a \\ a & a & a+l \end{vmatrix} = l^2(3a+l)$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} a+l & a & a \\ a & a+l & a \\ a & a & a+l \end{vmatrix}$$

$$= \begin{vmatrix} 3a+l & a & a \\ 3a+l & a+l & a \\ 3a+l & a & a+l \end{vmatrix}$$

By $C_1 + (C_2 + C_3)$

$$= (3a+l) \begin{vmatrix} 1 & a & a \\ 1 & a+l & a \\ 1 & a & a+l \end{vmatrix}$$

By taking $(3a+l)$ Common from C_1

$$= (3a+l) \begin{vmatrix} 1 & a & a \\ 0 & l & 0 \\ 0 & 0 & l \end{vmatrix}$$

By $R_2 + (-R_1)$ and $R_3 + (-R_1)$ By expanding from R_1

$$\begin{aligned} &= (3a+l) \left[1 \begin{vmatrix} l & 0 \\ 0 & l \end{vmatrix} - a \begin{vmatrix} 0 & 0 \\ 0 & l \end{vmatrix} + a \begin{vmatrix} 0 & l \\ 0 & 0 \end{vmatrix} \right] \\ &= (3a+l) [(l^2 - 0) - a(0) + a(0)] \\ &= (3a+l)(l^2 - 0 + 0) \\ &= l^2(3a+l) = \text{R.H.S} \end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} a+l & a & a \\ a & a+l & a \\ a & a & a+l \end{vmatrix} = l^2(3a+l)$$

$$(iv) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ yz & zx & xy \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ yz & zx & xy \end{vmatrix}$$

Multiply and divided by x, y and z to C_1, C_2 and C_3 respectively

$$= \frac{1}{x} \cdot \frac{1}{y} \cdot \frac{1}{z} \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ xyz & xyz & xyz \end{vmatrix}$$

$$= \frac{xyz}{xyz} \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix}$$

By taking xyz common from R_3

$$= (-1) \begin{vmatrix} x & y & z \\ 1 & 1 & 1 \\ x^2 & y^2 & z^2 \end{vmatrix}$$

By $R_2 \leftrightarrow R_3$

$$= (-1)(-1) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

By $R_1 \leftrightarrow R_2$

$$= \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = \text{R.H.S}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ yz & zx & xy \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$(v) \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= (b+c) \begin{vmatrix} c+a & b \\ c & a+b \end{vmatrix} - a \begin{vmatrix} b & b \\ c & a+b \end{vmatrix} + a \begin{vmatrix} b & c+a \\ c & c \end{vmatrix} \\ &= (b+c)[(a+b)(a+c) - bc] - a[b(a+b) - bc] + a[bc - c(a+c)] \\ &= (b+c)(a^2 + ac + ba + bc - bc) - a(ab + b^2 - bc) + a(bc - ac - c^2) \\ &= a^2b + abc + b^2a + bc^2 + abc - a^2b - ab^2 + abc + abc - ac^2 - ac^2 \\ &= 4abc = \text{R.H.S} \end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

$$(vi) \begin{vmatrix} b & -1 & a \\ a & b & 0 \\ 1 & a & b \end{vmatrix} = a^3 + b^3$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} b & -1 & a \\ a & b & 0 \\ 1 & a & b \end{vmatrix}$$

Expanding from C_3 we get

$$\begin{aligned} &= a \begin{vmatrix} a & b \\ 1 & a \end{vmatrix} - 0 \begin{vmatrix} b & -1 \\ 1 & a \end{vmatrix} + b \begin{vmatrix} b & -1 \\ a & b \end{vmatrix} \\ &= a(a^2 - b) - 0(ab + 1) + b(b^2 + a) \\ &= a^3 - ab - 0 + b^3 + ab \\ &= a^3 + b^3 = \text{R.H.S} \end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} b & -1 & a \\ a & b & 0 \\ 1 & a & b \end{vmatrix} = a^3 + b^3$$

$$(vii) \begin{vmatrix} r\cos\phi & 1 & -\sin\phi \\ 0 & 1 & 0 \\ r\sin\phi & 0 & \cos\phi \end{vmatrix} = r$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} r\cos\phi & 1 & -\sin\phi \\ 0 & 1 & 0 \\ r\sin\phi & 0 & \cos\phi \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= r\cos\phi \begin{vmatrix} 1 & 0 \\ 0 & \cos\phi \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 \\ r\sin\phi & \cos\phi \end{vmatrix} - \sin\phi \begin{vmatrix} 0 & 1 \\ r\sin\phi & 0 \end{vmatrix} \\ &= r\cos\phi(\cos\phi - 0) - 1(0 - 0) - \sin\phi(0 - r\sin\phi) \\ &= r\cos^2\phi + r\sin^2\phi \\ &= r(\cos^2\phi + \sin^2\phi) \\ &= r(1) \end{aligned}$$

$$= r = \text{R.H.S}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} r\cos\phi & 1 & -\sin\phi \\ 0 & 1 & 0 \\ r\sin\phi & 0 & \cos\phi \end{vmatrix} = r$$

$$(viii) \begin{vmatrix} a & b+c & a+b \\ b & c+a & b+c \\ c & a+b & c+a \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

Proof:

$$\text{Consider L.H.S} = \begin{vmatrix} a & b+c & a+b \\ b & c+a & b+c \\ c & a+b & c+a \end{vmatrix}$$

$$= \begin{vmatrix} a+b+c & b+c & a+b \\ a+b+c & c+a & b+c \\ a+b+c & a+b & c+a \end{vmatrix}$$

By $C_1 + C_2$

$$= (a+b+c) \begin{vmatrix} 1 & b+c & a+b \\ 1 & c+a & b+c \\ 1 & a+b & c+a \end{vmatrix}$$

By taking common $a+b+c$ from C_1

$$= (a+b+c) \begin{vmatrix} 1 & b+c & a+b \\ 0 & a-b & c-a \\ 0 & a-c & c-b \end{vmatrix} \quad \text{By } R_2 - R_1 \text{ and } R_3 - R_1$$

Expanding from C_1

$$\begin{aligned} &= (a+b+c) \left[\begin{vmatrix} a-b & c-a \\ a-c & c-b \end{vmatrix} - 0 \begin{vmatrix} b+c & a+b \\ a-c & c-b \end{vmatrix} + 0 \begin{vmatrix} b+c & a+b \\ a-b & c-a \end{vmatrix} \right] \\ &= (a+b+c) [(a-b)(c-b) - (a-c)(c-a) - 0 + 0] \\ &= (a+b+c) [\cancel{ac} - ab - bc + b^2 - ac + a^2 + c^2 - \cancel{ac}] \\ &= (a+b+c) [a^2 + b^2 + c^2 - ab - bc - ca] \\ &\therefore a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\ &\Rightarrow a^3 + b^3 + c^3 - 3abc = \text{R.H.S} \end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} a & b+c & a+b \\ b & c+a & b+c \\ c & a+b & c+a \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

$$\text{(ix) } \begin{vmatrix} a+\lambda & b & c \\ a & b+\lambda & c \\ a & b & c+\lambda \end{vmatrix} = \lambda^2(a+b+c+\lambda)$$

Proof:

$$\begin{aligned} \text{Consider L.H.S} &= \begin{vmatrix} a+\lambda & b & c \\ a & b+\lambda & c \\ a & b & c+\lambda \end{vmatrix} \\ &= \begin{vmatrix} a+b+c+\lambda & b & c \\ a+b+c+\lambda & b+\lambda & c \\ a+b+c+\lambda & b & c+\lambda \end{vmatrix} \quad \text{By } C_1 + (C_2 + C_3) \\ &= (a+b+c+\lambda) \begin{vmatrix} 1 & b & c \\ 1 & b+\lambda & c \\ 1 & b & c+\lambda \end{vmatrix} \quad \text{Taking } (a+b+c+\lambda) \text{ common from } C_1 \\ &= (a+b+c+\lambda) \begin{vmatrix} 1 & b & c \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} \quad \text{By } R_2 + (-R_1) \text{ and } R_3 + (-R_1) \end{aligned}$$

Expanding from C_1

$$\begin{aligned}
&= (a+b+c+\lambda) \left[1 \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - b \begin{vmatrix} 0 & 0 \\ 0 & \lambda \end{vmatrix} + c \begin{vmatrix} 0 & \lambda \\ 0 & 0 \end{vmatrix} \right] \\
&= (a+b+c+\lambda) \left[(\lambda^2 - 0) - b(0) + c(0) \right] \\
&= (a+b+c+\lambda)(\lambda^2 - 0 + 0) \\
&= \lambda^2(a+b+c+\lambda) = \text{R.H.S}
\end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} a+\lambda & b & c \\ a & b+\lambda & c \\ a & b & c+\lambda \end{vmatrix} = \lambda^2(a+b+c+\lambda)$$

$$\text{(x)} \quad \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

Proof:

$$\begin{aligned}
\text{Consider L.H.S} &= \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \\
&= \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix} \quad \text{By } C_2 - C_1 \text{ and } C_3 - C_1
\end{aligned}$$

Expanding from R_1 , we get

$$\begin{aligned}
&= 1 \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} - 0 \begin{vmatrix} a & c-a \\ a^2 & c^2-a^2 \end{vmatrix} + 0 \begin{vmatrix} a & b-a \\ a^2 & b^2-a^2 \end{vmatrix} \\
&= 1 \begin{vmatrix} b-a & c-a \\ (b-a)(b+a) & (c-a)(c+a) \end{vmatrix} - 0 + 0 \\
&= (b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix} \quad \text{Taking common } b-a \text{ and } c-a \text{ from } C_1 \text{ and } C_2 \text{ respectively} \\
&= (b-a)(c-a)(c+a-b-a) \\
&= (b-a)(c-a)(c-b) \\
&= (-1)(a-b) \times (-1)(b-c)(c-a) \\
&= (a-b)(b-c)(c-a) = \text{R.H.S}
\end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$(xi) \quad \begin{vmatrix} b+c & a & a^2 \\ c+a & b & b^2 \\ a+b & c & c^2 \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$$

Proof:

$$\begin{aligned} \text{Consider L.H.S} &= \begin{vmatrix} b+c & a & a^2 \\ c+a & b & b^2 \\ a+b & c & c^2 \end{vmatrix} \\ &= \begin{vmatrix} a+b+c & a & a^2 \\ a+b+c & b & b^2 \\ a+b+c & c & c^2 \end{vmatrix} && \text{By } C_1 + C_2 \\ &= (a+b+c) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} && \text{By } R_2 - R_1 \text{ and } R_3 - R_1 \end{aligned}$$

Expanding from C_1 we get

$$\begin{aligned} &= (a+b+c) \left[1 \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix} - 0 + 0 \right] \\ &= (a+b+c) \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} b-a & (b-a)(b+a) \\ c-a & (c-a)(c+a) \end{vmatrix} \\ &= (a+b+c)(b-a)(c-a) \begin{vmatrix} 1 & b+a \\ 1 & c+a \end{vmatrix} \quad \text{Taking common } b-a \text{ and } c-a \text{ from } R_1 \text{ and } R_2 \text{ respectively} \\ &= (a+b+c)(b-a)(c-a)(c+a-b-a) \\ &= (a+b+c)(b-a)(c-a)(c-b) \\ &= (a+b+c) \times (-1)(a-b)(-1)(b-c)(c-a) \\ &= (a+b+c)(a-b)(b-c)(c-a) = \text{R.H.S} \end{aligned}$$

Since L.H.S = R.H.S

$$\text{So } \begin{vmatrix} b+c & a & a^2 \\ c+a & b & b^2 \\ a+b & c & c^2 \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$$

Q.4 If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ and

$B = \begin{bmatrix} 5 & -2 & 5 \\ 3 & -1 & 4 \\ -2 & 1 & -2 \end{bmatrix}$, then find

(i) A_{12}, A_{22}, A_{32} and $|A|$

Solution:

Consider $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

Then by using formula

$$A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{12} = (-1)^{1+2} M_{12}$$

$$= - \begin{vmatrix} 0 & 0 \\ -2 & 1 \end{vmatrix}$$

$$= -(0+0) = 0$$

$$A_{12} = 0$$

$$A_{22} = (-1)^{2+2} M_{22}$$

$$= \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix}$$

$$= (1-6) = -5$$

$$A_{22} = -5$$

$$A_{32} = (-1)^{3+2} M_{32}$$

$$= - \begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix}$$

$$= -(0-0)$$

$$A_{32} = 0$$

Now $|A| = \begin{vmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 1 \end{vmatrix}$

Expanding from C_2 we get

$$|A| = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32}$$

$$= 2(0) + (-2)(-5) + (-2)(0)$$

Hence $|A| = 10$

(ii) $B = \begin{bmatrix} 5 & -2 & 5 \\ 3 & -1 & 4 \\ -2 & 1 & -2 \end{bmatrix}$,

Solution:

Consider $B = \begin{bmatrix} 5 & -2 & 5 \\ 3 & -1 & 4 \\ -2 & 1 & -2 \end{bmatrix}$

Then by using formula

$$B_{ij} = (-1)^{i+j} M_{ij}$$

$$B_{21} = (-1)^{2+1} M_{21}$$

$$= - \begin{vmatrix} -2 & 5 \\ 1 & -2 \end{vmatrix}$$

$$= -(4-5) = 1$$

$$B_{21} = 1$$

$$B_{22} = (-1)^{2+2} M_{22}$$

$$= \begin{vmatrix} 5 & 5 \\ -2 & -2 \end{vmatrix}$$

$$= -10+10$$

$$B_{22} = 0$$

$$B_{23} = (-1)^{2+3} M_{23}$$

$$= - \begin{vmatrix} 5 & -2 \\ -2 & 1 \end{vmatrix}$$

$$= -(5-4)$$

$$B_{23} = -1$$

Now $|B| = \begin{vmatrix} 5 & -2 & 5 \\ 3 & -1 & 4 \\ -2 & 1 & -2 \end{vmatrix}$

Expanding from R_2 we get

$$|B| = b_{21}B_{21} + b_{22}B_{22} + b_{23}B_{23}$$

$$= 3(1) + (-1)(0) + 4(-1)$$

$$= 3+0-4$$

$$|B| = -1$$

Q.5 Without expansion verify that

$$(i) \begin{vmatrix} \alpha & \beta + \gamma & 1 \\ \beta & \gamma + \alpha & 1 \\ \gamma & \alpha + \beta & 1 \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} \alpha & \beta + \gamma & 1 \\ \beta & \gamma + \alpha & 1 \\ \gamma & \alpha + \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \alpha + \beta + \gamma & \beta + \gamma & 1 \\ \beta + \alpha + \gamma & \gamma + \alpha & 1 \\ \gamma + \alpha + \beta & \alpha + \beta & 1 \end{vmatrix} \quad \because C_1 + C_2$$

$$= (\alpha + \beta + \gamma) \begin{vmatrix} 1 & \beta + \gamma & 1 \\ 1 & \gamma + \alpha & 1 \\ 1 & \alpha + \beta & 1 \end{vmatrix} \quad \text{Taking } (\alpha + \beta + \gamma) \text{ common from } C_1$$

$$= 0 \quad \because C_1 \text{ and } C_3 \text{ are identical}$$

$$\text{Hence } \begin{vmatrix} \alpha & \beta + \gamma & 1 \\ \beta & \gamma + \alpha & 1 \\ \gamma & \alpha + \beta & 1 \end{vmatrix} = 0$$

$$(ii) \begin{vmatrix} 1 & 2 & 3x \\ 2 & 3 & 6x \\ 3 & 5 & 9x \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} 1 & 2 & 3x \\ 2 & 3 & 6x \\ 3 & 5 & 9x \end{vmatrix}$$

$$= 3x \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 5 & 3 \end{vmatrix}$$

By taking $3x$ common from C_3

$$= 3x(0)$$

Because C_1 and C_3 are identical.

$$= 0$$

$$\text{Hence } \begin{vmatrix} 1 & 2 & 3x \\ 2 & 3 & 6x \\ 3 & 5 & 9x \end{vmatrix} = 0$$

$$(iii) \begin{vmatrix} 1 & a^2 & \frac{a}{bc} \\ 1 & b^2 & \frac{b}{ca} \\ 1 & c^2 & \frac{c}{ab} \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} 1 & a^2 & \frac{a}{bc} \\ 1 & b^2 & \frac{b}{ca} \\ 1 & c^2 & \frac{c}{ab} \end{vmatrix}$$

$$= \frac{abc}{abc} \begin{vmatrix} 1 & a^2 & \frac{a}{bc} \\ 1 & b^2 & \frac{b}{ca} \\ 1 & c^2 & \frac{c}{ab} \end{vmatrix}$$

Multiply and dividing by abc

$$= \frac{1}{abc} \begin{vmatrix} 1 & a^2 & abc \times \frac{a}{bc} \\ 1 & b^2 & abc \times \frac{b}{ca} \\ 1 & c^2 & abc \times \frac{c}{ab} \end{vmatrix}$$

Multiply C_3 by abc

$$= \frac{1}{abc} \begin{vmatrix} 1 & a^2 & a^2 \\ 1 & b^2 & b^2 \\ 1 & c^2 & c^2 \end{vmatrix}$$

$$= \frac{1}{abc} (0)$$

Q C_2 and C_3 are identical

$$= 0$$

$$\text{Hence} \begin{vmatrix} 1 & a^2 & \frac{a}{bc} \\ 1 & b^2 & \frac{b}{ca} \\ 1 & c^2 & \frac{c}{ab} \end{vmatrix} = 0$$

$$(iv) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

$$= \begin{vmatrix} a-b+b-c+c-a & b-c & c-a \\ b-c+c-a+a-b & c-a & a-b \\ c-a+a-b+b-c & a-b & b-c \end{vmatrix}$$

$QC_1 + (C_2 + C_3)$

$$= \begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix}$$

$$= 0$$

$\therefore$ All the entries of C_1 are zero

$$\text{Hence} \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

$$(v) \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a & b & c \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a & b & c \end{vmatrix}$$

$$= \frac{abc}{abc} \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a & b & c \end{vmatrix}$$

Multiplying and dividing by abc

$$= \frac{1}{abc} \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} \times abc & \frac{1}{b} \times abc & \frac{1}{c} \times abc \\ a & b & c \end{vmatrix}$$

Multiplying by abc to R_2

$$= \frac{1}{abc} \begin{vmatrix} bc & ca & ab \\ bc & ac & ab \\ a & b & c \end{vmatrix}$$

$$= \frac{1}{abc} (0)$$

$\because R_1$ and R_2 are identical

Hence $\begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a & b & c \end{vmatrix} = 0$

(vi) $\begin{vmatrix} mn & l & l^2 \\ nl & m & m^2 \\ lm & n & n^2 \end{vmatrix} = \begin{vmatrix} 1 & l^2 & l^3 \\ 1 & m^2 & m^3 \\ 1 & n^2 & n^3 \end{vmatrix}$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} mn & l & l^2 \\ nl & m & m^2 \\ lm & n & n^2 \end{vmatrix}$$

Multiplying R_1, R_2 and R_3 by l, m and n respectively and dividing the determined by lmn we get

$$= \frac{1}{l} \cdot \frac{1}{m} \cdot \frac{1}{n} \begin{vmatrix} lmn & l^2 & l^3 \\ lmn & m^2 & m^3 \\ lmn & n^2 & n^3 \end{vmatrix}$$

$$= \frac{1}{lmn} \cdot lmn \begin{vmatrix} 1 & l^2 & l^3 \\ 1 & m^2 & m^3 \\ 1 & n^2 & n^3 \end{vmatrix}$$

Taking common lmn from C_1 we get

$$= \begin{vmatrix} 1 & l^2 & l^3 \\ 1 & m^2 & m^3 \\ 1 & n^2 & n^3 \end{vmatrix} = \text{R.H.S}$$

L.H.S = R.H.S

Hence $\begin{vmatrix} mn & l & l^2 \\ nl & m & m^2 \\ lm & n & n^2 \end{vmatrix} = \begin{vmatrix} 1 & l^2 & l^3 \\ 1 & m^2 & m^3 \\ 1 & n^2 & n^3 \end{vmatrix}$

$$(vii) \begin{vmatrix} 2a & 2b & 2c \\ a+b & 2b & b+c \\ a+c & b+c & 2c \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} 2a & 2b & 2c \\ a+b & 2b & b+c \\ a+c & b+c & 2c \end{vmatrix}$$

$$= 2 \begin{vmatrix} a & b & c \\ a+b & 2b & b+c \\ a+c & b+c & 2c \end{vmatrix}$$

Taking 2 common from R_1

$$= 2 \begin{vmatrix} a & b & c \\ b & b & b \\ c & c & c \end{vmatrix}$$

By $R_2 - R_1$ and $R_3 - R_1$

$$= 2bc \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

By taking common in b from R_2 and

c from R_3

$$= 2bc(0)$$

$\therefore R_2$ and R_3 are identical

$$0 = \text{R.H.S}$$

$$\text{Since L.H.S} = \text{R.H.S so } \begin{vmatrix} 2a & 2b & 2c \\ a+b & 2b & b+c \\ a+c & b+c & 2c \end{vmatrix} = 0$$

$$(viii) \begin{vmatrix} 7 & 2 & 6 \\ 6 & 3 & 2 \\ -3 & 5 & 1 \end{vmatrix} = \begin{vmatrix} 7 & 2 & 7 \\ 6 & 3 & 5 \\ -3 & 5 & -3 \end{vmatrix} + \begin{vmatrix} 7 & 2 & -1 \\ 6 & 3 & -3 \\ -3 & 5 & 4 \end{vmatrix}$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} 7 & 2 & 6 \\ 6 & 3 & 2 \\ -3 & 5 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 7 & 2 & (7-1) \\ 6 & 3 & (5-3) \\ -3 & 5 & (-3+4) \end{vmatrix}$$

$$= \begin{vmatrix} 7 & 2 & 7 \\ 6 & 3 & 5 \\ -3 & 5 & -3 \end{vmatrix} + \begin{vmatrix} 7 & 2 & -1 \\ 6 & 3 & -3 \\ -3 & 5 & 4 \end{vmatrix}$$

By using the property of each entry if a row (or column) of a square matrix consist of two terms, then its determinant can be written as the sum of two determinants

$$= \text{R.H.S}$$

$$\text{L.H.S} = \text{R.H.S}$$

Hence

$$\begin{vmatrix} 7 & 2 & 6 \\ 6 & 3 & 2 \\ -3 & 5 & 1 \end{vmatrix} = \begin{vmatrix} 7 & 2 & 7 \\ 6 & 3 & 5 \\ -3 & 5 & -3 \end{vmatrix} + \begin{vmatrix} 7 & 2 & -1 \\ 6 & 3 & -3 \\ -3 & 5 & 4 \end{vmatrix}$$

$$\text{(ix)} \quad \begin{vmatrix} -a & 0 & c \\ 0 & a & -b \\ b & -c & 0 \end{vmatrix} = 0$$

Solution: Consider

$$\text{L.H.S} = \begin{vmatrix} -a & 0 & c \\ 0 & a & -b \\ b & -c & 0 \end{vmatrix}$$

Multiply and divided by b, c and a to R_1, R_2 and R_3 respectively

$$= \frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c} \begin{vmatrix} -ab & 0 & bc \\ 0 & ac & -bc \\ ab & -ac & 0 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} -ab+0+ab & 0+ac-ac & bc-bc+0 \\ 0 & ac & -bc \\ ab & -ac & 0 \end{vmatrix} \quad \text{By } R_1 + (R_2 + R_3)$$

$$= \frac{1}{abc} \begin{vmatrix} 0 & 0 & 0 \\ 0 & ac & -bc \\ ab & -ac & 0 \end{vmatrix}$$

$$= \frac{1}{abc} (0)$$

$\therefore$ Each entry of R_1 is zero

$$0 = \text{R.H.S}$$

$$\text{Since L.H.S.} = \text{R.H.S so } \begin{vmatrix} -a & 0 & c \\ 0 & a & -b \\ b & -c & 0 \end{vmatrix} = 0$$

Q.6 Find the values of x if

$$(i) \begin{vmatrix} 3 & 1 & x \\ -1 & 3 & 4 \\ x & 1 & 0 \end{vmatrix} = -30$$

Solution:

$$\text{Consider } \begin{vmatrix} 3 & 1 & x \\ -1 & 3 & 4 \\ x & 1 & 0 \end{vmatrix} = -30$$

Expanding from R_1 , we get

$$3 \begin{vmatrix} 3 & 4 \\ 1 & 0 \end{vmatrix} - 1 \begin{vmatrix} -1 & 4 \\ x & 0 \end{vmatrix} + x \begin{vmatrix} -1 & 3 \\ x & 1 \end{vmatrix} = -30$$

$$3(0-4) - 1(0-4x) + x(-1-3x) = -30$$

$$-12 + 4x - x - 3x^2 = -30$$

$$-3x^2 + 3x - 12 + 30 = 0$$

$$-3x^2 + 3x + 18 = 0$$

$$-3(x^2 - x - 6) = 0$$

$$\Rightarrow x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$x(x-3) + 2(x-3) = 0$$

$$(x-3)(x+2) = 0$$

$$\Rightarrow \text{Either } x-3=0 \quad \text{or} \quad x+2=0$$

$$\Rightarrow x=3 \quad \text{or} \quad x=-2$$

So $x=3$ and $x=-2$ are the required values of x

$$(ii) \begin{vmatrix} 1 & x-1 & 3 \\ -1 & x+1 & 2 \\ 2 & -2 & x \end{vmatrix} = 0$$

Solution:

$$\text{Consider } \begin{vmatrix} 1 & x-1 & 3 \\ -1 & x+1 & 2 \\ 2 & -2 & x \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x-1 & 3 \\ 0 & 2x & 5 \\ 0 & -2x & x-6 \end{vmatrix} = 0$$

By $R_2 + R_1, R_3 - 2R_1$

Expanding from C_1 we get

$$1 \begin{vmatrix} 2x & 5 \\ -2x & x-6 \end{vmatrix} - 0 \begin{vmatrix} x-1 & 3 \\ -2x & x-6 \end{vmatrix} + 0 \begin{vmatrix} x-1 & 3 \\ 2x & 5 \end{vmatrix} = 0$$

$$2x(x-6) + 10x = 0$$

$$2x^2 - 12x + 10x = 0$$

$$2x^2 - 2x = 0$$

$$2x(x-1) = 0$$

$$\Rightarrow \text{Either } 2x = 0 \quad \text{or} \quad x-1 = 0$$

$$\Rightarrow x = 0 \quad \text{or} \quad x = 1$$

So $x = 0$ and $x = 1$ are the required values of x .

$$\text{(iii)} \quad \begin{vmatrix} 1 & 2 & 1 \\ 2 & x & 2 \\ 3 & 6 & x \end{vmatrix} = 0$$

Solution:

$$\text{Consider } \begin{vmatrix} 1 & 2 & 1 \\ 2 & x & 2 \\ 3 & 6 & x \end{vmatrix} = 0$$

Expanding from R_1 we get

$$1 \begin{vmatrix} x & 2 \\ 6 & x \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 \\ 3 & x \end{vmatrix} + 1 \begin{vmatrix} 2 & x \\ 3 & 6 \end{vmatrix} = 0$$

$$x^2 - 12 - 2(2x - 6) + 12 - 3x = 0$$

$$x^2 - 12 - 4x + 12 + 12 - 3x = 0$$

$$x^2 - 7x + 12 = 0$$

$$x^2 - 3x - 4x + 12 = 0$$

$$x(x-3) - 4(x-3) = 0$$

$$(x-3)(x-4) = 0$$

$$\Rightarrow \text{Either } x-3 = 0 \quad \text{or} \quad x-4 = 0$$

$$x = 3 \quad \text{or} \quad x = 4$$

So $x = 3$ and $x = 4$ are required values of x

Q.7 Evaluate the following determinants

$$(i) \begin{vmatrix} 3 & 4 & 2 & 7 \\ 2 & 5 & 0 & 3 \\ 1 & 2 & -3 & 5 \\ 4 & 1 & -2 & 6 \end{vmatrix}$$

Solution:

$$\text{Consider } \begin{vmatrix} 3 & 4 & 2 & 7 \\ 2 & 5 & 0 & 3 \\ 1 & 2 & -3 & 5 \\ 4 & 1 & -2 & 6 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 & 2 & 4 \\ 2 & 5 & 0 & 3 \\ 1 & 2 & -3 & 5 \\ 4 & 1 & -2 & 6 \end{vmatrix}$$

By $R_1 - R_2$

$$= \begin{vmatrix} 1 & -1 & 2 & 4 \\ 0 & 7 & -4 & -5 \\ 0 & 3 & -5 & 1 \\ 0 & 5 & -10 & -10 \end{vmatrix}$$

By $R_2 - 2R_1, R_3 - R_1, R_4 - 4R_1$

Expanding from C_1

$$= 1 \begin{vmatrix} 7 & -4 & -5 \\ 3 & -5 & 1 \\ 5 & -10 & -10 \end{vmatrix} - 0 \begin{vmatrix} -1 & 2 & 4 \\ 3 & -5 & 1 \\ 5 & -10 & -10 \end{vmatrix} + 0 \begin{vmatrix} -1 & 2 & 4 \\ 7 & -4 & -5 \\ 5 & -10 & -10 \end{vmatrix} - 0 \begin{vmatrix} -1 & 2 & 4 \\ 7 & -4 & -5 \\ 3 & -5 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 7 & -4 & -5 \\ 3 & -5 & 1 \\ 5 & -10 & -10 \end{vmatrix}$$

Expanding from R_1

$$= 7(50 + 10) + 4(-30 - 5) - 5(-30 + 25)$$

$$= 7(60) + 4(-35) - 5(-5)$$

$$= 420 - 140 + 25$$

$$= 280 + 25$$

$$= 305$$

$$\text{Hence } \begin{vmatrix} 3 & 4 & 2 & 7 \\ 2 & 5 & 0 & 3 \\ 1 & 2 & -3 & 5 \\ 4 & 1 & -2 & 6 \end{vmatrix} = 305$$

$$\text{(ii)} \quad \begin{vmatrix} 2 & 3 & 1 & -1 \\ 4 & 0 & 2 & 1 \\ 5 & 2 & -1 & 6 \\ 3 & -7 & 2 & -2 \end{vmatrix}$$

Solution:

$$\text{Consider } \begin{vmatrix} 2 & 3 & 1 & -1 \\ 4 & 0 & 2 & 1 \\ 5 & 2 & -1 & 6 \\ 3 & -7 & 2 & -2 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 3 & 1 & -1 \\ 0 & -6 & 0 & 3 \\ 7 & 5 & 0 & 5 \\ -1 & -13 & 0 & 0 \end{vmatrix}$$

By $R_2 - 2R_1, R_3 + R_1, R_4 - 2R_1$

Expanding from C_3

$$= 1 \begin{vmatrix} 0 & -6 & 3 \\ 7 & 5 & 5 \\ -1 & -13 & 0 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 & -1 \\ 7 & 5 & 5 \\ -1 & -13 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 3 & -1 \\ 0 & -6 & 3 \\ -1 & -13 & 0 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 & -1 \\ 0 & -6 & 3 \\ 7 & 5 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -6 & 3 \\ 7 & 5 & 5 \\ -1 & -13 & 0 \end{vmatrix}$$

Expanding from R_1

$$= 0(0+65) + 6(0+5) + 3(-91+5)$$

$$= 0 + 30 + 3(-86)$$

$$= 30 - 258$$

$$= -228$$

$$\text{Hence } \begin{vmatrix} 2 & 3 & 1 & -1 \\ 4 & 0 & 2 & 1 \\ 5 & 2 & -1 & 6 \\ 3 & -7 & 2 & -2 \end{vmatrix} = -228$$

$$(iii) \begin{vmatrix} -3 & 9 & 1 & 1 \\ 0 & 3 & -1 & 2 \\ 9 & 7 & -1 & 1 \\ -2 & 0 & 1 & -1 \end{vmatrix}$$

Solution:

$$\text{Consider } \begin{vmatrix} -3 & 9 & 1 & 1 \\ 0 & 3 & -1 & 2 \\ 9 & 7 & -1 & 1 \\ -2 & 0 & 1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} -3 & 9 & 1 & 1 \\ -3 & 12 & 0 & 3 \\ 6 & 16 & 0 & 2 \\ 1 & -9 & 0 & -2 \end{vmatrix}$$

By $R_2 + R_1, R_3 + R_1, R_4 - R_1$

Expanding from C_3

$$= 1 \begin{vmatrix} -3 & 12 & 3 \\ 6 & 16 & 2 \\ 1 & -9 & -2 \end{vmatrix} - 0 \begin{vmatrix} -3 & 9 & 1 \\ 6 & 16 & 2 \\ 1 & -9 & -2 \end{vmatrix} + 0 \begin{vmatrix} -3 & 9 & 1 \\ -3 & 12 & 3 \\ 1 & -9 & -2 \end{vmatrix} - 0 \begin{vmatrix} -3 & 9 & 1 \\ -3 & 12 & 3 \\ 6 & 16 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} -3 & 12 & 3 \\ 6 & 16 & 2 \\ 1 & -9 & -2 \end{vmatrix}$$

Expanding from R_1

$$= -3(-32+18) - 12(-12-2) + 3(-54-16)$$

$$= 42 + 168 + 3(-70)$$

$$= 0$$

$$\text{Hence } \begin{vmatrix} -3 & 9 & 1 & 1 \\ 0 & 3 & -1 & 2 \\ 9 & 7 & -1 & 1 \\ -2 & 0 & 1 & -1 \end{vmatrix} = 0$$

Q.8 Show that
$$\begin{vmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{vmatrix} = (x+3)(x-1)^3$$

Proof:

$$\text{L.H.S} = \begin{vmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{vmatrix}$$

$$= \begin{vmatrix} x+3 & 1 & 1 & 1 \\ x+3 & x & 1 & 1 \\ x+3 & 1 & x & 1 \\ x+3 & 1 & 1 & x \end{vmatrix}$$

By $C_1 + (C_2 + C_3 + C_4)$

$$= (x+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{vmatrix}$$

By taking common $(x+3)$ from C_1

$$= (x+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & x-1 & 0 & 0 \\ 0 & 0 & x-1 & 0 \\ 0 & 0 & 0 & x-1 \end{vmatrix}$$

By $R_2 - R_1, R_3 - R_1, R_4 - R_1$

$$= (x+3) [1 \times (x-1) \times (x-1) \times (x-1)]$$

$\therefore$ Determinant of a diagonal matrix is the product of

diagonal entries

$$= (x+3)(x-1)^3 = \text{R.H.S}$$

Hence
$$\begin{vmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{vmatrix} = (x+3)(x-1)^3$$

Q.9 Find $|AA'|$ and $|A'A|$

(i) $A = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$

Solution:

if $A = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$ then

$$A' = \begin{bmatrix} 3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Now

$$\begin{aligned}
 AA^t &= \begin{bmatrix} 3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 9+4+1 & 6+2-3 \\ 6+2-3 & 4+1+9 \end{bmatrix} \\
 &= \begin{bmatrix} 14 & 5 \\ 5 & 14 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } |AA^t| &= \begin{vmatrix} 14 & 5 \\ 5 & 14 \end{vmatrix} \\
 &= 196 - 25 \\
 |AA^t| &= 171
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } A^tA &= \begin{bmatrix} 3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 & -1 \\ 2 & 1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 6+4 & 9+2 & -3+6 \\ 4+2 & 6+1 & -2+3 \\ -2+6 & -3+3 & 1+9 \end{bmatrix} \\
 &= \begin{vmatrix} 10 & 11 & 3 \\ 6 & 7 & 1 \\ 4 & 0 & 10 \end{vmatrix} \\
 &= 10(70-0) - 11(60-4) + 3(0-28) \\
 &= 700 - 11(56) - 84 \\
 &= 700 - 616 - 84 \\
 &= 700 - 700 = 0 \\
 \text{Hence } |A^tA| &= 0
 \end{aligned}$$

$$\text{(ii) If } A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}, \text{ then}$$

Solution:

$$A^t = \begin{bmatrix} 3 & 2 & 1 & 2 \\ 4 & 1 & 1 & 3 \end{bmatrix}$$

$$\text{Now } AA^t = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 & 2 \\ 4 & 1 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9+16 & 6+4 & 3+4 & 6+12 \\ 6+4 & 4+1 & 2+1 & 4+3 \\ 3+4 & 2+1 & 1+1 & 2+3 \\ 6+12 & 4+3 & 2+3 & 4+9 \end{bmatrix}$$

$$AA^t = \begin{bmatrix} 25 & 10 & 7 & 18 \\ 10 & 5 & 3 & 7 \\ 7 & 3 & 2 & 5 \\ 18 & 7 & 5 & 13 \end{bmatrix}$$

$$|AA^t| = \begin{vmatrix} 25 & 10 & 7 & 18 \\ 10 & 5 & 3 & 7 \\ 7 & 3 & 2 & 5 \\ 18 & 7 & 5 & 13 \end{vmatrix}$$

$$= \begin{vmatrix} 25 & 10 & 7 & 18 \\ 3 & 2 & 1 & 2 \\ 7 & 3 & 2 & 5 \\ 18 & 7 & 5 & 13 \end{vmatrix}$$

By $R_2 - R_3$

$$= \begin{vmatrix} 4 & -4 & 0 & 4 \\ 3 & 2 & 1 & 2 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 0 & 3 \end{vmatrix}$$

By $R_1 - 7R_2, R_3 - 2R_2, R_4 - 5R_2$

$$= 4 \begin{vmatrix} 1 & -1 & 0 & 1 \\ 3 & 2 & 1 & 2 \\ 1 & -1 & 0 & 1 \\ 3 & -3 & 0 & 3 \end{vmatrix}$$

$$= 4(0)$$

$\therefore R_1$ and R_3 are identical

$$= 0$$

$$\text{Hence } |AA^t| = 0$$

$$\text{Now } A^t A = \begin{bmatrix} 3 & 2 & 1 & 2 \\ 4 & 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9+4+1+4 & 12+2+1+6 \\ 12+2+1+6 & 16+1+1+9 \end{bmatrix}$$

$$A^t A = \begin{bmatrix} 18 & 21 \\ 21 & 27 \end{bmatrix}$$

$$\text{Now } |A^t A| = \begin{vmatrix} 18 & 21 \\ 21 & 27 \end{vmatrix}$$

$$= 486 - 441$$

$$= 45$$

$$\text{Hence } |A^t A| = 45$$

Q.10 If A is a square matrix of order 3, then show that $|kA| = k^3 |A|$

Proof: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix of order 3 and k be any scalar then by scalar

multiplication

$$kA = k \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & ka_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{bmatrix}$$

Now

$$|kA| = \begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{vmatrix}$$

$$= k.k.k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Taking common k from R_1, R_2 and R_3 , respectively

$$= k^3 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k^3 |A|$$

$$\text{Hence } |kA| = k^3 |A|$$

Q.11 Find the values of λ if A and B are singular

$$(i) \quad A = \begin{bmatrix} 4 & \lambda & 3 \\ 7 & 3 & 6 \\ 2 & 3 & 1 \end{bmatrix}$$

Solution: If A is singular, then $|A| = 0$

$$|A| = \begin{vmatrix} 4 & \lambda & 3 \\ 7 & 3 & 6 \\ 2 & 3 & 1 \end{vmatrix} = 0$$

Expanding from R_1 , we get

$$4 \begin{vmatrix} 3 & 6 \\ 3 & 1 \end{vmatrix} - \lambda \begin{vmatrix} 7 & 6 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 7 & 3 \\ 2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 4(3-18) - \lambda(7-12) + 3(21-6) = 0$$

$$\Rightarrow -60 + 5\lambda + 45 = 0$$

$$\Rightarrow 5\lambda - 15 = 0$$

$$\Rightarrow \lambda = 3$$

Hence A will be singular if

$$\lambda = 3$$

$$(ii) \quad B = \begin{bmatrix} 5 & 1 & 2 & 0 \\ 8 & 2 & 5 & 1 \\ 3 & 2 & 0 & 1 \\ 2 & \lambda & -1 & 3 \end{bmatrix}$$

Solution: If B is singular, then $|B| = 0$

$$\begin{vmatrix} 5 & 1 & 2 & 0 \\ 8 & 2 & 5 & 1 \\ 3 & 2 & 0 & 1 \\ 2 & \lambda & -1 & 3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 5 & 1 & 2 & 0 \\ 8 & 2 & 5 & 1 \\ -5 & 0 & -5 & 0 \\ -22 & \lambda - 6 & -16 & 0 \end{vmatrix} = 0$$

By $R_3 - R_2, R_4 - 3R_2$

Expanding from C_4 , we get

$$-0 \begin{vmatrix} 8 & 2 & 5 \\ -5 & 0 & -5 \\ -22 & \lambda - 6 & -16 \end{vmatrix} + 1 \begin{vmatrix} 5 & 1 & 2 \\ -5 & 0 & -5 \\ -22 & \lambda - 6 & -16 \end{vmatrix} - 0 \begin{vmatrix} 5 & 1 & 2 \\ 8 & 2 & 5 \\ -22 & \lambda - 6 & -16 \end{vmatrix} + 0 \begin{vmatrix} 5 & 1 & 2 \\ 8 & 2 & 5 \\ -5 & 0 & -5 \end{vmatrix} = 0$$

$$\begin{vmatrix} 5 & 1 & 2 \\ -5 & 0 & -5 \\ -22 & \lambda - 6 & -16 \end{vmatrix} = 0$$

Expanding from R_1 we get

$$5 \begin{vmatrix} 0 & -5 \\ \lambda - 6 & -16 \end{vmatrix} - 1 \begin{vmatrix} -5 & -5 \\ -22 & -16 \end{vmatrix} + 2 \begin{vmatrix} -5 & 0 \\ -22 & \lambda - 6 \end{vmatrix} = 0$$

$$5(0 - (-5)(\lambda - 6)) - 1(80 - 110) + 2((-5)(\lambda - 6) - 0) = 0$$

$$5(5\lambda - 30) - 1(-30) + 2(-5\lambda + 30) = 0$$

$$25\lambda - 150 + 30 - 10\lambda + 60 = 0$$

$$15\lambda - 60 = 0$$

$$15\lambda = 60$$

$$\lambda = 4$$

Hence B is singular if $\lambda = 4$

Q.12 Which of the following matrices are singular and which of them are non-singular.

(i) $\begin{bmatrix} 1 & 0 & 3 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}$

Solution:

Consider $\begin{bmatrix} 1 & 0 & 3 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}$, then

$$\begin{vmatrix} 1 & 0 & 3 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{vmatrix}$$

Expanding from R_1 , we get

$$\begin{aligned} &= 1 \begin{vmatrix} 1 & -1 \\ 0 & 4 \end{vmatrix} - 0 \begin{vmatrix} 3 & -1 \\ 0 & 4 \end{vmatrix} + 3 \begin{vmatrix} 3 & 1 \\ 0 & 2 \end{vmatrix} \\ &= 1(4 - 0) - 0(12 - 0) + 3(6 - 0) \\ &= 4 - 0 + 18 \\ &= 22 \neq 0 \end{aligned}$$

Since $\begin{vmatrix} 1 & 0 & 3 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{vmatrix} = 22 \neq 0$ so

$$\begin{bmatrix} 1 & 0 & 3 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix} \text{ is non-singular matrix.}$$

$$(ii) \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$$

Solution:

$$\text{Consider } \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix} \text{ then}$$

$$\begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix}$$

Expanding from R_1 we get

$$\begin{aligned} &= 2 \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix} - 3 \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} \\ &= 2(5) - 3(5 - 0) - 1(-3 - 2) \\ &= 10 - 15 + 5 \\ &= 0 \end{aligned}$$

$$\text{Since } \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix} = 0 \text{ so}$$

$$\begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix} \text{ is a singular matrix}$$

$$(iii) \begin{bmatrix} 1 & 1 & 2 & -1 \\ 1 & 2 & -1 & -3 \\ 2 & 3 & 1 & 2 \\ 3 & -1 & 3 & 4 \end{bmatrix}$$

Solution:

$$\text{Consider } \begin{bmatrix} 1 & 1 & 2 & -1 \\ 1 & 2 & -1 & -3 \\ 2 & 3 & 1 & 2 \\ 3 & -1 & 3 & 4 \end{bmatrix} \text{ then}$$

$$\begin{vmatrix} 1 & 1 & 2 & -1 \\ 1 & 2 & -1 & -3 \\ 2 & 3 & 1 & 2 \\ 3 & -1 & 3 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 2 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 1 & -3 & 4 \\ 0 & -4 & -3 & 7 \end{vmatrix}$$

By $R_2 - R_1, R_3 - 2R_1, R_4 - 3R_1$

Expanding from C_1 we get

$$= 1 \begin{vmatrix} 1 & -3 & -2 \\ 1 & -3 & 4 \\ -4 & -3 & 7 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & 4 \\ -4 & -3 & 7 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & -2 \\ -4 & -3 & 7 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & -2 \\ 1 & -3 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -3 & -2 \\ 1 & -3 & 4 \\ -4 & -3 & 7 \end{vmatrix}$$

Expanding from R_1 , we get

$$\begin{aligned} &= 1 \begin{vmatrix} -3 & 4 \\ -3 & 7 \end{vmatrix} + 3 \begin{vmatrix} 1 & 4 \\ -4 & 7 \end{vmatrix} - 2 \begin{vmatrix} 1 & -3 \\ -4 & -3 \end{vmatrix} \\ &= 1(-21+12) + 3(7+16) - 2(-3-12) \\ &= -9 + 69 + 30 \\ &= 90 \neq 0 \end{aligned}$$

$$\text{Since } \begin{vmatrix} 1 & 1 & 2 & -1 \\ 1 & 2 & -1 & -3 \\ 2 & 3 & 1 & 2 \\ 3 & -1 & 3 & 4 \end{vmatrix} = 90 \neq 0 \text{ so}$$

$$\begin{bmatrix} 1 & 1 & 2 & -1 \\ 1 & 2 & -1 & -3 \\ 2 & 3 & 1 & 2 \\ 3 & -1 & 3 & 4 \end{bmatrix} \text{ is non-singular matrix}$$

Q.13 Find the inverse of $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$ and show that $A^{-1}A = I_3$

Solution: Since $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$

We know that

$$A^{-1} = \frac{\text{Adj}A}{|A|}, |A| \neq 0$$

So,

$$|A| = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix}$$

$$= 2(5-0) - 1(5-0) + 0(-3-2)$$

$$|A| = 5 \neq 0$$

$$\text{Now } AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t \text{ and } A_{ij} = (-1)^{i+j} M_{ij} \text{ so}$$

$$A_{11} = (-1)^{1+1} M_{11} = \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix} = 5 - 0 = 5$$

$$A_{12} = (-1)^{1+2} M_{12} = - \begin{vmatrix} 1 & 0 \\ 2 & 5 \end{vmatrix} = -(5-0) = -5$$

$$A_{13} = (-1)^{1+3} M_{13} = \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} = -3 - 2 = -5$$

$$A_{21} = (-1)^{2+1} M_{21} = - \begin{vmatrix} 1 & 0 \\ -3 & 5 \end{vmatrix} = -(5-0) = -5$$

$$A_{22} = (-1)^{2+2} M_{22} = \begin{vmatrix} 2 & 0 \\ 2 & 5 \end{vmatrix} = 10 - 0 = 10$$

$$A_{23} = (-1)^{2+3} M_{23} = - \begin{vmatrix} 2 & 1 \\ 2 & -3 \end{vmatrix} = -(-6-2) = 8$$

$$A_{31} = (-1)^{3+1} M_{31} = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$A_{32} = (-1)^{3+2} M_{32} = - \begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

$$A_{33} = (-1)^{3+3} M_{33} = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = 2 - 1 = 1$$

$$\text{Now } AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\text{Adj}A = \begin{bmatrix} 5 & -5 & -5 \\ -5 & 10 & 8 \\ 0 & 0 & 1 \end{bmatrix}^t$$

$$\text{Adj}A = \begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}$$

$$\text{Now } A^{-1} = \frac{\text{Adj}A}{|A|}$$

$$A^{-1} = \frac{\begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}}{5} = \frac{1}{5} \begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix}$$

$$\text{Now } A^{-1}A = \frac{1}{5} \begin{bmatrix} 5 & -5 & 0 \\ -5 & 10 & 0 \\ -5 & 8 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 10-5+0 & 5-5+0 & 0+0+0 \\ -10+10+0 & -5+10+0 & 0+0+0 \\ -10+8+2 & -5+8-3 & 0+0+5 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

Hence

$$A^{-1}A = I_3$$

Q.14 Verify that $(AB)^{-1} = B^{-1}A^{-1}$ if

$$(i) \quad A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, B = \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix}$$

Solution:

$$|A| = \begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} \quad \text{and} \quad |B| = \begin{vmatrix} -3 & 1 \\ 4 & -1 \end{vmatrix}$$

$$|A| = 2 \quad \text{and} \quad |B| = -1$$

$$\text{Adj}A = \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \text{Adj}B = \begin{bmatrix} -1 & -1 \\ -4 & -3 \end{bmatrix}$$

So

$$A^{-1} = \frac{\text{Adj}A}{|A|} \quad ; \quad B^{-1} = \frac{\text{Adj}B}{|B|}$$

$$A^{-1} = \frac{\begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}}{2} \quad ; \quad B^{-1} = \frac{\begin{bmatrix} -1 & -1 \\ -4 & -3 \end{bmatrix}}{-1}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} \quad ; \quad B^{-1} = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 0+1 & -2+1 \\ 0+3 & -8+3 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 3 & -5 \end{bmatrix} \quad (i)$$

Now

$$AB = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -3+8 & 1-2 \\ 3+0 & -1+0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 5 & -1 \\ 3 & -1 \end{bmatrix}$$

So

$$|AB| = \begin{vmatrix} 5 & -1 \\ 3 & -1 \end{vmatrix} = -5+3$$

$$|AB| = -2$$

$$\text{Adj}(AB) = \begin{bmatrix} -1 & 1 \\ -3 & 5 \end{bmatrix}$$

So

$$(AB)^{-1} = \frac{\text{Adj}(AB)}{|AB|}$$

$$= \frac{\begin{bmatrix} -1 & 1 \\ -3 & 5 \end{bmatrix}}{-2}$$

$$(AB)^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 3 & -5 \end{bmatrix} \quad (\text{ii})$$

From (i) and (ii)

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(\text{ii}) \quad \mathbf{A} = \begin{bmatrix} 5 & 1 \\ 2 & 2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$$

Solution:

$$\mathbf{A} = \begin{bmatrix} 5 & 1 \\ 2 & 2 \end{bmatrix} \quad ; \quad \mathbf{B} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$$

$$|\mathbf{A}| = \begin{vmatrix} 5 & 1 \\ 2 & 2 \end{vmatrix} \quad ; \quad |\mathbf{B}| = \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix}$$

$$|\mathbf{A}| = 8 \quad ; \quad |\mathbf{B}| = -2$$

$$\text{Adj}\mathbf{A} = \begin{bmatrix} 2 & -1 \\ -2 & 5 \end{bmatrix} \quad ; \quad \text{Adj}\mathbf{B} = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$$

So

$$\mathbf{A}^{-1} = \frac{\text{Adj}\mathbf{A}}{|\mathbf{A}|} \quad ; \quad \mathbf{B}^{-1} = \frac{\text{Adj}\mathbf{B}}{|\mathbf{B}|}$$

$$\mathbf{A}^{-1} = \frac{1}{8} \begin{bmatrix} 2 & -1 \\ -2 & 5 \end{bmatrix} \quad ; \quad \mathbf{B}^{-1} = -\frac{1}{2} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$$

So

$$\begin{aligned} \mathbf{B}^{-1}\mathbf{A}^{-1} &= \left(-\frac{1}{2}\right)\left(\frac{1}{8}\right) \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -2 & 5 \end{bmatrix} \\ &= -\frac{1}{16} \begin{bmatrix} 2+6 & -1-15 \\ -4-8 & 2+20 \end{bmatrix} \\ &= -\frac{1}{16} \begin{bmatrix} 8 & -16 \\ -12 & 22 \end{bmatrix} \\ \mathbf{B}^{-1}\mathbf{A}^{-1} &= -\frac{1}{8} \begin{bmatrix} 4 & -8 \\ -6 & 11 \end{bmatrix} \quad (\text{i}) \end{aligned}$$

Consider

$$\mathbf{AB} = \begin{bmatrix} 5 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} 20+2 & 15+1 \\ 8+4 & 6+2 \end{bmatrix} \\
 AB &= \begin{bmatrix} 22 & 16 \\ 12 & 8 \end{bmatrix} \\
 |AB| &= \begin{vmatrix} 22 & 16 \\ 12 & 8 \end{vmatrix} = 176 - 192 \\
 |AB| &= -16 \\
 \text{Adj}(AB) &= \begin{bmatrix} 8 & -16 \\ -12 & 22 \end{bmatrix}
 \end{aligned}$$

So

$$\begin{aligned}
 (AB)^{-1} &= \frac{\text{Adj}(AB)}{|AB|} \\
 &= -\frac{1}{16} \begin{bmatrix} 8 & -16 \\ -12 & 22 \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} 4 & -8 \\ -6 & 11 \end{bmatrix} \quad \text{(ii)}
 \end{aligned}$$

From (i) and (ii)

$$(AB)^{-1} = B^{-1}A^{-1}$$

Q.15 Verify that $(AB)^t = B^t A^t$, if $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 3 & 2 \\ 0 & -1 \end{bmatrix}$

Solution:

$$\text{If } A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}, \text{ then } A^t = \begin{bmatrix} 1 & 0 \\ -1 & 3 \\ 2 & 1 \end{bmatrix}$$

$$\text{and } B = \begin{bmatrix} 1 & 1 \\ 3 & 2 \\ 0 & -1 \end{bmatrix}, \text{ then } B^t = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\text{Consider } AB = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-3+0 & 1-2-2 \\ 0+9+0 & 0+6-1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -2 & -3 \\ 9 & 5 \end{bmatrix}$$

$$(AB)^t = \begin{bmatrix} -2 & 9 \\ -3 & 5 \end{bmatrix} \quad (i)$$

$$\text{Consider } B^t A^t = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 3 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-3+0 & 0+9+0 \\ 1-2-2 & 0+6-1 \end{bmatrix}$$

$$B^t A^t = \begin{bmatrix} -2 & 9 \\ -3 & 5 \end{bmatrix} \quad (ii)$$

From (i) and (ii)

$$(AB)^t = B^t A^t$$

Q.16 If $A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$ verify that $(A^{-1})^t = (A^t)^{-1}$

Solution:

$$\text{If } A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$$

$$\Rightarrow A^t = \begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix}$$

$$\text{And } A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{\begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}}{2+3} = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix}$$

$$(A^{-1})^t = \frac{1}{5} \begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix} \quad (i)$$

$$\text{Consider } (A^t)^{-1} = \frac{\text{Adj}A^t}{|A^t|} = \frac{\begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix}}{2+3}$$

$$(A^t)^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -3 \\ 1 & 2 \end{bmatrix} \quad (ii)$$

From (i) and (ii)

$$(A^{-1})^t = (A^t)^{-1}$$

Q.17 If A and B are non-singular matrices, then show that

(i) $(AB)^{-1} = B^{-1}A^{-1}$

Solution:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Since A and B are non singular matrices, so they are invertible now consider

$$\begin{aligned} & (AB)(B^{-1}A^{-1}) \\ &= A(BB^{-1})A^{-1} && \because \text{Using associative property} \\ &= A(I)A^{-1} && \because BB^{-1} = B^{-1}B = I \\ &= AA^{-1} \\ &= I \\ & (AB)(B^{-1}A^{-1}) = I \end{aligned} \tag{i}$$

Again consider

$$\begin{aligned} & (B^{-1}A^{-1})(AB) \\ &= B^{-1}(A^{-1}A)B && \because \text{Using associative property} \\ &= B^{-1}(I)B \\ &= B^{-1}B && \because IB = B \\ &= I \\ & (B^{-1}A^{-1})(AB) = I \end{aligned} \tag{ii}$$

From (i) and (ii) it is clear that AB and $B^{-1}A^{-1}$ are multiplicative inverse of each other

So, we can say that $B^{-1}A^{-1}$ is the multiplicative inverse of AB hence

$$(AB)^{-1} = B^{-1}A^{-1}$$

(ii) $(A^{-1})^{-1} = A$

Consider

$$\begin{aligned} & (A)(A^{-1}) \\ &= AA^{-1} \\ &= I \\ & AA^{-1} = I \end{aligned} \tag{i}$$

Again $(A^{-1})(A)$

$$\begin{aligned} &= A^{-1}A \\ &= I \\ & A^{-1}A = I \end{aligned} \tag{ii}$$

From (i) and (ii) it is clear that

$$AA^{-1} = A^{-1}A = I$$

A^{-1} is the multiplicative inverse of A

$$(A^{-1})^{-1} = A$$

Elementary row and Column operations on a Matrix:

Following are the row or column operations.

- (i) Interchanging of two rows or columns.
- (ii) Multiplying a row or column by a non zero number.
- (iii) Adding a multiple of one row or column to another row or column respectively.

Upper Triangular Matrix:

A square matrix $A = [a_{ij}]$ is called **upper triangular** if all elements below the principal diagonal are zero, that is, $a_{ij} = 0$ for all $i > j$.

Lower Triangular Matrix:

A square matrix $A = [a_{ij}]$ is said to be **lower triangular** if all elements above the principal diagonal are zero, that is, $a_{ij} = 0$ for all $i < j$.

Triangular Matrix:

A square matrix A is named as **triangular** whether it is upper triangular or lower triangular matrix. For example the matrices.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 6 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -1 & 2 & 3 & 1 \end{bmatrix} \text{ are triangular matrices.}$$

Symmetric Matrix:

A square matrices $A = [a_{ij}]_{n \times n}$ is called **symmetric matrix** if $A^t = A$.

Skew Symmetric Matrix:

A square matrix $A = [a_{ij}]_{n \times n}$ is called **skew symmetric** or anti-symmetric if $A^t = -A$.

Hermitian Matrix:

A square matrix $A = [a_{ij}]_{n \times n}$ with complex entries, is called **hermitian** if $(\bar{A})^t = A$.

Skew Hermitian Matrix:

A square matrix $A = [a_{ij}]_{n \times n}$ with complex entries, is called **skew – hermitian** or **anti-hermitian** if $(\bar{A})^t = -A$.

Echelon and Reduced Echelon Forms of Matrix:

An $m \times n$ matrix A is called in **(row) echelon form** if

- (i) In each successive non-zero row, the number of zeros before the leading entry is greater than the number of such zeros in the preceding row,
- (ii) Every non-zero row in A precedes every zero row (if any).

Reduced Echelon Form of a Matrix:

A matrix is in reduced (row) echelon form if it satisfies the following conditions.

- It is in Row echelon form.
- Every leading entry is 1 and is the only non-zero entry in its column.

The matrices $\begin{bmatrix} 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ are in (row) reduced echelon form.

Rank of Matrix:

Let A be a non-zero matrix, if r is the number of non-zero rows when it is reduced to the reduced echelon form, then r is called the (row) **rank** of the matrix A .

EXERCISE 3.4

Q.1 If $A = \begin{bmatrix} 1 & -2 & 5 \\ -2 & 3 & -1 \\ 5 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 1 & -2 \\ 1 & 0 & -1 \\ -2 & -1 & 2 \end{bmatrix}$, then show that $A+B$ is symmetric.

Solution:

$$\begin{aligned} A+B &= \begin{bmatrix} 1 & -2 & 5 \\ -2 & 3 & -1 \\ 5 & -1 & 0 \end{bmatrix} + \begin{bmatrix} -3 & 1 & -2 \\ 1 & 0 & -1 \\ -2 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1-3 & -2+1 & 5-2 \\ -2+1 & 3+0 & -1-1 \\ 5-2 & -1-1 & 0+2 \end{bmatrix} \\ &= \begin{bmatrix} -2 & -1 & 3 \\ -1 & 3 & -2 \\ 3 & -2 & 2 \end{bmatrix} \end{aligned}$$

Now

$$\begin{aligned} (A+B)^t &= \begin{bmatrix} -2 & -1 & 3 \\ -1 & 3 & -2 \\ 3 & -2 & 2 \end{bmatrix}^t \\ &= \begin{bmatrix} -2 & -1 & 3 \\ -1 & 3 & -2 \\ 3 & -2 & 2 \end{bmatrix} = A+B \end{aligned}$$

$$(A+B)^t = A+B$$

Hence $A+B$ is symmetric

Q.2 If $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & -1 \\ -1 & 3 & 2 \end{bmatrix}$, show that

(i) $A+A^t$ is symmetric

Solution:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & -1 \\ -1 & 3 & 2 \end{bmatrix}, \text{ then } A^t = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

Now let

$$U = A + A^t = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & -1 \\ -1 & 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 & -1 \\ 2 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1+1 & 2+3 & 0-1 \\ 3+2 & 2+2 & -1+3 \\ -1+0 & 3-1 & 2+2 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 & 5 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 4 \end{bmatrix}$$

Now

$$U^t = (A + A^t)^t = \begin{bmatrix} 2 & 5 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 4 \end{bmatrix}^t$$

$$U^t = \begin{bmatrix} 2 & 5 & -1 \\ 5 & 4 & 2 \\ -1 & 2 & 4 \end{bmatrix} = U$$

$$(A + A^t)^t = A + A^t$$

Hence $A + A^t$ is symmetric

(ii) $A - A^t$ is skew symmetric

Solution:

$$\text{Let } V = A - A^t = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 2 & -1 \\ -1 & 3 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 & -1 \\ 2 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

$$V = A - A^t = \begin{bmatrix} 1-1 & 2-3 & 0-(-1) \\ 3-2 & 2-2 & -1-3 \\ -1-0 & 3-(-1) & 2-2 \end{bmatrix}$$

$$V = A - A^t = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -4 \\ -1 & 4 & 0 \end{bmatrix}$$

$$\text{Now } V^t = (A - A^t)^t = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -4 \\ -1 & 4 & 0 \end{bmatrix}^t = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 4 \\ 1 & -4 & 0 \end{bmatrix}$$

$$V^t = (A - A^t)^t = - \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -4 \\ -1 & 4 & 0 \end{bmatrix} = -(A - A^t)$$

Hence $A - A^t$ is skew symmetric

Q.3 If A is any matrix of order 3, show that

(i) $A + A^t$ is symmetric

Solution:

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix of order 3. then

$$A^t = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

$$\text{Now } A + A^t = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} + a_{11} & a_{12} + a_{21} & a_{13} + a_{31} \\ a_{21} + a_{12} & a_{22} + a_{22} & a_{23} + a_{32} \\ a_{31} + a_{32} & a_{32} + a_{23} & a_{33} + a_{33} \end{bmatrix}$$

$$A + A^t = \begin{bmatrix} 2a_{11} & a_{12} + a_{21} & a_{13} + a_{31} \\ a_{21} + a_{12} & 2a_{22} & a_{23} + a_{32} \\ a_{31} + a_{32} & a_{32} + a_{23} & 2a_{33} \end{bmatrix}$$

Now

$$\begin{aligned} (A + A^t)^t &= \begin{bmatrix} 2a_{11} & a_{12} + a_{21} & a_{13} + a_{31} \\ a_{21} + a_{12} & 2a_{22} & a_{23} + a_{32} \\ a_{31} + a_{32} & a_{32} + a_{23} & 2a_{33} \end{bmatrix}^t \\ &= \begin{bmatrix} 2a_{11} & a_{12} + a_{21} & a_{13} + a_{31} \\ a_{21} + a_{12} & 2a_{22} & a_{23} + a_{32} \\ a_{31} + a_{32} & a_{32} + a_{23} & 2a_{33} \end{bmatrix} = A + A^t \end{aligned}$$

$$(A + A^t)^t = A + A^t$$

Hence $A + A^t$ is symmetric.

(ii) $A - A^t$ is skew symmetric

Solution:

$$\text{Now } A - A^t = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} - \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} - a_{11} & a_{12} - a_{21} & a_{13} - a_{31} \\ a_{21} - a_{12} & a_{22} - a_{22} & a_{23} - a_{32} \\ a_{31} - a_{32} & a_{32} - a_{23} & a_{33} - a_{33} \end{bmatrix}$$

$$A - A^t = \begin{bmatrix} 0 & a_{12} - a_{21} & a_{13} - a_{31} \\ a_{21} - a_{12} & 0 & a_{23} - a_{32} \\ a_{31} - a_{32} & a_{32} - a_{23} & 0 \end{bmatrix}$$

Now

$$(A - A^t)^t = \begin{bmatrix} 0 & a_{12} - a_{21} & a_{13} - a_{31} \\ a_{21} - a_{12} & 0 & a_{23} - a_{32} \\ a_{31} - a_{32} & a_{32} - a_{23} & 0 \end{bmatrix}^t = \begin{bmatrix} 0 & a_{21} - a_{12} & a_{31} - a_{32} \\ a_{12} - a_{21} & 0 & a_{32} - a_{23} \\ a_{13} - a_{31} & a_{23} - a_{32} & 0 \end{bmatrix}$$

$$= - \begin{bmatrix} 0 & a_{21} - a_{12} & a_{31} - a_{32} \\ a_{12} - a_{21} & 0 & a_{32} - a_{23} \\ a_{13} - a_{31} & a_{23} - a_{32} & 0 \end{bmatrix} = A - A^t$$

Since $(A - A^t)^t = -(A - A^t)$ so $A - A^t$ is skew symmetric.

Q.4 If the matrices A and B are symmetric and $AB = BA$, show that AB is symmetric.

Solution: Let A and B are symmetric, then

$$A^t = A \text{ ——— (i)}$$

$$B^t = B \text{ ——— (ii)}$$

Now we have to show that AB is symmetric, for this

$$\begin{aligned} \text{Consider } (AB)^t &= B^t A^t & \therefore (AB)^t &= B^t A^t \\ &= BA & \therefore B^t &= B, A^t = A \\ &= AB & \therefore AB &= BA \end{aligned}$$

$$(AB)^t = AB$$

Hence AB is symmetric

Q.5 Show that AA^t and $A^t A$ are symmetric for any matrix of order 2×3

Solution: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ be a matrix of order 2×3 , then $A^t = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$

Now

$$AA^t = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}^2 + a_{12}^2 + a_{13}^2 & a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23} \\ a_{21}a_{11} + a_{22}a_{12} + a_{23}a_{13} & a_{21}^2 + a_{22}^2 + a_{23}^2 \end{bmatrix}$$

$$\begin{aligned}
 (AA^t)^t &= \begin{bmatrix} a_{11}^2 + a_{12}^2 + a_{13}^2 & a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23} \\ a_{21}a_{11} + a_{22}a_{12} + a_{23}a_{13} & a_{21}^2 + a_{22}^2 + a_{23}^2 \end{bmatrix}^t \\
 &= \begin{bmatrix} a_{11}^2 + a_{12}^2 + a_{13}^2 & a_{21}a_{11} + a_{22}a_{12} + a_{23}a_{13} \\ a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23} & a_{21}^2 + a_{22}^2 + a_{23}^2 \end{bmatrix} \\
 &= AA^t
 \end{aligned}$$

Hence AA^t is symmetric

Now

$$\begin{aligned}
 \text{Consider } A^t A &= \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \\
 A^t A &= \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{21} + a_{21}a_{22} & a_{11}a_{13} + a_{21}a_{23} \\ a_{21}a_{11} + a_{22}a_{21} & a_{12}^2 + a_{22}^2 & a_{12}a_{13} + a_{22}a_{23} \\ a_{13}a_{11} + a_{23}a_{21} & a_{13}a_{12} + a_{23}a_{22} & a_{13}^2 + a_{23}^2 \end{bmatrix} \\
 (A^t A)^t &= \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{21} + a_{21}a_{22} & a_{11}a_{13} + a_{21}a_{23} \\ a_{21}a_{11} + a_{22}a_{21} & a_{12}^2 + a_{22}^2 & a_{12}a_{13} + a_{22}a_{23} \\ a_{13}a_{11} + a_{23}a_{21} & a_{13}a_{12} + a_{23}a_{22} & a_{13}^2 + a_{23}^2 \end{bmatrix}^t \\
 &= \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{21}a_{11} + a_{22}a_{21} & a_{13}a_{11} + a_{23}a_{21} \\ a_{11}a_{21} + a_{21}a_{22} & a_{12}^2 + a_{22}^2 & a_{13}a_{12} + a_{23}a_{22} \\ a_{11}a_{13} + a_{21}a_{23} & a_{12}a_{13} + a_{22}a_{23} & a_{13}^2 + a_{23}^2 \end{bmatrix} \\
 (A^t A)^t &= A^t A
 \end{aligned}$$

Hence $A^t A$ is symmetric

Q.6 If $A = \begin{bmatrix} i & 1+i \\ 1 & -i \end{bmatrix}$, show that

(i) $A + (\bar{A})^t$ is hermitian

Solution: Consider $A = \begin{bmatrix} i & 1+i \\ 1 & -i \end{bmatrix}$

$$\text{then } \bar{A} = \begin{bmatrix} -i & 1-i \\ 1 & i \end{bmatrix}$$

$$(\bar{A})^t = \begin{bmatrix} -i & 1 \\ 1-i & i \end{bmatrix}$$

$$\text{Let } U = A + (\bar{A})^t = \begin{bmatrix} i & 1+i \\ 1 & -i \end{bmatrix} + \begin{bmatrix} -i & 1 \\ 1-i & i \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & 2+i \\ 2-i & 0 \end{bmatrix}$$

$$\bar{U} = \begin{bmatrix} 0 & 2-i \\ 2+i & 0 \end{bmatrix}$$

$$(\bar{U})^t = \begin{bmatrix} 0 & 2-i \\ 2+i & 0 \end{bmatrix}$$

$$(\bar{U})^t = \begin{bmatrix} 0 & 2+i \\ 2-i & 0 \end{bmatrix} = U$$

$$(\bar{U})^t = U$$

$$\left(\overline{A + (\bar{A})^t} \right)^t = A + (\bar{A})^t \quad \because U = A + (\bar{A})^t$$

So $A + (\bar{A})^t$ is a hermitian matrix.

(ii) $A - (\bar{A})^t$ is skew hermitian

Solution: Let $V = A - (\bar{A})^t$ so

$$V = A - (\bar{A})^t = \begin{bmatrix} i & 1+i \\ 1 & -i \end{bmatrix} - \begin{bmatrix} -i & 1 \\ 1-i & i \end{bmatrix}$$

$$V = \begin{bmatrix} 2i & i \\ i & -2i \end{bmatrix}$$

$$\text{So } \bar{V} = \begin{bmatrix} -2i & -i \\ -i & 2i \end{bmatrix}$$

$$(\bar{V})^t = \begin{bmatrix} -2i & -i \\ -i & 2i \end{bmatrix}$$

$$= \begin{bmatrix} -2i & -i \\ -i & 2i \end{bmatrix}$$

$$= - \begin{bmatrix} 2i & i \\ i & -2i \end{bmatrix} = -V$$

$$(\bar{V})^t = -V$$

$$\left(\overline{A - (\bar{A})^t} \right)^t = - \left(A - (\bar{A})^t \right) \quad \because V = A - (\bar{A})^t$$

Hence $A - (\bar{A})^t$ is skew hermitian

Q.7 If A is symmetric or skew symmetric show that A^2 is symmetric.

Proof: Case I: when A is symmetric

Since A is symmetric so $A^t = A$

Now $A^2 = A.A$

$$(A^2)^t = (A.A)^t$$

$$= A^t.A^t$$

$$= A.A \quad \because A^t = A$$

$$= A^2$$

$$(A^2)^t = A^2$$

Hence if A is symmetric, then A^2 is symmetric.

Case II: when A is skew symmetric

Since A is skew symmetric so $A^t = -A$

$$(A^2)^t = (A.A)^t$$

$$= A^t.A^t \quad \because A^t = -A$$

$$= (-A)(-A)$$

$$= A^2$$

$$(A^2)^t = A^2$$

Hence if A is skew symmetric, then A^2 is symmetric

Q.8 If $A = \begin{bmatrix} 1 \\ 1+i \\ i \end{bmatrix}$ find $A(\bar{A})^t$

Solution: Consider $A = \begin{bmatrix} 1 \\ 1+i \\ i \end{bmatrix}$

$$\bar{A} = \begin{bmatrix} 1 \\ 1-i \\ -i \end{bmatrix}$$

$$(\bar{A})^t = [1 \quad 1-i \quad -i]$$

$$A(\bar{A})^t = \begin{bmatrix} 1 \\ 1+i \\ i \end{bmatrix} [1 \quad 1-i \quad -i]$$

$$\begin{aligned}
 A(\bar{A})^t &= \begin{bmatrix} 1 & 1-i & -i \\ 1+i & 1-i^2 & -i(1+i) \\ i & i(1-i) & -i^2 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1-i & -i \\ 1+i & 1+1 & -i-i^2 \\ i & i-i^2 & -i^2 \end{bmatrix} \qquad \because i^2 = -1
 \end{aligned}$$

$$A(\bar{A})^t = \begin{bmatrix} 1 & 1-i & -i \\ 1+i & 2 & -i+1 \\ i & 1+i & 1 \end{bmatrix}$$

Q.9 Find the inverses of the following matrices. Also find their inverses by row and column operations.

(i) $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$

Solution: Let $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$

Inverse of any square matrix A is given by

$$A^{-1} = \frac{AdjA}{|A|}$$

Therefore $|A| = \begin{vmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{vmatrix}$

$$\begin{aligned}
 &= 1 \begin{vmatrix} -2 & 0 \\ -2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 0 & 0 \\ -2 & 2 \end{vmatrix} - 3 \begin{vmatrix} 0 & -2 \\ -2 & -2 \end{vmatrix} \\
 &= 1(-4-0) - 2(0-0) - 3(0-4)
 \end{aligned}$$

$$|A| = -4 + 12 = 8 \neq 0$$

Since $|A| \neq 0$ so A^{-1} exists

Now

$$AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\text{Adj}A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

From $A_{ij} = (-1)^{i+j} M_{ij}$ we get

$$A_{11} = (-1)^{1+1} M_{11} = M_{11} = \begin{vmatrix} -2 & 0 \\ -2 & 2 \end{vmatrix} = -4$$

$$A_{12} = (-1)^{1+2} M_{12} = -M_{12} = -\begin{vmatrix} 0 & 0 \\ -2 & 2 \end{vmatrix} = 0$$

$$A_{13} = (-1)^{1+3} M_{13} = M_{13} = \begin{vmatrix} 0 & -2 \\ -2 & -2 \end{vmatrix} = -4$$

$$A_{21} = (-1)^{2+1} M_{21} = -M_{21} = -\begin{vmatrix} 2 & -3 \\ -2 & 2 \end{vmatrix} = -(4-6) = 2$$

$$A_{22} = (-1)^{2+2} M_{22} = M_{22} = \begin{vmatrix} 1 & -3 \\ -2 & 2 \end{vmatrix} = 2-6 = -4$$

$$A_{23} = (-1)^{2+3} M_{23} = -M_{23} = -\begin{vmatrix} 1 & 2 \\ -2 & -2 \end{vmatrix} = -(-2+4) = -2$$

$$A_{31} = (-1)^{3+1} M_{31} = M_{31} = \begin{vmatrix} 2 & -3 \\ -2 & 0 \end{vmatrix} = 0-6 = -6$$

$$A_{32} = (-1)^{3+2} M_{32} = -M_{32} = -\begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{33} = (-1)^{3+3} M_{33} = M_{33} = \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} = -2-0 = -2$$

$$\begin{aligned} \text{So } \text{Adj}A &= \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \\ &= \begin{bmatrix} -4 & 2 & -6 \\ 0 & -4 & 0 \\ -4 & -2 & -2 \end{bmatrix} \end{aligned}$$

$$\text{Now } A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{8} \begin{bmatrix} -4 & 2 & -6 \\ 0 & -4 & 0 \\ -4 & -2 & -2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

Inverse of A by Row operations

Appending I_3 on the right of matrix A

$$\begin{bmatrix} 1 & 2 & -3 & : & 1 & 0 & 0 \\ 0 & -2 & 0 & : & 0 & 1 & 0 \\ -2 & -2 & 2 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_1 \rightarrow R'_3$$

$$\begin{bmatrix} 1 & 2 & -3 & : & 1 & 0 & 0 \\ 0 & -2 & 0 & : & 0 & 1 & 0 \\ 0 & 2 & -4 & : & 2 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{2}R_2 \rightarrow R'_2$$

$$\begin{bmatrix} 1 & 2 & -3 & : & 1 & 0 & 0 \\ 0 & 1 & 0 & : & 0 & -\frac{1}{2} & 0 \\ 0 & 2 & -4 & : & 2 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2 \rightarrow R'_1$$

$$\begin{bmatrix} 1 & 0 & -3 & : & 1 & 1 & 0 \\ 0 & 1 & 0 & : & 0 & -\frac{1}{2} & 0 \\ 0 & 2 & -4 & : & 2 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 3R_2 \rightarrow R'_3$$

$$\begin{bmatrix} 1 & 0 & -3 & : & 1 & 1 & 0 \\ 0 & 1 & 0 & : & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & : & -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

$$R_1 \rightarrow R_1 + 3R_3 \rightarrow R'_1$$

$$\begin{bmatrix} 1 & 0 & 0 & : & -\frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & 1 & 0 & : & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & : & -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

Thus the inverse of A is

$$\begin{bmatrix} -\frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

Inverse of A by column operation,

Appending I_3 below the matrix A

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ -2 & 2 & -4 \\ \vdots & \vdots & \vdots \\ 1 & -2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{By } C_2 - 2C_1 \rightarrow C'_2, C_3 + 3C_1 \rightarrow C'_3$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & -4 \\ \vdots & \vdots & \vdots \\ 1 & 1 & 3 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{By } -\frac{1}{2}C_2 \rightarrow C'_2$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 1 & -\frac{3}{4} \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{4} \end{bmatrix}$$

$$\text{By } \left(-\frac{1}{4}\right)C_3 \rightarrow C'_3$$

$$C \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ -\frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

$$\text{By } C_1 + 2C_3 \rightarrow C'_1, C_2 + C_3 \rightarrow C'_2$$

Hence inverse of A is

$$\begin{bmatrix} -\frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

$$(ii) \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\text{Solution: Let } A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 1 & 0 & 2 \end{bmatrix}$$

Inverse of any square matrix A is given by

$$A^{-1} = \frac{\text{Adj}A}{|A|}$$

$$\begin{aligned} \text{Therefore } |A| &= \begin{vmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 1 & 0 & 2 \end{vmatrix} \\ &= 1 \begin{vmatrix} -1 & 3 \\ 0 & 2 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} \\ &= 1(-2-0) - 2(0-3) - 1(0+1) \\ &= -2+6-1 \\ |A| &= 3 \neq 0 \end{aligned}$$

Since $|A| \neq 0$ so A^{-1} exists

Now

$$AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Now by using $A_{ij} = (-1)^{i+j} M_{ij}$

$$A_{11} = (-1)^{1+1} M_{11} = M_{11} = \begin{vmatrix} -1 & 3 \\ 0 & 2 \end{vmatrix} = -2$$

$$A_{12} = (-1)^{1+2} M_{12} = -M_{12} = -\begin{vmatrix} 0 & 3 \\ 1 & 2 \end{vmatrix} = -(0-3) = 3$$

$$A_{13} = (-1)^{1+3} M_{13} = M_{13} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = (0+1) = 1$$

$$A_{21} = (-1)^{2+1} M_{21} = -M_{21} = -\begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix} = -(4-0) = -4$$

$$A_{22} = (-1)^{2+2} M_{22} = M_{22} = \begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} = (2+1) = 3$$

$$A_{23} = (-1)^{2+3} M_{23} = -\begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -(0-2) = 2$$

$$A_{31} = (-1)^{3+1} M_{31} = M_{31} = \begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix} = (6-1) = 5$$

$$A_{32} = (-1)^{3+2} M_{32} = -M_{32} = -\begin{vmatrix} 1 & -1 \\ 0 & 3 \end{vmatrix} = -(3-0) = -3$$

$$A_{33} = (-1)^{3+3} M_{33} = M_{33} = \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} = (-1-0) = -1$$

$$\begin{aligned} \text{Now } AdjA &= \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \\ &= \begin{bmatrix} -2 & -4 & 5 \\ 3 & 3 & -3 \\ 1 & 2 & -1 \end{bmatrix} \end{aligned}$$

$$A^{-1} = \frac{AdjA}{|A|} = \frac{1}{3} \begin{bmatrix} -2 & -4 & 5 \\ 3 & 3 & -3 \\ 1 & 2 & -1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -\frac{2}{3} & -\frac{4}{3} & \frac{5}{3} \\ 1 & 1 & -1 \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

Now inverse of A by row operation Appending I_3 to the right of matrix A

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_1 \rightarrow R'_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 1 & 0 \\ 0 & -2 & 3 & -1 & 0 & 1 \end{array} \right]$$

$$R_2 \rightarrow (-1)R_2 \rightarrow R'_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & -1 & 0 \\ 0 & -2 & 3 & -1 & 0 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_2 \rightarrow R'_1, R_3 \rightarrow R_3 + 2R_2 \rightarrow R'_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 & -1 & 0 \\ 0 & 0 & -3 & -1 & -2 & 1 \end{array} \right]$$

$$R_3 \rightarrow -\frac{1}{3}R_3 \rightarrow R'_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right]$$

$$R_1 \rightarrow R_1 - 5R_3 \rightarrow R'_1, R_2 \rightarrow R_2 + 3R_3 \rightarrow R'_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{2}{3} & -\frac{4}{3} & \frac{5}{3} \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right]$$

$$R_2 \rightarrow R_2 + 3R_3 \rightarrow R'_2, R_1 \rightarrow R_1 - 5R_3 \rightarrow R'_1$$

Thus inverse of A is

$$A^{-1} = \begin{bmatrix} -\frac{2}{3} & -\frac{4}{3} & \frac{5}{3} \\ 1 & 1 & -1 \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

Inverse by column operation:

Appending I_3 below the given matrix

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 1 & 0 & 2 \\ : & : & : \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 3 \\ 1 & -2 & 3 \\ : & : & : \\ 1 & -2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

By

$$C_2 - 2C_1 \rightarrow C'_2, C_3 + C_1 \rightarrow C'_3$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 1 & 2 & 3 \\ : & : & : \\ 1 & 2 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

By $(-1)C_2 \rightarrow C'_2$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & -3 \\ : & : & : \\ 1 & 2 & -5 \\ 0 & -1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

By $C_3 - 3C_2 \rightarrow C'_3$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 2 & \frac{5}{3} \\ 0 & -1 & -1 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}$$

$$\text{By } \left(-\frac{1}{3}\right)C_3 \rightarrow C'_3$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ -\frac{2}{3} & -\frac{4}{3} & \frac{5}{3} \\ 1 & 1 & -1 \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

By

$$C_1 - C_3 \rightarrow C'_1, C_2 - 2C_3 \rightarrow C'_2$$

$$\text{Thus inverse of } A \text{ is } \begin{bmatrix} -\frac{2}{3} & -\frac{4}{3} & \frac{5}{3} \\ 1 & 1 & -1 \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

$$\text{(iii)} \quad \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\text{Solution: Let } A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Inverse of any square matrix A is given by

$$A^{-1} = \frac{\text{Adj}A}{|A|}$$

$$\text{Therefore } |A| = \begin{vmatrix} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$\begin{aligned}
&= 1 \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} \\
&= 1(1-0) + 3(2-0) + 2(-2-0) \\
|A| &= 1 + 6 - 4 = 3 \neq 0
\end{aligned}$$

Since $|A| \neq 0$ so A^{-1} exists

Now

$$AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Now by using $A = (-1)^{i+j} M_{ij}$ we get

$$A_{11} = (-1)^{1+1} M_{11} = M_{11} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = (1-0) = 1$$

$$A_{12} = (-1)^{1+2} M_{12} = -M_{12} = -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -(2-0) = -2$$

$$A_{13} = (-1)^{1+3} M_{13} = M_{13} = \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} = -2-0 = -2$$

$$A_{21} = (-1)^{2+1} M_{21} = -M_{21} = -\begin{vmatrix} -3 & 2 \\ -1 & 1 \end{vmatrix} = -(-3+2) = 1$$

$$A_{22} = (-1)^{2+2} M_{22} = M_{22} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = (1-0) = 1$$

$$A_{23} = (-1)^{2+3} M_{23} = -M_{23} = -\begin{vmatrix} 1 & -3 \\ 0 & -1 \end{vmatrix} = -(-1-0) = 1$$

$$A_{31} = (-1)^{3+1} M_{31} = M_{31} = \begin{vmatrix} -3 & 2 \\ 1 & 0 \end{vmatrix} = (0-2) = -2$$

$$A_{32} = (-1)^{3+2} M_{32} = -M_{32} = -\begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} = -(0-4) = 4$$

$$A_{33} = (-1)^{3+3} M_{33} = M_{33} = \begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix} = (1+6) = 7$$

$$\begin{aligned}\text{Thus } AdjA &= \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & -2 \\ -2 & 1 & 4 \\ -2 & 1 & 7 \end{bmatrix}\end{aligned}$$

Now

$$\begin{aligned}A^{-1} &= \frac{AdjA}{|A|} = \frac{1}{3} \begin{bmatrix} 1 & 1 & -2 \\ -2 & 1 & 4 \\ -2 & 1 & 7 \end{bmatrix} \\ A^{-1} &= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{4}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{bmatrix}\end{aligned}$$

Inverse of the matrix by row operation

Appending I_3 to the right of the matrix

$$\begin{aligned}&\begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 2 & 1 & 0 & : & 0 & 1 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \\ R_1 &\rightarrow \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 7 & -4 & : & -2 & 1 & 0 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \\ R_2 &\rightarrow \begin{bmatrix} 1 & -3 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & 2 & : & -2 & 1 & 6 \\ 0 & -1 & 1 & : & 0 & 0 & 1 \end{bmatrix} \\ R_1 &\rightarrow \begin{bmatrix} 1 & 0 & 8 & : & -5 & 3 & 18 \\ 0 & 1 & 2 & : & -2 & 1 & 6 \\ 0 & 0 & 3 & : & -2 & 1 & 7 \end{bmatrix}\end{aligned}$$

By $R_2 - 2R_1 \rightarrow R'_2$

By $R_2 + 6R_3 \rightarrow R'_2$

By

$$R_1 + 3R'_2 \rightarrow R'_1, R_3 + R_2 \rightarrow R'_3$$

$$R_{\square} \left[\begin{array}{ccc|ccc} 1 & 0 & 8 & : & -5 & 3 & 18 \\ 0 & 1 & 2 & : & -2 & 1 & 6 \\ 0 & 0 & 1 & : & -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{array} \right]$$

$$\text{By } \left(\frac{1}{3} R_3 \right) \rightarrow R'_3$$

$$R_{\square} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & : & \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & 0 & : & -\frac{2}{3} & \frac{1}{3} & \frac{4}{3} \\ 0 & 0 & 1 & : & -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{array} \right]$$

$$\text{By } R_2 - 2R_3 \rightarrow R'_2, R_1 - 8R_3 \rightarrow R'_1$$

This the inverse of given matrix is

$$\left[\begin{array}{ccc} \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{4}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{array} \right]$$

Inverse by column operation:

Appending I_3 below the given matrix

$$\left[\begin{array}{ccc} 1 & -3 & 2 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \\ : & : & : \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

$$C_{\square} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 7 & -4 \\ 0 & -1 & 1 \\ : & : & : \\ 1 & 3 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

$$\text{By } C_2 + 3C_1 \rightarrow C'_2, C_3 - 2C_1 \rightarrow C'_3$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & -4 \\ 0 & 1 & 1 \\ : & : & : \\ 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

$$\text{By } C_2 + 2C_3 \rightarrow C'_2$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & -4 \\ 0 & -1 & 1 \\ : & : & : \\ 1 & 1 & -2 \\ 0 & -1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\text{By } (-1)C_2 \rightarrow C'_2$$

$$C_{\square} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & -3 \\ : & : & : \\ 1 & 1 & 2 \\ 2 & -1 & -4 \\ 4 & -2 & -7 \end{bmatrix}$$

$$\text{By}$$

$$C_1 - 2C_2 \rightarrow C'_1, C_3 + 4C_2 \rightarrow C'_3$$

$$C_{\square} \begin{bmatrix} 1 & 0 & -0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \\ : & : & : \\ -1 & 1 & -\frac{2}{3} \\ 2 & -1 & \frac{4}{3} \\ 4 & -2 & \frac{7}{3} \end{bmatrix}$$

$$\text{By } \left(-\frac{1}{3}\right)C_3 \rightarrow C'_3$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{4}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{bmatrix}$$

$$\text{By } C_2 + C_3 \rightarrow C'_2, C_1 - 2C_3 \rightarrow C'_1$$

$$\text{Thus inverse of } A \text{ is } \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{4}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{7}{3} \end{bmatrix}$$

Q.10 Find the rank of the following matrices

$$(i) \begin{bmatrix} 1 & -1 & 2 & 1 \\ 2 & -6 & 5 & 1 \\ 3 & 5 & 4 & -3 \end{bmatrix}$$

Solution: Consider $\begin{bmatrix} 1 & -1 & 2 & 1 \\ 2 & -6 & 5 & 1 \\ 3 & 5 & 4 & -3 \end{bmatrix}$ we reduced it to reduced echelon form by using elementary row operation.

$$\sim R \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & -4 & 1 & -1 \\ 0 & 8 & -2 & -6 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 3R_1 \rightarrow R'_3$$

$$\sim R \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 8 & -2 & -6 \end{bmatrix}$$

$$\text{By } \left(\frac{-1}{4}\right)R_2 \rightarrow R'_2$$

$$\sim R \begin{bmatrix} 1 & 0 & \frac{7}{4} & \frac{5}{4} \\ 0 & 1 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

$$\text{By } R_1 + R_2 \rightarrow R'_1, R_3 - 8R_2 \rightarrow R'_3$$

$$R \sim \begin{bmatrix} 1 & 0 & \frac{7}{4} & \frac{5}{4} \\ 0 & 1 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{By } \left(-\frac{1}{8}\right)R_3 \rightarrow R'_3$$

Number of non-zero rows are 3 hence rank of the given matrix is 3.

$$(ii) \begin{bmatrix} 1 & -4 & -7 \\ 2 & -5 & 1 \\ 1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix}$$

Solution: Consider $\begin{bmatrix} 1 & -4 & -7 \\ 2 & -5 & 1 \\ 1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix}$ we reduced it to reduced echelon form by using elementary row operation.

$$R \sim \begin{bmatrix} 1 & -4 & -7 \\ 0 & 3 & 15 \\ 0 & 2 & 10 \\ 0 & 5 & 25 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R_2, R_3 - R_1 \rightarrow R'_3, R_4 - 3R_1 \rightarrow R'_4$$

$$R \sim \begin{bmatrix} 1 & -4 & -7 \\ 0 & 1 & 5 \\ 0 & 2 & 10 \\ 0 & 5 & 25 \end{bmatrix}$$

$$\text{By } R_2 - R_3 \rightarrow R'_2$$

$$R \sim \begin{bmatrix} 1 & 0 & 13 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_1 + 4R_2 \rightarrow R'_1, R_3 - 2R_2 \rightarrow R'_3, R_4 - 5R_2 \rightarrow R'_4$$

As the number of non-zero rows in reduce echelon form are 2, hence rank of the matrix is 2.

$$(iii) \begin{bmatrix} 3 & -1 & 3 & 0 & -1 \\ 1 & 2 & -1 & -3 & -2 \\ 2 & 3 & 4 & 2 & 5 \\ 2 & 5 & -2 & -3 & 3 \end{bmatrix}$$

Solution: Consider $\begin{bmatrix} 3 & -1 & 3 & 0 & -1 \\ 1 & 2 & -1 & -3 & -2 \\ 2 & 3 & 4 & 2 & 5 \\ 2 & 5 & -2 & -3 & 3 \end{bmatrix}$ we reduced it to reduced echelon form by using elementary row operation.

$$\begin{array}{l}
 R \sim \begin{bmatrix} 1 & 2 & -1 & -3 & -2 \\ 3 & -1 & 3 & 0 & -1 \\ 2 & 3 & 4 & 2 & 5 \\ 2 & 5 & -2 & -3 & 3 \end{bmatrix} \quad \text{By } R_1 \leftrightarrow R_2 \\
 R \sim \begin{bmatrix} 1 & 2 & -1 & -3 & -2 \\ 0 & -7 & 6 & 9 & 5 \\ 0 & -1 & 6 & 8 & 9 \\ 0 & 1 & 0 & 3 & 7 \end{bmatrix} \quad \text{By } R_2 - 3R_1 \rightarrow R'_2, R_3 - 2R_1 \rightarrow R'_3, R_4 - 2R_1 \rightarrow R'_4 \\
 R \sim \begin{bmatrix} 1 & 2 & -1 & -3 & -2 \\ 0 & 1 & 0 & 3 & 7 \\ 0 & -1 & 6 & 8 & 9 \\ 0 & -7 & 6 & 9 & 5 \end{bmatrix} \quad \text{By } R_2 \leftrightarrow R_4 \\
 R \sim \begin{bmatrix} 1 & 0 & -1 & -9 & -16 \\ 0 & 1 & 0 & 3 & 7 \\ 0 & 0 & 6 & 11 & 16 \\ 0 & 0 & 6 & 30 & 54 \end{bmatrix} \quad \text{By } R_1 - 2R_2 \rightarrow R'_1, R_3 + R_2 \rightarrow R'_3, R_4 + 7R_2 \rightarrow R'_4 \\
 R \sim \begin{bmatrix} 1 & 0 & -1 & -9 & -16 \\ 0 & 1 & 0 & 3 & 7 \\ 0 & 0 & 1 & \frac{11}{6} & \frac{8}{3} \\ 0 & 0 & 6 & 30 & 54 \end{bmatrix} \quad \text{By } \frac{1}{6}R_3 \rightarrow R'_3 \\
 R \sim \begin{bmatrix} 1 & 0 & -1 & -9 & -16 \\ 0 & 1 & 0 & 3 & 7 \\ 0 & 0 & 1 & \frac{11}{6} & \frac{8}{3} \\ 0 & 0 & 0 & 19 & 38 \end{bmatrix} \quad \text{By } R_4 - 6R_3 \rightarrow R'_4 \\
 R \sim \begin{bmatrix} 1 & 0 & -1 & -9 & -16 \\ 0 & 1 & 0 & 3 & 7 \\ 0 & 0 & 1 & \frac{11}{6} & \frac{8}{3} \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad \text{By } \frac{1}{19}R_4 \rightarrow R'_4
 \end{array}$$

As the number of non-zero rows are 4 in reduce echelon form. So, the rank of the matrix is 4.

Exercise No 3.5

Q.1 Solve the following systems of linear equations by Cramer's Rule.

$$(i) \quad \left. \begin{array}{l} 2x + 2y + z = 3 \\ 3x - 2y - 2z = 1 \\ 5x + y - 3z = 2 \end{array} \right\}$$

Solution: Here $|A| = \begin{vmatrix} 2 & 2 & 1 \\ 3 & -2 & -2 \\ 5 & 1 & -3 \end{vmatrix}$

$$= 2(6+2) - 2(-9+10) + 1(3+10)$$

$$= 16 - 2 + 13 = 27 \neq 0$$

Since $|A| \neq 0$ so solution of system is possible. Now by Cramer's Rule

$$x = \frac{\begin{vmatrix} 3 & 2 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & -3 \end{vmatrix}}{|A|} = \frac{3(6+2) - 2(-3+4) + 1(1+4)}{27}$$

$$= \frac{24 - 2 + 5}{27}$$

$$= \frac{27}{27}$$

$$\Rightarrow x = 1$$

$$y = \frac{\begin{vmatrix} 2 & 3 & 1 \\ 3 & 1 & -2 \\ 5 & 2 & -3 \end{vmatrix}}{|A|} = \frac{2(-3+4) - 3(-9+10) + 1(6-5)}{27}$$

$$= \frac{2 - 3 + 1}{27} = \frac{0}{27}$$

$$\Rightarrow y = 0$$

$$z = \frac{\begin{vmatrix} 2 & 2 & 3 \\ 3 & -2 & 1 \\ 5 & 1 & 2 \end{vmatrix}}{|A|} = \frac{2(-4-1) - 2(6-5) + 3(3+10)}{27}$$

$$= \frac{-10 - 2 + 39}{27}$$

$$= \frac{27}{27}$$

$$\Rightarrow z = 1$$

$$\text{Hence } x = 1, y = 0, z = 1$$

$$(ii) \quad \left. \begin{array}{l} 2x_1 - x_2 + x_3 = 5 \\ 4x_1 + 2x_2 + 3x_3 = 8 \\ 3x_1 - 4x_2 - x_3 = 3 \end{array} \right\}$$

$$\text{Solution: Here } |A| = \begin{vmatrix} 2 & -1 & 1 \\ 4 & 2 & 3 \\ 3 & -4 & -1 \end{vmatrix} = 2(-2+12) + 1(-4-9) + 1(-16-6)$$

$$= 2(10) + 1(-13) + 1(-22)$$

$$= 20 - 13 - 22$$

$$\Rightarrow |A| = -15 \neq 0$$

So, solution is possible.

Now by Cramer's rule

$$x_1 = \frac{\begin{vmatrix} 5 & -1 & 1 \\ 8 & 2 & 3 \\ 3 & -4 & -1 \end{vmatrix}}{|A|} = \frac{5(-2+12) + 1(-8-9) + 1(-32-6)}{-15}$$

$$= \frac{50 - 17 - 38}{-15}$$

$$x_1 = \frac{1}{3}$$

$$x_2 = \frac{\begin{vmatrix} 2 & 5 & 1 \\ 4 & 8 & 3 \\ 3 & 3 & -1 \end{vmatrix}}{|A|} = \frac{2(-8-9) - 5(-4-9) + 1(12-24)}{-15}$$

$$= \frac{-34 + 65 - 12}{-15}$$

$$x_2 = \frac{-19}{15}$$

$$x_3 = \frac{\begin{vmatrix} 2 & -1 & 5 \\ 4 & 2 & 8 \\ 3 & -4 & 3 \end{vmatrix}}{|A|} = \frac{2(6+32) + 1(12-24) + 5(-16-6)}{-15}$$

$$= \frac{76-12-110}{-15}$$

$$x_3 = \frac{46}{15}$$

$$\text{Hence } x_1 = \frac{1}{3}, x_2 = \frac{-19}{15}, x_3 = \frac{46}{15}$$

$$\text{(iii) } \left. \begin{array}{l} 2x_1 - x_2 + x_3 = 8 \\ x_1 + 2x_2 + 2x_3 = 6 \\ x_1 - 2x_2 - x_3 = 1 \end{array} \right\}$$

$$\text{Solution: Here } |A| = \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & 2 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= 2(-2+4) + 1(-1-2) + 1(-2-2)$$

$$= 4 - 3 - 4$$

$$|A| = -3 \neq 0$$

Since $|A| \neq 0$ so solution is possible now by Cramer's Rule

$$x_1 = \frac{\begin{vmatrix} 8 & -1 & 1 \\ 6 & 2 & 2 \\ 1 & -2 & -1 \end{vmatrix}}{|A|} = \frac{8(-2+4) + 1(-6-2) + 1(-12-2)}{-3}$$

$$= \frac{16-8-14}{-3}$$

$$= \frac{-6}{-3} = 2$$

$$x_1 = 2$$

$$x_2 = \frac{\begin{vmatrix} 2 & 8 & 1 \\ 1 & 6 & 2 \\ 1 & 1 & -1 \end{vmatrix}}{|A|} = \frac{2(-6-2) - 8(-1-2) + 1(1-6)}{-3}$$

$$= \frac{-16+24-5}{-3} = \frac{3}{-3} = -1$$

$$x_2 = -1$$

$$\begin{aligned}
 x_3 &= \frac{\begin{vmatrix} 2 & -1 & 8 \\ 1 & 2 & 6 \\ 1 & -2 & 1 \end{vmatrix}}{|A|} = \frac{2(2+12)+1(1-6)+8(-2-2)}{|A|} \\
 &= \frac{28-5-32}{-3} \\
 &= \frac{-9}{-3} = 3
 \end{aligned}$$

$$x_3 = 3$$

Hence $x_1 = 2, x_2 = -1, x_3 = 3$

Q.2 Use matrices to solve the following systems.

$$\text{(i) } \begin{cases} x - 2y + z = -1 \\ 3x + y - 2z = 4 \\ y - z = 1 \end{cases}$$

Solution:

The matrix form of the given system is

$$\begin{bmatrix} 1 & -2 & 1 \\ 3 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B$$

(i)

$$\text{Where } A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$$

$$\begin{aligned}
 |A| &= \begin{vmatrix} 1 & -2 & 1 \\ 3 & 1 & -2 \\ 0 & 1 & -1 \end{vmatrix} \\
 &= 1(-1+2)+2(-3-0)+1(3-0) \\
 |A| &= 1(1)-6+3 = -2 \neq 0
 \end{aligned}$$

Since $|A| \neq 0$ so inverse of A is possible and given by $A^{-1} = \frac{AdjA}{|A|}$

$$\text{Now } AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\begin{aligned}
&= \begin{bmatrix} \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 3 & -2 \\ 0 & -1 \end{vmatrix} & \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} \\ -\begin{vmatrix} -2 & 1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} \\ \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} \end{bmatrix}^t \\
&= \begin{bmatrix} 1 & 3 & 3 \\ -1 & -1 & -1 \\ 3 & 5 & 7 \end{bmatrix}^t \\
&= \begin{bmatrix} 1 & -1 & 3 \\ 3 & -1 & 5 \\ 3 & -1 & 7 \end{bmatrix} \\
A^{-1} &= \frac{AdjA}{|A|} = \frac{-1}{2} \begin{bmatrix} 1 & -1 & 3 \\ 3 & -1 & 5 \\ 3 & -1 & 7 \end{bmatrix}
\end{aligned}$$

Put the values of A^{-1} and B in (i) we get

$$\begin{aligned}
X = A^{-1}B &= -\frac{1}{2} \begin{bmatrix} 1 & -1 & 3 \\ 3 & -1 & 5 \\ 3 & -1 & 7 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} \\
&= \frac{-1}{2} \begin{bmatrix} -1-4+3 \\ -3-4+5 \\ -3-4+7 \end{bmatrix} \\
&= \frac{-1}{2} \begin{bmatrix} -2 \\ -2 \\ 0 \end{bmatrix} \\
\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}
\end{aligned}$$

Hence by definition of equal matrices $x=1, y=1, z=0$

$$\text{(ii) } \left. \begin{aligned} 2x_1 + x_2 + 3x_3 &= 3 \\ x_1 + x_2 - 2x_3 &= 0 \\ -3x_1 - x_2 + 2x_3 &= -4 \end{aligned} \right\}$$

Solution: The matrix form of given system of linear equation is

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & -2 \\ -3 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B \quad (i)$$

Where

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & -2 \\ -3 & -1 & 2 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix}$$

Now

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 1 & 3 \\ 1 & 1 & -2 \\ -3 & -1 & 2 \end{vmatrix} = 2(2-2) - 1(2-6) + 3(-1+3) \\ &= 0 + 4 + 6 \\ |A| &= 10 \neq 0 \end{aligned}$$

Since $|A| \neq 0$ so inverse of A is possible and given by $A^{-1} = \frac{AdjA}{|A|}$

$$\text{Now } AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} \begin{vmatrix} 1 & -2 \\ -1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & -2 \\ -3 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -3 & -1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ -3 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ -3 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & 3 \\ 1 & -2 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} \end{bmatrix}^t = \begin{bmatrix} 0 & 4 & 2 \\ -5 & 13 & -1 \\ -5 & 7 & 1 \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} 0 & -5 & -5 \\ 4 & 13 & 7 \\ 2 & -1 & 1 \end{bmatrix}$$

$$\text{Now } A^{-1} = \frac{AdjA}{|A|} = \frac{1}{10} \begin{bmatrix} 0 & -5 & -5 \\ 4 & 13 & 7 \\ 2 & -1 & 1 \end{bmatrix}$$

Put the values of A^{-1} and B in (i) we get

$$\begin{aligned} X &= A^{-1}B \\ &= \frac{1}{10} \begin{bmatrix} 0 & -5 & -5 \\ 4 & 13 & 7 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{10} \begin{bmatrix} 0+0+20 \\ 12+0-28 \\ 6+0-4 \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \frac{1}{10} \begin{bmatrix} 20 \\ -16 \\ 2 \end{bmatrix} \\
 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 2 \\ -\frac{8}{5} \\ \frac{1}{5} \end{bmatrix}
 \end{aligned}$$

Hence by definition of equal matrices $x_1 = 2, x_2 = -\frac{8}{5}$ and $x_3 = \frac{1}{5}$

Note: Answer of the question does not match with the answer of the text book.

$$\begin{aligned}
 &\left. \begin{aligned} \mathbf{x} + \mathbf{y} &= \mathbf{2} \\ \mathbf{2x} - \mathbf{z} &= \mathbf{1} \\ \mathbf{2y} - \mathbf{3z} &= \mathbf{-1} \end{aligned} \right\} \\
 \text{(iii)} &
 \end{aligned}$$

The matrix form of the given system of linear equation is

$$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B$$

(i)

Where

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 2 & -3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$\begin{aligned}
 \text{So } |A| &= \begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 2 & -3 \end{vmatrix} = 1(0+2) - 1(-6-0) + 0(4-0) \\
 &= 2+6=8 \neq 0 \\
 |A| &= 8 \neq 0
 \end{aligned}$$

Since $|A| \neq 0$ so inverse of A is possible and given by $A^{-1} = \frac{AdjA}{|A|}$

$$\text{Now } AdjA = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

$$\text{Now } AdjA = \begin{bmatrix} \begin{vmatrix} 0 & -1 \\ 2 & -3 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 0 & -3 \end{vmatrix} & \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} \\ -\begin{vmatrix} 1 & 0 \\ 2 & -3 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 0 & -3 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} \\ \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} 2 & 6 & 4 \\ 3 & -3 & -2 \\ -1 & 1 & -2 \end{bmatrix}^t$$

$$AdjA = \begin{bmatrix} 2 & 3 & -1 \\ 6 & -3 & 1 \\ 4 & -2 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{AdjA}{|A|} = \frac{1}{8} \begin{bmatrix} 2 & 3 & -1 \\ 6 & -3 & 1 \\ 4 & -2 & -2 \end{bmatrix}$$

Put the values of A^{-1} and B in (i) we get

$$\begin{aligned} X &= A^{-1}B \\ &= \frac{1}{8} \begin{bmatrix} 2 & 3 & -1 \\ 6 & -3 & 1 \\ 4 & -2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \\ &= \frac{1}{8} \begin{bmatrix} 4+3+1 \\ 12-3-1 \\ 8-2+2 \end{bmatrix} \\ \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{8} \begin{bmatrix} 8 \\ 8 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

Hence by definition of equal matrices $x=1, y=1$ and $z=1$

Q.3 Solve the following system by reducing their augmented matrix to the echelon form and reduced echelon form:

$$(i) \quad \left. \begin{aligned} x_1 - 2x_2 - 2x_3 &= -1 \\ 2x_1 + 3x_2 + x_3 &= 1 \\ 5x_1 - 4x_2 - 3x_3 &= 1 \end{aligned} \right\}$$

Solution: The augmented matrix of the given system is

$$\left[\begin{array}{ccc|c} 1 & -2 & -2 & -1 \\ 2 & 3 & 1 & 1 \\ 5 & -4 & -3 & 1 \end{array} \right]$$

We reduce the above matrix to echelon and reduced echelon form by applying elementary row operations, that is,

$$\left[\begin{array}{ccc|c} 1 & -2 & -2 & -1 \\ 2 & 3 & 1 & 1 \\ 5 & -4 & -3 & 1 \end{array} \right]$$

$$R_{\square} \left[\begin{array}{ccc|c} 1 & -2 & -2 & -1 \\ 0 & 7 & 5 & 3 \\ 0 & 6 & 7 & 6 \end{array} \right]$$

$$\text{By } R_2 + (-2)R_1 \rightarrow R'_2, R_3 + (-5)R_1 \rightarrow R'_3$$

$$R_{\square} \left[\begin{array}{ccc|c} 1 & -2 & -2 & -1 \\ 0 & 1 & -2 & -3 \\ 0 & 6 & 7 & 6 \end{array} \right]$$

$$\text{By } R_2 + (-1)R_3 \rightarrow R'_2$$

$$R_{\square} \left[\begin{array}{ccc|c} 1 & 0 & -6 & -7 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 19 & 24 \end{array} \right]$$

$$\text{By } R_1 + 2R_2 \rightarrow R'_1, R_3 + (-6)R_2 \rightarrow R'_3$$

$$R_{\square} \left[\begin{array}{ccc|c} 1 & 0 & -6 & -7 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & \frac{24}{19} \end{array} \right]$$

$$(I) \quad \text{By } \frac{1}{19}(R_3) \rightarrow R'_3$$

$$R_{\square} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{11}{19} \\ 0 & 1 & 0 & -\frac{9}{19} \\ 0 & 0 & 1 & \frac{24}{19} \end{array} \right]$$

$$(II) \quad \text{By } R_2 + 2R_3 \rightarrow R'_2, R_1 + 6R_3 \rightarrow R'_1$$

(I) is in echelon form the equivalent system is

$$x_1 - 6x_3 = -7 \quad (i)$$

$$x_2 - 2x_3 = -3 \quad (ii)$$

$$x_3 = \frac{24}{19} \quad (iii)$$

From (ii)

$$x_2 - 2\left(\frac{24}{19}\right) = -3$$

$$\Rightarrow x_2 = -3 + \frac{48}{19} = \frac{-9}{19}$$

$$\Rightarrow x_2 = \frac{-9}{19}$$

From (i)

$$x_1 - 6\left(\frac{24}{19}\right) = -7$$

$$\Rightarrow x_1 = -7 + \frac{144}{19}$$

$$\Rightarrow x_1 = \frac{-133 + 144}{19}$$

$$\Rightarrow x_1 = \frac{11}{19}$$

$$\text{Hence } x_1 = \frac{11}{19}, x_2 = \frac{-9}{19} \text{ and } x_3 = \frac{24}{19}$$

Note: Answer in the text books is different

$$x_1 = \frac{11}{19}, x_2 = \frac{-9}{19} \text{ and } x_3 = \frac{-24}{19}$$

(II) is in reduced echelon form the equivalent system in (row) reduced echelon form

$$x_1 = \frac{11}{19}$$

$$x_2 = -\frac{9}{19}$$

$$x_3 = \frac{24}{19}$$

$$(ii) \quad \left. \begin{array}{l} x + 2y + z = 2 \\ 2x + y + 2z = -1 \\ 2x + 3y - z = 9 \end{array} \right\}$$

Solution: The augmented matrix of the given system is

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & : & 2 \\ 2 & 1 & 2 & : & -1 \\ 2 & 3 & -1 & : & 9 \end{array} \right]$$

We reduce the above matrix to echelon and reduced echelon form by applying elementary row operations, that is,

$$R_1 \left[\begin{array}{cccc|c} 1 & 2 & 1 & : & 2 \\ 0 & -3 & 0 & : & -5 \\ 0 & -1 & -3 & : & 5 \end{array} \right]$$

$$\text{By } R_2 + (-2)R_1 \rightarrow R'_2, R_3 + (-2)R_1 \rightarrow R'_3$$

$$R_1 \left[\begin{array}{cccc|c} 1 & 2 & 1 & : & 2 \\ 0 & 1 & 0 & : & \frac{5}{3} \\ 0 & -1 & -3 & : & 5 \end{array} \right]$$

$$\text{By } -\frac{1}{3}(R_2) \rightarrow R'_2$$

$$R_1 \left[\begin{array}{cccc|c} 1 & 0 & 1 & : & -\frac{4}{3} \\ 0 & 1 & 0 & : & \frac{5}{3} \\ 0 & 0 & -3 & : & \frac{20}{3} \end{array} \right]$$

$$\text{By } R_1 + (-2)R_2 \rightarrow R'_1, R_3 + R_2 \rightarrow R'_3$$

$$R_1 \left[\begin{array}{cccc|c} 1 & 0 & 1 & : & -\frac{4}{3} \\ 0 & 1 & 0 & : & \frac{5}{3} \\ 0 & 0 & 1 & : & -\frac{20}{9} \end{array} \right]$$

$$(I) \quad \text{By } \left(\frac{-1}{3} \right) R_3 \rightarrow R'_3$$

$$R_1 \left[\begin{array}{cccc|c} 1 & 0 & 0 & : & \frac{8}{9} \\ 0 & 1 & 0 & : & \frac{5}{3} \\ 0 & 0 & 1 & : & -\frac{20}{9} \end{array} \right]$$

$$(II) \quad \text{By } R_1 + (-1)R_3 \rightarrow R'_1$$

(I) is in echelon form, the equivalent system is

$$x + z = -\frac{4}{3} \quad (i)$$

$$y = \frac{5}{3} \quad (ii)$$

$$z = -\frac{20}{9} \quad (iii)$$

Put the value of z in (i)

$$x - \frac{20}{9} = \frac{-4}{3}$$

$$\Rightarrow x = -\frac{4}{3} + \frac{20}{9} = \frac{-12 + 20}{9} = \frac{8}{9}$$

$$\text{Hence } x = \frac{8}{9}, y = \frac{5}{3} \text{ and } z = -\frac{20}{9}$$

(II) is in reduce echelon form, the equivalent system is

$$x = \frac{8}{9}, y = \frac{5}{3}, z = -\frac{20}{9}$$

$$(iii) \quad \left. \begin{aligned} x_1 + 4x_2 + 2x_3 &= 2 \\ 2x_1 + x_2 - 2x_3 &= 9 \\ 3x_1 + 2x_2 - 2x_3 &= 12 \end{aligned} \right\}$$

Solution: The augmented matrix of the given system is

$$\left[\begin{array}{cccc|c} 1 & 4 & 2 & : & 2 \\ 2 & 1 & -2 & : & 9 \\ 3 & 2 & -2 & : & 12 \end{array} \right]$$

We reduce the above matrix to echelon and reduced echelon form by applying elementary row operation, that is

$$R \rightarrow \left[\begin{array}{cccc|c} 1 & 4 & 2 & : & 2 \\ 0 & -7 & -6 & : & 5 \\ 0 & -10 & -8 & : & 6 \end{array} \right]$$

$$\text{By } R_2 + (-2)R_1 \rightarrow R'_2, R_3 + (-3)R_1 \rightarrow R'_3$$

$$R \rightarrow \left[\begin{array}{cccc|c} 1 & 4 & 2 & : & 2 \\ 0 & 1 & \frac{6}{7} & : & -\frac{5}{7} \\ 0 & -10 & -8 & : & 6 \end{array} \right]$$

$$\text{By } -\frac{1}{7}(R_2) \rightarrow R'_2$$

$$R \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & -\frac{10}{7} & : & \frac{34}{7} \\ 0 & 1 & \frac{6}{7} & : & -\frac{5}{7} \\ 0 & 0 & \frac{4}{7} & : & -\frac{8}{7} \end{array} \right]$$

$$\text{By } R_1 + (-4)R_2 \rightarrow R'_1, R_3 + 10R_2 \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & 0 & -\frac{10}{7} & : & \frac{34}{7} \\ 0 & 1 & \frac{6}{7} & : & -\frac{5}{7} \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$(I) \quad \text{By } \frac{7}{4}(R_3) \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$(II) \text{ By } R_2 + \left(\frac{-6}{7}\right)R_3 \rightarrow R'_2, R_1 + \frac{10}{7}R_3 \rightarrow R'_1$$

(I) is in echelon form. The equivalent system is

$$x_1 + \left(-\frac{10}{7}\right)x_3 = \frac{34}{7} \quad (i)$$

$$x_2 + \frac{6}{7}x_3 = -\frac{5}{7} \quad (ii)$$

$$x_3 = -2 \quad (iii)$$

Put $x_3 = -2$ in (ii) we get

$$x_2 + \frac{6}{7}(-2) = -\frac{5}{7}$$

$$\Rightarrow x_2 = \frac{-5}{7} + \frac{12}{7} = \frac{7}{7} = 1$$

$$\Rightarrow x_2 = 1$$

Put $x_3 = -2$ in (i) we get

$$x_1 + \left(-\frac{10}{7}\right)(-2) = \frac{34}{7}$$

$$\Rightarrow x_1 = \frac{34}{7} - \frac{20}{7} = \frac{14}{7} = 2$$

Hence $x_1 = 2, x_2 = 1$ and $x_3 = -2$ are the required values.

(II) is in reduce echelon form. The equivalent system is $x_1 = 2, x_2 = 1$ and $x_3 = -2$

Q.4 Solve the following system of homogenous linear equations.

$$(i) \quad \left. \begin{aligned} x + 2y - 2z &= 0 \\ 2x + y + 5z &= 0 \\ 5x + 4y + 8z &= 0 \end{aligned} \right\}$$

Solution: Coefficient matrix 'A' is

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 5 \\ 5 & 4 & 8 \end{bmatrix}$$

We convert 'A' into echelon form.

$$\underset{\sim}{R} \begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 9 \\ 0 & -6 & 18 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 5R_1 \rightarrow R'_3$$

$$\underset{\sim}{R} \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & -6 & 18 \end{bmatrix}$$

$$\text{By } -\frac{1}{3}R_2 \rightarrow R'_2$$

$$\underset{\sim}{R} \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 + 6R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equations is

$$x + 2y - 2z = 0 \quad (i)$$

$$y - 3z = 0 \quad (ii)$$

Let $z = t$ then (ii) becomes $y = 3t$. Put the value of $y = 3t$ and $z = t$ in (i) we get

$$x + 6t - 2t = 0$$

$$x = -4t$$

Hence solution set is $x = -4t, y = 3t, z = t, \quad t \in R, \quad t \neq 0$

$$(ii) \quad \left. \begin{aligned} x_1 + 4x_2 + 2x_3 &= 0 \\ 2x_1 + x_2 - 3x_3 &= 0 \\ 3x_1 + 2x_2 - 4x_3 &= 0 \end{aligned} \right\}$$

Solution: Coefficient matrix 'A' is

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 1 & -3 \\ 3 & 2 & -4 \end{bmatrix}$$

Now we convert 'A' into echelon form ; so here

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & -7 & -7 \\ 0 & -10 & -10 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 3R_1 \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & -10 & -10 \end{bmatrix}$$

$$\text{By } -\frac{1}{7}R_2 \rightarrow R'_2$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 + 10R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equations is

$$x_1 + 4x_2 + 2x_3 = 0 \quad (i)$$

$$x_2 + x_3 = 0 \quad (ii)$$

Let $x_3 = t$ then (ii) becomes $x_2 = -t$

Put the value of $x_2 = -t$ and $x_3 = t$ in (i) we get

$$x_1 + 4(-t) + 2t = 0$$

$$x_1 = 2t$$

Hence solution set is $x_1 = 2t, x_2 = -t, x_3 = t, \quad t \in R, \quad t \neq 0$

$$(iii) \quad \left. \begin{aligned} x_1 - 2x_2 - x_3 &= 0 \\ x_1 + x_2 + 5x_3 &= 0 \\ 2x_1 - x_2 + 4x_3 &= 0 \end{aligned} \right\}$$

Solution: The coefficient matrix 'A' is

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 1 & 1 & 5 \\ 2 & -1 & 4 \end{bmatrix}$$

We convert 'A' into echelon form

$$R \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 6 \\ 0 & 3 & 6 \end{bmatrix}$$

$$\text{By } R_2 - R_1 \rightarrow R'_2, R_3 - 2R_1 \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{bmatrix}$$

$$\text{By } \frac{1}{3}R_2 \rightarrow R'_2$$

$$R \rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 - 3R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equations is

$$x_1 - 2x_2 - x_3 = 0 \quad (i)$$

$$x_2 + 2x_3 = 0 \quad (ii)$$

Let $x_3 = t$ then (ii) becomes $x_2 = -2t$

Put the value of $x_2 = -2t$ and $x_3 = t$ in (i) we get

$$x_1 - 2(-2t) - t = 0$$

$$x_1 + 4t - t = 0$$

$$x_1 = -3t$$

Hence solution set $x_1 = -3t, x_2 = -2t, x_3 = t, \quad t \in R, t \neq 0$

Q.5 Find the value of λ for which the following system have non trivial solutions. Also solve the system for the value of λ

$$(i) \quad \left. \begin{aligned} x + y + z &= 0 \\ 2x + y - \lambda z &= 0 \\ x + 2y - 2z &= 0 \end{aligned} \right\}$$

Solution: Let coefficient matrix represented is by A so

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -\lambda \\ 1 & 2 & -2 \end{bmatrix}$$

For non-trivial solution take $|A| = 0$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -\lambda \\ 1 & 2 & -2 \end{vmatrix} = 0$$

Expanding form R_1 we get;

$$1 \begin{vmatrix} 1 & -\lambda \\ 2 & -2 \end{vmatrix} - 1 \begin{vmatrix} 2 & -\lambda \\ 1 & -2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 0$$

$$-2 + 2\lambda - 1(-4 + \lambda) + 1(4 - 1) = 0$$

$$-2 + 2\lambda + 4 - \lambda + 3 = 0$$

$$\lambda + 5 = 0$$

$$\lambda = -5$$

Put $\lambda = -5$ in given system of equations we get;

$$\left. \begin{aligned} x + y + z &= 0 \\ 2x + y + 5z &= 0 \\ x + 2y - 2z &= 0 \end{aligned} \right\}$$

$$\text{Here } A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 5 \\ 1 & 2 & -2 \end{bmatrix}$$

Now we convert 'A' into echelon form, so here

$$R \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 3 \\ 0 & 1 & -3 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - R_1 \rightarrow R'_3$$

$$R_{\square} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{bmatrix}$$

$$\text{By } (-1) \times R_2 \rightarrow R'_2$$

$$R_{\square} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 - R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equation is

$$x + y + z = 0 \quad (i)$$

$$y - 3z = 0 \quad (ii)$$

Let $z = t$ then (ii) becomes $y = 3t$

Put the values of $y = 3t$ and $z = t$ in (i) we get

$$x + 3t + t = 0 \Rightarrow x = -4t$$

Hence solution set $x = -4t, y = 3t, z = t, \quad t \in R, \quad t \neq 0$

$$(ii) \quad \left. \begin{aligned} x_1 + 4x_2 + \lambda x_3 &= 0 \\ 2x_1 + x_2 - 3x_3 &= 0 \\ 3x_1 + \lambda x_2 - 4x_3 &= 0 \end{aligned} \right\}$$

Solution: Let coefficient matrix represented is by A so

$$A = \begin{bmatrix} 1 & 4 & \lambda \\ 2 & 1 & -3 \\ 3 & \lambda & -4 \end{bmatrix}$$

For non-trivial solution put $|A| = 0$

$$\begin{vmatrix} 1 & 4 & \lambda \\ 2 & 1 & -3 \\ 3 & \lambda & -4 \end{vmatrix} = 0$$

$$1 \begin{vmatrix} 1 & -3 \\ \lambda & -4 \end{vmatrix} - 4 \begin{vmatrix} 2 & -3 \\ 3 & -4 \end{vmatrix} + \lambda \begin{vmatrix} 2 & 1 \\ 3 & \lambda \end{vmatrix} = 0$$

$$(-4 + 3\lambda) - 4(-8 + 9) + \lambda(2\lambda - 3) = 0$$

$$-4 + 3\lambda - 4 + 2\lambda^2 - 3\lambda = 0$$

$$2\lambda^2 = 8$$

$$\lambda^2 = 4$$

$$\lambda = \pm 2$$

When $\lambda = 2$ given system of equation becomes;

$$\left. \begin{aligned} x_1 + 4x_2 + 2x_3 &= 0 \\ 2x_1 + x_2 - 3x_3 &= 0 \\ 3x_1 + 2x_2 - 4x_3 &= 0 \end{aligned} \right\}$$

Here

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 1 & -3 \\ 3 & 2 & -4 \end{bmatrix}$$

Now we convert 'A' into echelon form;

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & -7 & -7 \\ 0 & -10 & -10 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 3R_1 \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & -10 & -10 \end{bmatrix}$$

$$\text{By } \frac{-R_2}{7} \rightarrow R'_2$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 + 10R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equations is

$$x_1 + 4x_2 + 2x_3 = 0 \quad (i)$$

$$x_2 + x_3 = 0 \quad (ii)$$

Let $x_3 = t$ then (ii) becomes $x_2 = -t$

Put $x_2 = -t$ and $x_3 = t$ in (i) we get

$$x_1 + 4(-t) + 2t = 0$$

$$x_1 - 2t = 0 \Rightarrow x_1 = 2t$$

So when $\lambda = 2$ then solution set $x_1 = 2t, x_2 = -t, x_3 = t, t \in R, t \neq 0$

When $\lambda = -2$ given system of equation becomes

$$x_1 + 4x_2 - 2x_3 = 0$$

$$2x_1 + x_2 - 3x_3 = 0$$

$$3x_1 - 2x_2 - 4x_3 = 0$$

Here

$$A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 1 & -3 \\ 3 & -2 & -4 \end{bmatrix}$$

Now we convert 'A' into echelon form

$$R \rightarrow \begin{bmatrix} 1 & 4 & -2 \\ 0 & -7 & 1 \\ 0 & -14 & 2 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 3R_1 \rightarrow R'_3$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & -2 \\ 0 & 1 & -\frac{1}{7} \\ 0 & -14 & 2 \end{bmatrix}$$

$$\text{By } \frac{R_2}{-7} \rightarrow R'_2$$

$$R \rightarrow \begin{bmatrix} 1 & 4 & -2 \\ 0 & 1 & -\frac{1}{7} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{By } R_3 + 14R_2 \rightarrow R'_2$$

Since matrix is in echelon form so equivalent system of equations is

$$x_1 + 4x_2 - 2x_3 = 0 \quad (i)$$

$$x_2 - \frac{1}{7}x_3 = 0 \quad (ii)$$

Let $x_3 = 7t$ then (ii) becomes $x_2 = t$

Put $x_2 = t$ and $x_3 = 7t$ in (i) we get

$$x_1 + 4t - 14t = 0$$

$$x_1 = 10t$$

Hence solutions set $x_1 = 10t, x_2 = t, x_3 = 7t, t \in R, t \neq 0$

Q.6 Find the value of λ for which the following system does not possess a unique solution. Also solve the system for the value of λ .

$$\left. \begin{aligned} x_1 + 4x_2 + \lambda x_3 &= 2 \\ 2x_1 + x_2 - 2x_3 &= 11 \\ 3x_1 + 2x_2 - 2x_3 &= 16 \end{aligned} \right\}$$

Solution: Let coefficient matrix be represented by A then

$$A = \begin{bmatrix} 1 & 4 & \lambda \\ 2 & 1 & -2 \\ 3 & 2 & -2 \end{bmatrix}$$

Non-homogeneous system of linear equation doesn't possess unique solution when

$$|A| = 0$$

$$|A| = \begin{vmatrix} 1 & 4 & \lambda \\ 2 & 1 & -2 \\ 3 & 2 & -2 \end{vmatrix} = 0$$

Expanding form R_1

$$1 \begin{vmatrix} 1 & -2 \\ 2 & -2 \end{vmatrix} - 4 \begin{vmatrix} 2 & -2 \\ 3 & -2 \end{vmatrix} + \lambda \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = 0$$

$$1(-2+4) - 4(-4+6) + \lambda(4-3) = 0$$

$$2 - 8 + \lambda = 0$$

$$-6 + \lambda = 0$$

$$\lambda = 6$$

Put $\lambda = 6$ in give system of equation we get;

$$x_1 + 4x_2 + 6x_3 = 2$$

$$2x_1 + x_2 - 2x_3 = 11$$

$$3x_1 + 2x_2 - 2x_3 = 16$$

Augmented matrix A_b is

$$A_b = \left[\begin{array}{ccc|c} 1 & 4 & 6 & 2 \\ 2 & 1 & -2 & 11 \\ 3 & 2 & -2 & 16 \end{array} \right]$$

Now we convert into echelon form

$$\sim R \begin{bmatrix} 1 & 4 & 6 & | & 2 \\ 0 & -7 & -14 & | & 7 \\ 0 & -10 & -20 & | & 10 \end{bmatrix}$$

$$\text{By } R_2 - 2R_1 \rightarrow R'_2, R_3 - 3R_1 \rightarrow R'_3$$

$$\sim R \begin{bmatrix} 1 & 4 & 6 & | & 2 \\ 0 & 1 & 2 & | & -1 \\ 0 & -10 & -20 & | & 10 \end{bmatrix}$$

$$\text{By } -\frac{1}{7}R_2 \rightarrow R'_2$$

$$\square R \begin{bmatrix} 1 & 4 & 6 & | & 2 \\ 0 & 1 & 2 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\text{By } R_3 + 10R_2 \rightarrow R'_3$$

Since matrix is in echelon form so equivalent system of equation is

$$x_1 + 4x_2 + 6x_3 = 2 \quad (i)$$

$$x_2 + 2x_3 = -1 \quad (ii)$$

Let $x_3 = t$ then equation (ii) becomes

$$x_2 + 2t = -1$$

$$x_2 = -2t - 1$$

Put value of x_2 and x_3 in equation (i)

$$x_1 + 4(-2t - 1) + 6(t) = 2$$

$$x_1 - 8t - 4 + 6t = 2$$

$$x_1 = 6 + 2t$$

$$\Rightarrow x_1 = 2t + 6$$

Hence

Solution set $x_1 = 2t + 6, x_2 = -2t - 1, x_3 = t, t \in R, t \neq 0$